

MGM University

Chhatrapati Sambhajinagar

MGMUNIVERSITY

Name of Faculty – Engineering & Technology

Name of College – School of Engineering & Technology

Name of Department – Electronics and Telecommunication Engineering

Name of Programme – B.Tech Electronics Engineering (VLSI Design and Technology)

CURRICULUM BOOKLET

(With effect from Academic Year 2023-24)

MGM University

Vision

- To ensure sustainable human development which encourages self-reliant and self-content society.
- To promote activities related to community services, social welfare and also Indian heritage and culture.
- To inculcate the culture of non-violence and truthfulness through vipassanna meditation and Gandhian Philosophy.
- To develop the culture of simple living and high thinking

Mission

MGMUNIVERSITY

- To impart state of art education and technical expertise to students and give necessary training to teachers to create self-reliant society for future.
- To encourage students to participate in Indian and International activities in sports, literature, etc. so that future generation becomes base for free and liberal society
- To educate students in areas like Management, Finance, Human relations to inculcate philosophy of simple living and high thinking value of simple economic society.
- To inculcate culture of non-violence and truthfulness through Vipassana.

  To sustain activities of Indian culture (viz. classical dance, music and fine arts) through establishing institutes like Mahagami, Naturopathy, etc.

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MGMUNIVERSITY

School of Engineering and Technology (SOET) at a Glance

The School of Engineering and Technology (SoET), a self-aided, non-minority, conducted college of MGM University is the first institution to provide integrated B.Tech and M.Tech programs in the Marathwada region of Maharashtra. A sister institution of the illustrious Jawaharlal Nehru Engineering College (JNEC), Aurangabad, it was established as MGM Polytechnic College in 2010. However, recognizing the rising need of quality engineering education after schooling in the region and beyond, it was transformed into the School of Engineering and Technology (SOET) in 2022.

Inheriting a 39 years rich legacy of imparting world-class professional education from MGM Trust, SoET provides ultramodern research and innovation facilities to its students. Its integrated programs eliminate the need to appear for 12th standard and various engineering entrance tests, thus saving valuable time and energy of the students. It allows the students to utilize the same time to gain more practical knowledge of the field and a strong edge over the competition.

The college offers a wide variety of Undergraduate (Including Six Years Integrated Programs after 10th), Postgraduate (Including Five Years Integrated Programs after 12th) and PG Diploma programs to fulfil the dreams of the students. All these are advanced engineering programs founded on industry-oriented curriculum.

The Institution employs many modern teaching-learning methods like Project-based Learning, Research-based Learning and Collaborative Learning to develop comprehensive understanding and real-life skills in the students. The School has also signed several memoranda of understanding with renowned institutions, associations and industries to provide internship and project work opportunities to the students. It also joined hands with NPTEL, SWAYAM, ATAL and other MOOC platforms to facilitate blended learning facilities for the students.

Unique in its structure, methods and goals, the institution is strongly rooted in the philosophy of training and research that enhances the relationship between knowledge and its application and seeks to promote the creation of an ideal society through holistic education.

Vision

Enrich the students in technical education to work globally with human values.

Mission

- To inculcate technical skills by continuous assessment practices.
- To strengthen interaction between industry and institute.
- To involve the students in social activities with environmental awareness.

Programs offered at SOET

Undergraduate Programmes	Postgraduate Programmes	Ph.D. Programmes	Certificate Programmes
B.Tech (Integrated) Civil Engineering with Computer Applications	M. Tech (Integrated) Computer Science & Engineering (AI & Machine Learning)	Ph.D. Energy Engineering	Diploma In Electric Vehicle Technology
B.Tech (Integrated) Computer Science & Engineering (Data Science)	M.Tech Energy Engineering	Ph.D. Medical Technology	Diploma In Drone Technology
B.Tech (Integrated) Computer Science & Engineering (IoT, Cyber Security Including Block Chain Technology)	M.Tech Medical Technology	Ph.D. Environmental Engineering	
B.Tech (Integrated) Advanced Mechatronics & Industrial Automation	M.Tech Environmental Engineering	Ph.D. Design Engineering	
B.Tech (Integrated) Robotics & AI	M.Tech Design Engineering		
B.Tech (Integrated) Electrical & Computer Engineering			
B.Tech (Integrated) Information Technology			
B.Tech Agriculture Engineering			
B.Tech Computer Science & Engineering (IoT, Cyber Security Including Block Chain Technology)			
B.Tech Electrical & Instrumentation Engineering			
B.Tech Electronics Engineering (VLSI Design & Technology)			
B.Tech Computer Science & Design			
B.Tech Robotics & Automation			

MGMUNIVERSITY

Department of Electronics and Telecommunication Engineering

The Bachelor of Technology (B.Tech) in Electronics Engineering (VLSI Design & Technology) degree program started in the year 2023 and offers Choice Based Credit System education. In addition to core courses, students can opt for discipline specific elective subjects, open elective subjects from different institutes of the University. In addition, this program is uniquely designed to increase the employability and to prepare students to work in a Multi-disciplinary work environment. The program offers Major degrees in Electronics Engineering and is open to students opting for minor specializations as per their interests. Pedagogies concentrating on student's active participation are extensively used in the teaching-learning process.

Vision

To provide update technology in Electronics and Telecommunication Engineering to fulfill needs of industry and society.

Mission

- To impart technical knowledge and skills by involving students in educational activities.
- To enable students to be entrepreneurs enhancing interaction between institute and industry.
- To nurture moral and professional values to build up society and nation.

Eligibility –

1. Maharashtra State Candidate.

(i) The Candidate should be an Indian National and having domicile of Maharashtra state and/or born in Maharashtra state.

(ii) Passed HSC or its equivalent examination with Physics and Mathematics as compulsory subjects along with one of the Chemistry or Biotechnology or Biology or Technical Vocational subject or Computer Science or Information Technology or Informatics Practices or Agriculture or Engineering Graphics or Business Studies, and obtained at least 45% marks (at least 40% marks, in case of Backward class categories and Persons with Disability candidates belonging to Maharashtra State only) in the above subjects taken together and the candidate should have appeared in MGMU-CET / MHT-CET / PERA CET / JEE (Main) Paper-I and should obtain non-zero score in MGMU-CET / MHT-CET / PERA CET / JEE (Main) Paper-I. However, preference shall be given to the candidate obtaining non-zero positive score in MGMU-CET over the candidates who obtained non-zero score in MHT-CET / PERA CET.

OR

(ii)Passed Diploma in Engineering and Technology and obtained at least 45% marks (at least 40% marks, in case of Backward class categories and Persons with Disability candidates belonging to Maharashtra State only).

2. All India Candidates –

(i) The Candidate should be an Indian National.

(ii) Passed HSC or its equivalent examination with Physics and Mathematics as compulsory subjects along with one of the Chemistry or Biotechnology or Biology or Technical Vocational subject or Computer Science or Information Technology or Informatics Practices or Agriculture or Engineering Graphics or Business Studies , and obtained at least 45% marks (at least 40% marks, in case of Backward class categories and Persons with Disability candidates belonging to Maharashtra State only) in the above subjects taken together and candidate should have appeared in MGMU-CET / MHT-CET / PERA CET / JEE (Main)

Paper-I and should obtain non-zero score in MGMU-CET / MHT-CET / PERA CET / JEE (Main) Paper-I. However, preference shall be given to the candidate obtaining non-zero positive score in JEE Mains Paper-I over the candidates who obtained non-zero score in MGMU-CET / MHT-CET / PERA CET.

OR

(ii) Passed Diploma in Engineering and Technology and obtained at least 45% marks (at least 40% marks, in case of Backward class categories and Persons with Disability candidates belonging to Maharashtra State only)

MGMUNIVERSITY

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Electronics Engineering (VLSI Design &Technology) with Multidisciplinary Minor

Programme Type (UG/PG): UG

Duration: 04 Years

First Year - Semester I (Group A)												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs / Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
BSC	APS21BSL101	Single and Multivariable Calculus	Theory	4	4	-	60	40	100	-	16	40
BSC	APS21BSL102	Engineering Physics	Theory	3	3	-	60	40	100	-	16	40
ESC	APS21ESL101	Python Programming	Theory	2	2	-	60	40	100	-	16	40
ESC	APS21ESL102	Engineering Graphics	Theory	2	2	-	60	40	100	-	16	40
AEC	MGM54AEL101	Communicative English	Theory	1	1	-	30	20	50	-	8	20
VSEC	APS21VSP101	Engineering Exploration	Practical	2	-	4	60	40	100	-	16	40
BSC	APS21BSP101	Engineering Physics Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP101	Python Programming Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP102	Engineering Graphics Studio	Practical	2	-	4	30	20	50	-	8	20
ESC	APS21ESP103	Recent Trends in Integrated Technology	Practical	1	-	2	30	20	50	-	8	20
AEC	MGM54AEP101	Communicative English Lab	Practical	1	-	2	30	20	50	-	8	20
CCA	MGM82CCP101 MGM82CCP102 MGM82CCP103	NCC / Yoga / Sports	Practical	2	-	4	30	20	50	-	8	20
Total				**22**	**12**	**20**	**510**	**340**	**850**	**-**	**-**	**-**

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC-Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training /

Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

MGMUNIVERSITY

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Electronics Engineering (VLSI Design & Technology) with Multidisciplinary Minor

Programme Type (UG/PG): UG

Duration: 04 Years

First Year- Semester II (Group A)												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
BSC	APS21BSL103	Linear Algebra and Differential Equations	Theory	4	4	-	60	40	100	-	16	40
BSC	APS21BSL104	Engineering Chemistry	Theory	3	3	-	60	40	100	-	16	40
ESC	APS21ESL103	Engineering Mechanics	Theory	2	2	-	60	40	100	-	16	40
ESC	APS21ESL104	Building Programming logic in C	Theory	1	1	-	30	20	50	-	8	20
PCC	APS21PCL101	Basics of Electrical and Electronics Engineering	Theory	2	2	-	60	40	100	-	16	40
IKS	APS21IKL1XX	Indian Knowledge System	Theory	2	2	-	60	40	100	-	16	40
VSEC	APS21VSP102	Workshop Practices	Practical	2	-	4	60	40	100	-	16	40
BSC	APS21BSP102	Engineering Chemistry Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP104	Engineering Mechanics Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP105	Building Programming logic in C Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	APS21PCP101	Electrical and Electronics Technology Lab	Practical	1	-	2	30	20	50	-	8	20
CCA	MGM82CCP104 MGM73CCP105 MGM73CCP106	NSS/ Fine Art/ Visual Art	Practical	2	-	4	30	20	50	-	8	20
Total				**22**	**14**	**16**	**540**	**360**	**900**	**-**	**-**	**-**

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC-Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

Level 4.5 Award of UG Certificate with 40 Credits and an additional 4-Credits Core NSQF Course / Internship OR continue with Major and Minor

MGMUNIVERSITY

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Electronics Engineering (VLSI Design & Technology) with Multidisciplinary Minor

Programme Type (UG/PG): UG

Duration: 04 Years

First Year - Semester I (Group B)												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs / Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
BSC	APS21BSL101	Single and Multivariable Calculus	Theory	4	4	-	60	40	100	-	16	40
BSC	APS21BSL104	Engineering Chemistry	Theory	3	3	-	60	40	100	-	16	40
ESC	APS21ESL101	Python Programming	Theory	2	2	-	60	40	100	-	16	40
ESC	APS21ESL103	Engineering Mechanics	Theory	2	2	-	60	40	100	-	16	40
AEC	MGM54AEL101	Communicative English	Theory	1	1	-	30	20	50	-	8	20
PCC	APS21PCL101	Basics of Electrical and Electronics Engineering	Theory	2	2	-	60	40	100	-	16	40
VSEC	APS21VSP102	Workshop Practices	Practical	2	-	4	60	40	100	-	16	40
BSC	APS21BSP102	Engineering Chemistry Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP101	Python Programming Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP104	Engineering Mechanics Lab	Practical	1	-	2	30	20	50	-	8	20
AEC	MGM54AEP101	Communicative English Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	APS21PCP101	Electrical and Electronics Technology Lab	Practical	1	-	2	30	20	50	-	8	20
CCA	MGM82CCP101 MGM82CCP102 MGM82CCP103	NCC / Yoga / Sports	Practical	2	-	4	30	20	50	-	8	20
Total				**23**	**14**	**18**	**570**	**380**	**950**	**-**	**-**	**-**

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC-Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

MGMUNIVERSITY

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Electronics Engineering (VLSI Design & Technology) with Multidisciplinary Minor

Programme Type (UG/PG): UG

Duration: 04 Years

First Year- Semester II (Group B)												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
BSC	APS21BSL103	Linear Algebra and Differential Equations	Theory	4	4	-	60	40	100	-	16	40
BSC	APS21BSL102	Engineering Physics	Theory	3	3	-	60	40	100	-	16	40
ESC	APS21ESL102	Engineering Graphics	Theory	2	2	-	60	40	100	-	16	40
IKS	APS21IKL1XX	Indian Knowledge System	Theory	2	2	-	60	40	100	-	16	40
ESC	APS21ESL104	Building Programming Logic in C	Theory	1	1	-	30	20	50	-	8	20
VSEC	APS21VSP101	Engineering Exploration	Practical	2	-	4	60	40	100	-	16	40
BSC	APS21BSP101	Engineering Physics Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP102	Engineering Graphics Studio	Practical	2	-	4	30	20	50	-	8	20
ESC	APS21ESP105	Building Programming Logic in C Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP103	Recent Trends in Integrated Technology	Practical	1	-	2	30	20	50	-	8	20
CCA	MGM82CCP104 MGM73CCP105 MGM73CCP106	NSS/ Fine Art/ Visual Art	Practical	2	-	4	30	20	50	-	8	20
Total				**21**	**12**	**18**	**480**	**320**	**800**	**-**	**-**	**-**

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC-Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

Level 4.5 Award of UG Certificate with 40 Credits and an additional 4-Credits Core NSQF Course / Internship OR continue with Major and Minor

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Electronics Engineering (VLSI Design & Technology) with Multidisciplinary Minor

Programme Type (UG/PG): UG

Duration: 04 Years

Semester III												
Course Category	Course Code	Course Title	Nature of Course	No of Credits	Teaching (Contact hrs/ week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	ELE24PCL201	Electronic Device & Circuits	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP201	Electronic Device & Circuits Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	ELE24PCL202	Digital System Design	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP202	Digital System Design Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	ELE24PCL203	Data Structure	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP203	Data Structure Lab	Practical	1	-	2	30	20	50	-	8	20
EEMC	ELE24HSL201	Business Management & Financial Accounting	Lecture	2	2	-	60	40	100	-	16	40
CEP/FP	ELE24CEP201	Community Engagement	Practical	2	-	4	50	-	50	20	-	20
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50	-	8	20
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50	-	8	20
VEC	MGM56VEL102	Constitution of India	Lecture	2	2	-	30	20	50	-	8	20
		TOTAL		**21**	**16**	**10**	**530**	**340**	**850**	**-**	**-**	**-**

Honor (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	Refer the Honor Basket Structure Appendix B											

Double Minor (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	Refer the Double Minor Basket Structure Appendix C											

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC- Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

MGMUNIVERSITY

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Electronics Engineering (VLSI Design & Technology) with Multidisciplinary Minor

Programme Type (UG/PG): UG

Duration: 04 Years

Semester IV												
Course Category	Course Code	Course Title	Nature of Course	No of Credits	Teaching (Contact hrs/ week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	ELE24PCL204	Network Theory	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP204	Network Theory Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	ELE24PCL205	Analog Circuits	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP205	Analog Circuits Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	ELE24PCL206	Analog & Digital Communication	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP206	Analog & Digital Communication Lab	Practical	1	-	2	30	20	50	-	8	20
VSEC	ELE24VSP201	Introduction to Microfabrication	Practical	2	-	4	60	40	100	-	16	40
EEMC	ELE24HSL202	Entrepreneurship Development	Lecture	2	2	-	60	40	100	-	16	40
AEC		Refer MGMU AEC Basket	Lecture	2	2	-	30	20	50	-	8	20
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50	-	8	20
VEC	MGM21VEL101	Environmental Studies	Lecture	2	2	-	30	20	50	-	8	20
		TOTAL		**21**	**16**	**10**	**540**	**360**	**900**	**-**	**-**	**-**

•Exit option to qualify for UG Diploma in Electronics Engineering (VLSI Design & Technology): After securing all credits of first and second year and provided the student secures additional 8 credits in skill-based vocational courses (skill-based courses, internship, mini projects etc) offered during summer vacation after the first year or second year. (As per Annexure - A)

• Skill-based Vocational Course : 08 Credits

Honor (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	Refer the Honor Basket Structure Appendix B											

Double Minor (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	Refer the Double Minor Basket Structure Appendix C											

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC- Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Electronics Engineering (VLSI Design & Technology) with Multidisciplinary Minor

Programme Type (UG/PG): UG

Duration: 04 Years

Semester V												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	ELE24PCL301	Power Electronics	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP301	Power Electronics Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	ELE24PCL302	Electronics Measurement and Instruments	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP302	Electronics Measurement and Instruments Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	ELE24PCL303	VLSI Design with Verilog	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP303	VLSI Design with Verilog Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	ELE24PCL304	Mobile Cellular Communication	Lecture	2	2	-	60	40	100	-	16	40
Program Elective - I												
PEC	ELE24PEL301	Group A : Embedded Systems Design	Lecture	3	3	-	60	40	100	-	16	40
	ELE24PEL302	Group B : Basics of IOT										
	ELE24PEL303	Group C: Python for AI & ML										
PEC	ELE24PEP301	Group A : Embedded Systems Design Lab	Practical	1	-	2	30	20	50	-	8	20
	ELE24PEP302	Group B : Basics of IOT Lab										
	ELE24PEP303	Group C: Python for AI & ML Lab										
MDM		Refer MGMU MDM Basket	Lecture	4	4	-	60	40	100	-	16	40
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50	-	8	20
		TOTAL		**21**	**17**	**8**	**510**	**340**	**850**	**-**	**-**	**-**

Honor (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	Refer the Honor Basket Structure Appendix B											

Double Minor (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	Refer the Double Minor Basket Structure Appendix C											

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC- Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Electronics Engineering (VLSI Design & Technology) with Multidisciplinary Minor

Programme Type (UG/PG): UG

Duration: 04 Years

Semester VI												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	ELE24PCL305	VLSI Verification and Testing	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP305	VLSI Verification and Testing Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	ELE24PCL306	Semiconductor Equipment Design and Technology	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP306	Semiconductor Equipment Design and Technology Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	ELE24PCL307	Sensor and Signal Conditioning	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP307	Sensor and Signal Conditioning Lab	Practical	1	-	2	30	20	50	-	8	20
Program Elective - II												
PEC	ELE24PEL304	Group A : Low Power VLSI Design	Lecture	3	3	-	60	40	100	-	16	40
	ELE24PEL305	Group B : Hardware and Communication Protocols in IOT										
	ELE24PEL306	Group C : Machine Learning										
PEC	ELE24PEP304	Group A : Low Power VLSI Design Lab	Practical	1	-	2	30	20	50	-	8	20
	ELE24PEP305	Group B : Hardware and Communication Protocols in IOT Lab										
	ELE24PEP306	Group C : Machine Learning Lab										
Program Elective - III												
PEC	ELE24PEL307	Robotics	Lecture	3	3	-	60	40	100	-	16	40
	ELE24PEL308	Automotive Electronics										
	ELE24PEL309	Mobile Computing										
PEC	ELE24PEP307	Robotics Lab	Practical	1	-	2	30	20	50	-	8	20

	ELE24PEP308	Automotive Electronics Lab										
	ELE24PEP309	Mobile Computing Lab										
VSEC	ELE24VSL301	Computer Hardware and Networking	Lecture	1	1	-	30	20	50	-	8	20
VSEC	ELE24VSP301	Computer Hardware and Networking Lab	Practical	1	-	2	30	20	50	-	8	20
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
		TOTAL		**21**	**15**	**12**	**570**	**380**	**950**	**-**	**-**	**-**

•Exit option to qualify for B.Voc in Electronics Engineering (VLSI Design & Technology): After securing all credits of first, second and third year with additional 6 credits in skill-based vocational courses (skill-based courses, internship, mini projects etc) offered during summer vacation after the sixth semester. (As per Annexure - A).

• Skill-based Vocational Course : 06 Credits

Honor (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	Refer the Honor Basket Structure Appendix B											

Double Minor (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	Refer the Double Minor Basket Structure Appendix C											

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC- Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Electronics Engineering (VLSI Design & Technology) with Multidisciplinary Minor

Programme Type (UG/PG): UG

Duration: 04 Years

Semester - VII												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs/ Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	ELE24PCL401	CMOS Digital VLSI Design	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCL402	Thin Films Characterization	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCL403	High Speed Interconnects for VLSI Design	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCL404	Introduction to MEMS	Lecture	2	2	-	60	40	100	-	16	40
PCC	ELE24PCP403	High Speed Interconnects for VLSI Design Lab	Practical	1	-	2	30	20	50	-	8	20
RM	ELE24RML401	Research Methodology	Lecture	4	4	-	60	40	100	-	16	40
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
Project	ELE24RPJ401	Project	Practical	4	-	8	60	40	100	-	16	40
		TOTAL		**21**	**16**	**10**	**510**	**340**	**850**	**-**	**-**	**-**

Double Minor (In addition to the List of Courses in the above Table)

Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs/ Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	Refer the Double Minor Basket Structure Appendix C											

Honor with Research (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
RP	ELE24 RPJ402	Research Project Part - I	Project	8	-	16	50	100	150	-	40	60

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC- Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

MGMUNIVERSITY

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Electronics Engineering (VLSI Design & Technology) with Multidisciplinary Minor

Programme Type (UG/PG): UG

Duration: 04 Years

Semester - VIII												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs/ Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
Internship	ELE24JTI401	Internship	Practical	12	-	24	100	50	150	-	20	60
Program Elective - IV												
PEC	ELE24PEL401	Group A : Digital Signal Processing	Lecture	2	2	-	60	40	100	-	16	40
	ELE24PEL402	Group B : Wireless Sensor Network										
	ELE24PEL403	Group C : Artificial Intelligence										
Program Elective - V												
PEC	ELE24PEL404	App Development	Lecture	3	3	-	60	40	100	-	16	40
	ELE24PEL405	Cloud Computing										
	ELE24PEL406	Industry 4.0										
Program Elective - VI												
PEC	ELE24PEL407	Data Science	Lecture	3	3	-	60	40	100	-	16	40
	ELE24PEL408	E-Vehicle										
	ELE24PEL409	Agriculture Electronics										
		TOTAL		**20**	**08**	**24**	**280**	**170**	**450**	**-**	**-**	**-**

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC-Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

Honor (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	Refer the Honor Basket Structure Appendix B											

Double Minor (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
PCC	Refer the Double Minor Basket Structure Appendix C											

Honor with Research (In addition to the List of Courses in the above Table)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs/ Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
RP	ELE24 RPJ403	Research Project Part - II	Project	10	-	20	50	100	150	-	40	60

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Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC- Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

Annexure – A

Exit Options after FY, SY and TY

1. **Exit option to qualify for Certification, common at Institute level: After securing all credits of First Year and securing 8 credits in work-based vocational courses or internship / Apprenticeship offered during summer vacation will be awarded One Year UG certificate in Tech(Discipline) (As per Annexure- A).**

   •Work-based vocational course: 08 credits

2. **Exit option to qualify for UG Diploma in Electronics Engineering (VLSI Design & Technologywith Multidisciplinary Minor: After securing all credits of first and second year and provided the student secures additional 8 credits in skill-based vocational courses (Skill-Based Courses, Internship, Mini Projects etc.) offered during summer vacation after the first year or second year.(As per Annexure – A)**

   • Skill-based vocational course: 08 credits

3. **Exit option to qualify for B.Voc in Electronics Engineering (VLSI Design & Technology) With Multidisciplinary Minor: After securing all credits of first, second and third year with additional 6 credits in skill-based vocational courses (Skill-Based Courses, Internship, Mini Projects etc.) offered during summer vacation after the sixth semester (As per Annexure – A).**

Annexture - A
Exit Options After FY, SY and TY

1) NAME OF THE PROGRAMME: One Year UG Certificate in Electronics Engineering (VLSI Design & Technology)
After Securing All Credits of First Year

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs / Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
VSC	ELE24 VSI101	Electrical Vehicles	Practical	8	-	16	60	40	100	-	16	40

2) NAME OF THE PROGRAMME: UG Diploma in Electronics Engineering (VLSI Design & Technology)

After Securing All Credits of Second Year

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs / Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
VSC	ELE24 VSI201	Home Appliances Maintainance and Repair	Practical	8	-	16	60	40	100	-	16	40

3)NAME OF THE PROGRAMME: B.Voc. in Electronics Engineering (VLSI Design & Technology)

After Securing All Credits of Third Year

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs / Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
VSC	ELE24 VSI301	Industrial Automation	Practical	6	-	12	60	40	100	-	16	40

Appendix - B

Honor Offered

Sr. No.	Title of Honors	Program Offering
1.	**Power System Simulation**	B.Tech Electronics Engineering (VLSI Design and Technology)
2.	**Cyber Security**	

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Basket of Honors

Students after FY can opt for B.Tech Degree with Honors. Eligibility for admission will be a minimum CGPA of 7.5 at First Year.

The students will have to take additional 5 Courses of 20 Credits in Electronics Engineering (VLSI Design and Technology) discipline distributed over semesters III to VIII and will be awarded **B.Tech in Electronics Engineering (VLSI Design and Technology)**with Multidisciplinary Minor and Honors in**Power System Simulation.**

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs / Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)			Semester
					L	P	Internal	External	Total	Internal	External	Total	
PCC	EIE24HNL201	Power System Simulation	Lecture	4	4	-	60	40	100	-	16	40	III
PCC	EIE24HNL202	Power System Protection	Lecture	4	4	-	60	40	100	-	16	40	IV
PCC	EIE24HNL301	Electrical Power System	Lecture	4	4	-	60	40	100	-	16	40	V
PCC	EIE24HNL302	Electrical System Design	Lecture	4	4	-	60	40	100	-	16	40	VI
PCC	EIE24HNL401	SCADA System and Application	Lecture	4	4	-	60	40	100	-	16	40	VIII
TOTAL				**20**	**20**	**0**	**300**	**200**	**500**	-	-	-	

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC- Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

Basket of Honors

The students will have to take additional 5 Courses of 20 Credits in Electronics Engineering (VLSI Design and Technology) discipline distributed over semesters III to VIII and will be awarded **B.Tech in Electronics Engineering (VLSI Design and Technology)**with Multidisciplinary Minor and Honors in**Cyber Security.**

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs / Week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)			Semester
					L	P	Internal	External	Total	Internal	External	Total	
PCC	ELE24HNL201	Introduction of Cyber Security	Lecture	4	4	-	60	40	100	-	16	40	III
PCC	ELE24HNL202	Risk Analysis and Policy Design In Cyber Security	Lecture	4	4	-	60	40	100	-	16	40	IV
PCC	ELE24HNL301	Ethical Hacking	Lecture	4	4	-	60	40	100	-	16	40	V
PCC	ELE24HNL302	Cloud Security	Lecture	4	4	-	60	40	100	-	16	40	VI
PCC	ELE24HNL401	Case Studies in Cyber Security	Lecture	4	4	-	60	40	100	-	16	40	VIII
TOTAL				**20**	**20**	**0**	**300**	**200**	**500**	-	-	-	-

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation.

Course Category: PCC-Program Core Course, PEC-Program Elective Course, MDM-Multidisciplinary Minor, OE-Open Elective, EEMC- Entrepreneurship/Economic/Management Course, VSEC-Vocational Skill and Skill Enhancement Course, AEC-Ability Enhancement Course, IKS-Indian Knowledge System, VEC-Value Education Course, OJT-On Job Training / Internship / Apprenticeship, FP-Field Project, CEP-Community Engagement and Service, CC-Curricular course, RM-Research Methodology, RP-Research Project

Syllabus
Semester –I

Course Code:APS21BSL101 **Course Name:** Single and Multivariable Calculus **Course Category:**

Credits: 4 **Teaching scheme:** L-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university mathematics.

Course Objectives:
1. To provide the basic tools of calculus for the purpose of modelling the engineering problems mathematically and obtaining solutions.
2. To convey a sense of the utility of calculus and develop technical competence of the student.

Course Outcomes:
On completion of the course, the student should be able to:
CO1. Compute derivatives in engineering problems.
CO2. Compute definite integrals arise in the problems such as arc length, surface of revolution, Work and Fluid Forces.
CO3. Find the partial derivatives and apply the knowledge of partial differentiation to find maxima and minima of functions, Jacobians, estimating error and approximation.
CO4. Calculate Area, Volume, Center of mass and Gravity using Double and Triple integral.
CO5. Solve the problems on fundamental theorems of vector calculus such as Green's, Stokes and Divergence theorems.

Contents:

Unit	Content	Teaching Hours
1	**Differentiation and its Applications:** Limit ,Continuity and Differentiation, Rate of Change in sciences and Engineering, Chain Rule and implicit differentiation, Related rates, Extreme value theorem, Rolle's Theorem, Lagrange's Mean value theorems, Nth derivatives, Taylor and Maclaurin series Expansions, Linear approximations and Differentials.	10
2	**Integration and its Applications:** Integration, Reduction formulae, Beta and gamma function, Properties, Evaluation of integrals using Beta and gamma functions, Application of Definite integrals to volume, arc length, surface of revolution, Work and Fluid Forces.	10
3	**Partial Differentiation and its Applications:** Limit and Continuity, Partial derivatives of first and higher orders, Total differentials, Errors and Approximations, Total derivative, Extreme values and saddle points, Method of Lagrange multipliers, Jacobians, Vector differentiation, Gradient, Curl and Divergence, directional derivatives.	10
4	**Multiple Integrals and its Applications:** Double integrals ,Evaluation of Double integrals**,** Change of order of	10

	integration in double integrals,Change of variables (Cartesian to polar), Applications: Area and Volume, Root mean square value, Center of mass and Gravity (constant and variable densities); Triple integrals (Cartesian),Volume by triple integration.	
5	**Vector Integration and its Applications:** Line integrals, Surface integrals, Green's Theorem, Stokes Theorems, Divergence theorems.	10

Text Books:

1. James Stewart, Calculus Early Transcendental, 7th edition, Cengage.

2. George B. Thomas, Ross L. Finney, Calculus and Analytical Geometry, 9th edition, Pearson.

3. Howard Anton, Irl Bivens, Stephens Davis, Calculus, 10th Edition, Wiley.

Reference Books:

1. Shanti Narayan, Differential Calculus, S. Chand & Co.

2. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, New York.

3. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi.

4. P. N. Wartikar, J. N. Wartikar, Applied Mathematics (Vol I & II) , Pune Vidyarthi Griha Prakashan, Pune.

5. H. K. Das and Rajnish Verma, Higher Engineering Mathematics , S. Chand & CO. Pvt. Ltd., New Delhi.

6. K. D Joshi, Calculus for Scientists and Engineers, CRC Press.

7. Prasad and Reena Garg, Advanced Engineering Mathematics, Khanna Publishing Company Private Limited, New Delhi.

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Course Code:APS21BSL102 **Course Name:**Engineering Physics **Course Category:**

Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Student should know Basic Physics and basic Mathematics

Course Objectives:

1. To impart knowledge in basic concepts of physics relevant to engineering applications
2. To introduce advances in technology for engineering applications.

Course Outcomes:

On completion of the course, the student should be able to:

CO.1 Summarize fundamentals of electron optics, modern physics and ultrasonic waves related to the engineering fields.

CO.2 Identify the importance of the optical phenomenon i.e. interference, diffraction and polarization in relevance with its engineering applications.

CO.3 Classify the material on the basis of electric conductivity as semiconductor and superconductors and dielectric materials this leads to their fascinating applications.

CO.4 Recognize the use of laser and optical fibers in various fields.

CO.5 Outline basics of crystallography and X- rays and demonstrate the applications of nano-materials relevant to engineering program.

Contents:

Unit	Content	Teaching Hours
1	**Modern Physics:** Electron Optics: e/m by Thomson's method, Positive ray, Bainbridge mass spectrograph. Quantum Mechanics: Role and concepts, De- Broglie's hypothesis, Uncertainty Principle, Fundamentals of quantum computing, Quantum features Ultrasonic Waves: Production of ultrasonic waves (Magnetostriction & Piezoelectric method), Applications. Numericals.	8
2	**Wave Optics:** Interference- Interference in thin films (reflected light), Newton's Rings, Engineering applications of Interference. Diffraction- Fresnel's and Fraunhofer Diffraction, Theory of plane transmission Grating. Polarization-Polarization by reflection and double refraction, Optical activity, Specific rotation, Construction and working of Laurent's half shade polarimeter, Engineering applications of Polarization. Numericals.	8
3	**Materials of Technological Importance:** Dielectric Materials: Introduction, Types of polarizations: Electronic and Ionic, Orientation Polarizations - Applications of Dielectrics Semiconducting Materials: Introduction, Fermi energy in Intrinsic	8

	semiconductors and extrinsic semiconductors, Hall effect, Applications of Semiconductors. Numericals. Superconducting Materials: Introduction, Type – I and Type – II superconductors, Meissner effect, BCS Theory, Application	
4	**Optoelectronic Materials and Devices:** LASER : Absorption, spontaneous and stimulated emission, population inversion n, pumping mechanism, Construction and working of Ruby laser, Construction and working of He-Ne laser. Lasers in various technological applications. Introduction to Optical Fibers-Introduction Acceptance Angle-Numerical Aperture, Applications of optical fibers. Numericals.	8
5	**Physics of Materials:** Crystal structure: Unit cell, Coordination number, atomic radius, packing density of cubic system. X-rays: Bragg's law, X-Ray Diffraction (XRD), Industrial Applications of X-Rays. Numericals. Particle detector: G.M. Counter Nano-Materials: Basic principles of nano science and technology, properties, applications of nanotechnology.	8

Text Books:

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1. Engineering Physics- H.K. Malik & A.K. Singh, McGraw Hill publication.
2. Engineering Physics - R.K. Gaur and S. L. Gupta. Dhanpat Rai Publications Pvt. Ltd.-New Delhi
3. M. N.Avadhanulu, P.G.Kshirsagar "A Text book of Engineering Physics"-S. Chand Publications.
4. B.K. Pandey and S. Chaturvedi, Engineering Physics, Cengage Learning, 2012.

Reference Books:

1. Fundamental of Physics - Halliday and Resnik. Willey Eastern Limited.
2. Introduction to Electrodynamics –David R. Griffiths.
3. Concept of Modern Physics – Arthur Beizer. Tata McGraw-Hill Publishing Company Limited.
4. Optics – AjoyGhatak.MacGraw Hill Education (India) Pvt. Ltd.
5. Science of Engineering Materials- C.M. Srivastava and C. Srinivasan. New Age International Pvt.Ltd.
6. Solid State Physics – A.J. Dekker. McMillan India –Limited.
7. The Feynman Lectures on Physics Vol I, II,III.
8. Introduction to solid state physics – Charles Kittel. John Willey and Sons
9. T Pradeep "A Text book of Nano Science and Nano Technology"-Tata Mc GrawHill 2019.

E-Resources:

1. https://nptel.ac.in/courses/122107035/ Engineering Physics
2. https://youtu.be/98xoZknQjl8 Polarization
3. https://youtu.be/yINtzw63KncMaxwell's Equations and EM theory
4. https://youtu.be/bwreHReBH2AMaxwell's Equations and EM theory

5. https://www.youtube.com/playlist?list=PLuv3GM6-gsE3-hVNaw-YEb7EeY5XVPZdz
6. Maxwell's Equations and EM theory(nptel)
7. https://nptel.ac.in/courses/115105120/ Experimental Physics
8. https://youtu.be/2CsMpEBl5QY Crystal Structure and X- rays
9. https://youtu.be/z_8aJPLr21E Crystal Structure and X- rays
10. https://youtu.be/_Ckh-60B6LY Condensed matter Physics
11. https://youtu.be/QQZ6EGf0Ju8 Magnetic Properties
12. https://youtu.be/DDLljK1ODeg Magnetic Materials
13. https://youtu.be/etjZmdmrjSU Dielectrics
14. https://youtu.be/k6ZxP9Yr02E Semiconductor
15. https://youtu.be/D-9M3GWoBrw Superconductivity
16. https://youtu.be/GglT1RoBPzg Superconductivity
17. https://youtu.be/VHp2Ff5N_bs Superconductivity
18. https://youtu.be/FNp81kkxj5c LASER
19. https://youtu.be/YvrwVK9ZqQY LASER
20. https://nptel.ac.in/courses/115107095/ Optic Fiber
21. https://youtu.be/cjBPnIXK60U Quantum Mechanics (Prof.H.C. Verma)
22. https://youtu.be/BDuqChhUhM0 Divergence and Curl(Prof.H.C. Verma)
23. https://youtu.be/sCviGSMaYfI Divergence and Curl (Prof.H.C. Verma)
24. https://youtu.be/SZCsFS9izfQ Divergence and Curl

And other related videos from following resources

25. www.nptel.ac.in ;www.swayam.gov.in;https://inflibnet.ac.in/
26. www.sciencedirect.com
27. http://vlabs.iitb.ac.in/vlab/
28. www.youtube.com

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Course Code:APS21ESL101 **Course Name:** Python Programming **Course Category:**

Credits: 3 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

1. **Pre-requisites:** Basic Computer Knowledge &Knowledge of any programming language(optional)

Course Objectives:

1. To understand fundamental concepts in Python Programming
2. To learn the different Conditional Loops and Iteration
3. To understand various data structures and packages

Course Outcomes:

CO1: Describe programming fundamentals of python.
CO2: Interpret the python syntax and semantics of control flow statements.
CO3: Identify the methods to create and manipulate programs with python data structures.
CO4: Use modular approach for problem solving.
CO5: Apply advanced features and packages of python programming required for data science.

Contents:

Unit	Content	Teaching Hours
1	**Python for everybody:** Why Program, Hardware Overview, Python as a Language, Why Python, Installation Python Jupyter notebook, Using the Python Playground ,how to write program and compile in Jupyter notebook. Writing input and output statements in Python, output formatting, Writing comments, keywords in Python .variables and variable assignments, Operators ,writing expressions	6
2	**Conditional Loops and Iteration:** Conditional Statements, in Python, Loops and Iteration, Definite Loops, Finding the Largest Value, Loop Idioms	6
3	**Data Structures in Python:** Strings, Manipulating Strings, Files, Processing Files, Dictionaries, sets, Tuples, Lists, Manipulating Lists, Lists and Strings, Strings, Manipulating Strings	6
4	**Functions, Modules and Packages:** Functions, Lambda functions, Recursive function, Types of functions, modules and packages.	6
5	**Packages in Python for Data Science:** Numpy introduction, Numerical operations on Numpy, Introduction of Matplotlib, getting started with Pandas, Data frames basics in Pandas, key operations on data frames. Introduction to Data Science, Binary search, finding elements in common in lists using Hash, Finding largest elements, Introduction of SQL	6

Text Books:

1. Kent D. Lee, “Python Programming Fundamentals”, SecondEdition, Springer Publication.

2. Wes McKinney, “Python for Data Analysis” O’Reilly Publication.

Reference Books:

1. The Python Language Reference: http://docs.python.org/2/reference/index.html

2. The Python Standard Library: http://docs.python.org/2/library/

3. https://docs.scipy.org/doc/scipy/reference/tutorial/stats.html

4. http://matplotlib.org/api/mlab_api.html#module-matplotlib.mlab

5. http://conference.scipy.org/proceedings/scipy2010/pdfs/seabold.pdf

6. http://seaborn.pydata.org

7. https://www.datacamp.com/community/data-science-cheatsheets

8. PEP 20 -- The Zen of Python: https://www.python.org/dev/peps/pep-0020/

9. https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html

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General Instructions:

The theory classes are to be conducted batch wise in Lab. Each class should be divided into four batches.

Course Code:APS21SL102**Course Name:** Engineering Graphics **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Nil

Course Objectives:

1. Understand the basic principles of engineering graphics and improve the visualization skills
2. To gain knowledge on projection of points, straight lines, planes, solids.
3. To understand the real life objects through Drawings.
4. To know the principles of orthographic and isometric projections.

Course Outcomes:

CO 1. Identify basic concepts in drawing and its application.
CO 2. Plan and prepare neat orthographic drawings of points, straight lines, planes and solids.
CO 3 To visualize and draw orthographic and isometric projection of solids.
CO 4. Acquire skill to draw real life engineering objects by using the engineering drawing.

Contents:

Unit	Content	Teaching Hours
1	**Projections of Straight Lines:** Introduction to Engineering Graphics, Need of Engineering Drawing, Drawing Instruments, BIS code of practice for general engineering drawing, Projections of Points in Four Quadrants, Projections of Points in Reference Plane, Line Parallel to both the Plane, Line Parallel to One Plane and Perpendicular to the other, Line Inclined to One Plane and Parallel to The Other, Line Inclined to Both the Reference Planes, Traces of Line (Only first quadrant to be considered)	5
2	**Engineering Curves:** Curves used in Engineering Practice, Conic sections, Construction of conics by different methods, Rectangular-hyperbola, Cycloidal curves, Epi and hypo-cycloids, Involutes and Archimedean spiral.	5
3	**Projections of Planes:** Plane with Surface Parallel to One Plane and Perpendicular to other, Plane Inclined to One Plane and Perpendicular to other, Projections of Planes Inclined to both the Planes	5
4	**Projections of Solids:** Introduction to Solids: Prisms, Pyramid, Cylinder, Cone, Cube, Tetrahedron, Sphere, Projections of above Solids with Axis inclined to one plane, Projections of above solids with Axis inclined to both the Planes, Projection of composite solids (different arrangement of Spheres with above Solids).	5
5	**Orthographic Projections:** Orthographic projections of different Machine Parts (First Angle Projection method only)	5
6	**Isometric Projections ,Introduction to CAD:**	5

	Introduction to Pictorial views, Converting Orthographic Projections into Isometric Projections and Isometric views. Listing the computer technologies that impact on graphical communication, Demonstrating knowledge of the theory of CAD software [such as: The Menu System, Toolbars (Standard, Object Properties, Draw, Modify and Dimension), Drawing Area (Background, Crosshairs, Coordinate System), Dialog boxes and windows, Shortcut menus (Button Bars), The Command Line (where applicable), The Status Bar, Different methods of zoom as used in CAD, Select and erase objects	

Text Books:

1. N. D. Bhatt and Panchal V. M., "Engineering Drawing", Charotar Publishing House, Anand
2. P. J. Shah , " A Text Book of Engineering Drawing", S.Chand, New Delhi
3. Shah M.B. & Rana B.C, "Engineering Drawing & Computer Graphics", Pearson Publications,
4. Agrawal B.& Agrawal C.M, "Engineering Graphics", TMH Publications
5. Narayana K.L. & P. Kannaiah, "Engineering Drawing", Scitech Publications
6. P.I Vargese, "Engineering Graphics", Mcgraw Hill Publications
7. D.A.Hindoliya," Engineering Graphics", B. S. Publications

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Reference Books:

1.Dabhade M. L., "Engineering Graphics", Vol.-I and Vol.-II, Vision Publications, Pune

2.K.Venugopal, "Engineering Drawing and Graphics" ,New Age International Publishers

Course Code:MGM54AEL101**Course Name:** Communicative English **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Basicknowledge of English

Course Objectives:
The course aims at grooming the professional ethics of the students through variouspersonalitytraits andbehavioral patternsfocusingon communicationskills.

Course Outcomes:

CO.1 communicate formally with enhancedcommunicationCompetency

CO.2 to adapt professional nonverbal communication

CO.3construct English formal syntax and apply corporate vocabulary in written and verbal communication

CO.4 acquire listening and drafting skills with professional competency

Contents:

Unit	Content	Teaching Hours
1	**Communication andCommunicationProcess:** Introduction to Communication,Forms and functions ofCommunication, Barriers toCommunication and overcomingthem,Ways of EffectiveCommunication.	5
2	**Non-verbalCommunication And its types:** Kinesiccs, Occulesics, Appearance, Proxemics, Chronemics, Paralanguage, Qualities of effective speech	3
3	**English Grammar:** Overview of basic Mid-levelgrammar, Tenses&conceptoftime,Sentence construction, Corporate vocabulary, Difference between formal and informal sentences, phrases and words	3
4	**ListeningSkills and WritingSkills:** Listening : Active andPassive Listeningwritingstyles layouts BusinessLetters- job application, resignation, resume	4

Text Books/ Reference Books:

1 AshrafRizvi,CommunicationSkillsforEngineers,TataMcGrawHill

2 SanjayKumar, PushpLata,Communication Skills, Oxford UniversityPress, 2016.

3 MeenakshiRaman,SangeetaSharma,CommunicationSkills,Oxford UniversityPress,2017.

4 TeriKwalGamble,MichaelGamble,Communication Works,TataMcGraw HillEducation,2010.

5 Anderson, Kenneth. Joan Maclean and Tossny Lynch. Study Speaking: A Course in Spoken EnglishforAcademicPurposes.Cambridge: CUP, 2004.

6 Bellare,Nirmala.ReadingStrategies.Vols.1and2.NewDelhi.OxfordUniversityPress,1998

7 Bhasker,W.W. S&Prabhu, N.S.:English throughReading, Vols.1and2.Macmillan,1975

8 BoveeCourtland,Land Thrill,John V.BusinessCommunication, TodayMcGraw Hill, NewYork, TaxmanPublication(1989).

9 Murphy,Raymond.Essential EnglishGrammar, Cambridge:UniversityPress(2000)

10 HewingsMartinAdvancedEnglishGrammarCambridge:Universitypress(2003)

11 BansalHarrison.Spoken English

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Course Code:APS21VSP101**Course Name:**Engineering Exploration**Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Nil

Course Objectives:

1. Understand the principles of Engineering Design and apply them to identify product requirements, objectives, and constraints through interactions with clients, users, and relevant stakeholders.
2. Develop critical thinking and problem-solving skills to analyze existing technologies, conduct surveys, study literature, and generate creative design concepts using various means and combinations.
3. Acquire proficiency in using appropriate tools and techniques to generate electronic and mechanical simulations, schematic diagrams, and product concept sketches or CAD models for effective communication of design ideas.
4. Demonstrate competency in categorizing inputs and outputs of systems in terms of materials, information, and energy, enabling a holistic understanding of engineering projects.
5. Cultivate effective teamwork and communication skills to collaborate with peers, mentors, and team members during the development and presentation of engineering prototypes.
6. Develop the ability to evaluate, defend, and communicate design decisions, progress, and project outcomes, showcasing awareness of course concepts and their application in real-world scenarios.

Course Outcomes:

On completion of the course, the student should be able to:

LO 1. Effectively interact with clients, users, and stakeholders to gather relevant information and derive product requirements, thereby demonstrating proficiency in Requirement Analysis and Client Interaction.

LO 2. Analyse and compare existing technologies, conduct surveys, and study literature to identify potential solutions, and develop objective trees and function trees, reflecting their competency in Conceptual Design and Analysis.

LO 3. Create product concept diagrams, combining different means, and prepare sketches or CAD models to present their design ideas aesthetically and coherently, indicating proficiency in Product Concept Development.

LO 4. Categorize system inputs and outputs in terms of materials, information, and energy, illustrating their understanding of System Categorization and Analysis.

LO 5. Exhibit hands-on skills in physical assembly, connection, and demonstration of engineering prototypes, showcasing Proficiency in Prototyping and Implementation.

LO 6. Explain and defend their design choices, project progress, and outcomes during presentations, demonstrating Communication Skills, Project Evaluation, and Awareness of Engineering Concepts.

Contents:

Sr. No.	List of Practical	Lab Hours
1	The course is conducted in the following modules: 1. Introduction to Engineering Exploration 2. Engineering Design 3. Platform Based Development 4. Mechanisms 5. Data acquisition and analysis 6. Engineering Ethics 7. Project Management	10
2	The following practical contents are delivered in an integrated mode along with theory: 1. Conceptualizing a product 2. Designing a product with constraints 3. Simulation of electronic circuits (at least 10 sets) 4. Implementation of electronic circuits (at least 5 sets) 5. Implementation of four bar chain mechanism 6. Conversion of problem statement to need statement 7. Identification of objectives, constraints and functions 8. Generation of black box, glass box and expanded class box 9. Generation of morphological chart 10. Generation of concepts 11. Comparison of concepts 12. Selection of concept 13. Implementation and testing of prototype	20

Reference Books:

- George E. Dieter and Linda C. Schmidt (2009), Engineering Design, 4ed, Mc Graw Hill Higher Education
- Clive L. Dym, Patrick Little, and Elizabeth J. Orwin (2014), Engineering Design: A project-based introduction, 4ed, John Wiley and Sons
- G. Pahl, W. Beitz, J. Feldhusen and K.-H. Grote (2007), Engineering Design: A systematic approach, 3ed, Springer, New York

Course Code:APS21BSP101 **Course Name:**Engineering Physics Lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Student should know the basic aspects of measurements like least count and range of instrument, scale identification, accuracy, error etc.

Course Objectives:
The Objective of this course is
1. To make the students gain practical knowledge to co-relate with the theoretical studies.
2. To achieve perfectness in experimental skills.
3. The study of practical applications will bring more confidence.

Course Outcomes:
At the end of the laboratory course, the students will be able to,
LO1: Plot the I-V characteristics of pn junction diode and determine the value of band gap energy which can be used for core engineering Courses.
LO2: Perform optical experiments; using light properties like interference, polarization, diffraction and their applications.
LO3: Interpret the results and analyze the data and use the experimental data to plot the graph for a best fit.
LO4: Discuss the characteristics of plateau region and determine operating voltage of G.M. counter
LO5: Determine the numerical aperture and bending loss of optical fibre cable

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Contents:

Sr. No.	List of Practical	Lab Hours
1	Thomson's Bar Magnetic method -Determination of e/m of electron	02
2	Newton's rings -Determination of radius of curvature of Plano convex lens.	02
3	Polarization - Half shade Polarimeter -Determination of specific rotation of optically active material.	02
4	Diffraction - Determination of wavelength of light by plane transmission grating.	02
5	Wedge Shaped film -Determination of thickness of thin wire	02
6	Semiconductors – To study of forward and reverse bias characteristic of Semiconductors diode	02
7	Fibre Optics – Determination of Bending losses in optical fibre	02
8	Fibre Optics – Determination of Numerical Aperture of given optical fibre	02
9	Determination of operating voltage of G.M. tube and characteristics of plateau region	02
10	Determination of wavelength of laser source	02

Course Code:APS21ESP101 **Course Name:**Python Programming Lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The Objective of this course is

Course Outcomes:
LO1 : Demonstrate python program using development environment.
LO2: Develop logical thinking to solve the problems using programming fundamental concepts.
LO3 : Construct python program using various data structures.
LO4 : Apply modularization approach for solving complex problem.
LO5: Make use of various packages in Python for data science.
LO6 : Implement different SQL commands in python.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Program to perform input/output operations Write a program to take input (integer, float, string) and print it.	02
2	Program based on operators 1.Write a program to simulate a simple calculator (+ - / * %) that takes two operands as input and displays the result 2.Write a program to find area and perimeter of geometric objects. 3.The distance between two cities (in km.) is input through the keyboard. Write a program to convert and print this distance in meters, feet, inches and centimeters. 4. Write a Program to interchange two numbers. 5.Write a program to compute Fahrenheit from centigrade	02
3	Programs based on Decision making. 1.Write a program to read marks from keyboard and your program should display equivalent grade according to following table(else-if) (ladder) Marks Grade 100 - 80 Distinction 79 - 60 First Class 59 - 40 Second Class < 40 Fail 2.Write a program to input basic salary of an employee and calculate gross salary according to given conditions. Basic Salary <= 10000 : HRA = 20%, DA = 80% Basic Salary is between 10001 to 20000 : HRA = 25%, DA = 90% Basic Salary >= 20001 : HRA = 30%, DA = 95% 3.If the ages of three brothers are input through the keyboard, write a C Program to determine the youngest and oldest of the three.	02

	4.Write a program to calculate overtime pay of employee. Overtime is paid at the rate of Rs. 12.00 per hour for every hour worked above 40 hours. Assume that employee do not work for fractional part of an hour. 5.Write a program for checking the speed of drivers. If speed is less than 70, it should print "Ok". Otherwise, for every 5km above the speed limit (70), it should give the driver one demerit point and print the total number of demerit points. For example, if the speed is 80, it should print: "Points: 2". If the driver gets more than 12 points, the function should print: "License suspended"	
4	Programs using while and for loops 1.WAP to find factorial of given number 2.WAP to check whether given number is Palindrome or not 3.WAP to check whether given number is Armstrong or not 4.WAP to print Fibonacci series 5.Write a Python program which iterates the integers from 1 to 50. For multiples of three print "Fizz" instead of the number and for the multiples of five print "Buzz". For numbers which are multiples of both three and five print "FizzBuzz". 6.WAP to check whether given number is Perfect number or not 7.WAP to check whether given number is Prime number or not 8.Write C program to print given star and number patterns and reverse it. • * 1 • ** 12 • *** 123 **** 1234	02
5	Programs on string 1.Write Python Program to find length of string without using len() function. 2.Count all letters, digits, and special symbols from a given string. 3.Python Program to Count the Number of Vowels in a String. 4.Python Program to Calculate the Number of Upper Case Letters and Lower Case Letters in a String. 5.Python Program to Check whether given string is palindrome or not	02
6	Programs on List and Tuple 1.Write a Python program to sum all the items in a list. 2.Write a Python program to multiply all the items in a list 3.Write a Python program to get the largest number from a list. 4.Write a Python program to get the smallest number from a list 5.Write a Python program to count all elements in list and count Occurrences Of A List Item In Python 6.Write a Python program to create a tuple with different data types 7. Write a Python program to check whether an element exists within a tuple 8.Write a Python program to reverse a tuple 9. Write a Python program calculate the product of all the numbers given in tuple. Original Tuple: (2, 4, 8, 8, 3, 2, 9) Product - multiplying all the numbers of the said tuple: 27648	02
7	Programs on set and dictionary	02

	1.Write a Python program to concatenate following dictionaries to create a new one. Sample Dictionary : dic1={1:10, 2:20} dic2={3:30, 4:40} dic3={5:50,6:60} Expected Result : {1: 10, 2: 20, 3: 30, 4: 40, 5: 50, 6: 60} 2.Write a Python program to check whether a given key already exists in a dictionary 3.Write a Python script to generate and print a dictionary that contains a number (between 1 and n) in the form (x, x*x) Sample Dictionary (n = 5) : Expected Output : {1: 1, 2: 4, 3: 9, 4: 16, 5: 25} 4.Write a Python program to merge two Python dictionaries 5.Write a Python program to get the maximum and minimum value in a dictionary 6.Write a Python program to create set difference, Union and intersection 7.Write a Python program to check if two given sets have no elements in common	
8	Programs using function 1. Write Functions to calculate your trip's costs: 2. Define a function called hotel_cost with one argument nights as input 3. Define a function called plane_ride_cost that takes a string, city, as input. 4. Define a function called rental_car_cost with an argument called days. 5. Define a function called trip_cost that takes two arguments, city and days. Like the example above, have your function return the sum of calling the rental_car_cost(days), hotel_cost(days), and plane_ride_cost(city) functions. 6. Write a program in to check a given number is even or odd using the function. 7. Write a function Exchange to interchange the values of two variables, say x and y. illustrate the use of this function in a calling function. 8. Write a program to find Sum of natural number using recursion. 9. Write a program to print Fibonacci series number using recursion	02
9	Program using NumPy, Matplotlib and Pandas library 1.Write a program to perform matrix addition, substraction, multiplication. 2.Plot all types of graph using Matplotlib. 3.Write a program which perform basic operation of Pandas.	02
10	Program on SQL Commands 1. Write a program of binary search 2. Write a program which perform basic SQL commands 3. Programs based on real life problems/GUI based programs	02

Course Code:APS21ESP102**Course Name:**Engineering Graphics Lab **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The Objective of this course is

Course Outcomes:
Students will be able to,
1. Develop competence in correct expression of the visualized objects
2. Dimension and annotate two-dimensional engineering drawings
3. Plan and prepare neat orthographic drawings of points, straight lines, planes and solids
4. Develop the ability to visualize and draw orthographic and isometric projection of solids

Contents:

Sr. No.	List of Practical	Lab Hours
1	Drawing three problems based on projections of lines on half imperial size drawing sheet	02
2	Drawing three problems based on engineering curves on half imperial size drawing sheet	02
3	Drawing three problems based on projections of planes on half imperial size drawing sheet	02
4	Drawing three problems based on projections of solids on half imperial size drawing sheet	02
5	Drawing three problems based on orthographic projections on half imperial size drawing sheet	02
6	Drawing three problems based on isometric projections on half imperial size drawing sheet	02
7	Demonstration of CAD software in CAD lab, drawing simple objects using various commands	02

Course Code:APS21ESP103**Course Name:**Recent Trends in Integrated Technologies Lab
Course Category:

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

1.To introduce students the basics of additive manufacturing/rapid prototyping and its applications in various fields, reverse engineering techniques.
2.To recognize industrial control problems suitable for Industrial Robotics.
3.To acquire basic skills in exploring the potential of the drone technology in professional activities
4.Ability to recognize industrial automation problems suitable for PLC control.

Course Outcomes:

LO1. Prepare 3D Model (slice & print) in either Stratasys or Zortrax & generate scan data through Hexagon portable scanning arm.
LO2.Describe basic industrial robotics & it's applications
LO3.Operate a small drone in a controlled environment
LO4.Explain principles of sensor, PLC & applications.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Scan to CAD, CAD to STL conversion& patching, machine setup & processes for printing	02
2	Programming the TATA Robot(pendant) for pick & place, programming the Yaskawa Robot(pendant) for motion planning, Demonstrate welding exercise by the instructor	02
3	Introduction to components of drone, Demonstration of assembly of drone, Demonstration of mission planning & flying the drone	02
4	Controller & sensor & their interfacing, basic ladder logic instructions, pneumatic& hydraulic actuator	02

Reference Books:

1. A Step-by-Step Guide For Beginners: Aircraft Design & Construction Design Guide by :Merlin Debrie
2. Industrial Automation & Robotics By A.K. Gupta & S.K.Arora
3. Additive Manufacturing Principles, Technologies & Applications By C.P.Paul (TMH)
4. Basics of unmanned aerial vehicle By Garvit Pandya (Motion press)

Course Code:MGM54AEP101**Course Name:**Communicative English Lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: BasicknowledgeofEnglish

Course Objectives:
Thecourseaims at groomingthe professionalethicsofthe studentsthroughvariouspersonalitytraits andbehavioural patternsfocusingon communicationskills.

Course Outcomes:
TheStudents willbeable to
LO.1 Introducethemselves formallyand informallythroughpractice.
LO.2 Pronounce English vowel and Consonant soundseffectively
LO.3 Participate effectively in G.Ds, Presentations, &Interviews
LO.4 Face Interviews competently
LO.5 Draft resume,businessletters,reports formally
LO.6 ComprehendthemeaningofEnglishtext bycomprehensiontechniques.

Contents:

Sr. No.	List of Practical	Lab Hours
1	SelfIntroduction	02
2	Pronunciation of Vowel sounds in English	02
3	Pronunciation of consonant sounds in English	02
4	Groupdiscussion	04
5	Presentation techniques	04
6	Interviewtechniques	04
7	Letter writing	04
8	Email writing, Agenda of the meeting and notices	02
9	Resume Writing	02
10	Report writing	02
11	Skimming & Scanning	02

Course Code:MGM82CCP101**Course Name:**National Cadet Corps **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
Nil

Course Outcomes:
Nil

Contents:

Unit No.	Content	Hours
1	**NCC General, National Integration and Awareness, Social Service and Community Development, Drill:** Aims, Objectives and Organization of NCC, Duties of NCC Cadet, National Integration Importance and Necessity, Factors Affecting National Integration, Foot Drill, Drill With Arms, Ceremonial Drill with Arms, Social Service and Community Development Activities- Pollution, Mission Indradhanush, Beti Bachao Beti Padhao, Tree Plantation, Digital Awareness.	12
2	**Personality Development, Disaster Management, Weapon Training, Map Reading :** Critical and Creative Thinking, Decision Making, Self Awareness, Public Speaking, Interview Skills, Types, Org, Capability and Role of NCC cadets, Initiative Trg, Organizing Skills, Do's & Don't , Introduction and Characteristics of .22 rifle, Handling of .22 rifle, Introduction to Map Reading, Conduct of MR- Google and Tourist Maps and Apps.	10
3	**Health and Hygiene, Environmental Awareness and Conservation, Adventure, Obstacle Training:** Hygiene & Sanitation (Personal & Camp Hygiene) Soch Vichar, First Aid in Common Medical Emergencies, Treatment & Care of Wounds, Introduction Yoga & Exercises, Water Conservation, Energy Conservation, Introduction Adventure Activities. Obstacle Course.	10
4	**Leadership, Introduction to Infantry Weapons and Equipments:** Traits, Indicators, Motivation, Ethics, Case Studies- Chhatrapati Shivaji Maharaj, Maharana Pratap, Jhasi ki Rani, Ratan Tata, Narayan Murty, Rabindra Nath Tagor, Organization of Infantry Battalion and its weapons.	08
5	**Armed Forces, Field Craft and Battle Craft, :** Armed Forces, Army, CAPF, Police, Modes of Entry to Army, CAPF, Police, Introduction to Field Craft, Indication of Landmark, Observation, Camouflage and Concealment, Fire and Move Capsule.	08

References:

- Cadet's Handbook- Common Subject, all wings by DG NCC, New Delhi.
- Cadet's Handbook- Common Subject by NCC Directorate- Bhubaneshwar.

- Cadet's Handbook- Specialised Subjects, Army, Navy, Air-Force by DG NCC, New Delhi.
- NCC OTA Precise by DG NCC, New Delhi.
- Chanakya's 7 Secrets of Leadership by Radhakrishanan Pillai and D. Shivnandhan.
- National Cadets Corps (India) by Lambert M. Suvarkar.

E-Resources:

1. National Cadet Corps, Youth in Action (Google eBook).

https://indiancc.nic.in/dg-ncc-lt-gen-gurbirpal-singh/

www.youtube.com

MGMUNIVERSITY

Course Code:MGM82CCP103**Course Name:**Sports**Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

1. To understand the importance of sports in physical and mental development.
2. To learn about the different types of sports.
3. To learn about the different types of indoor games.
4. To participate in indoor games.
5. To learn about the different types of outdoor games.
6. To participate in outdoor games.

Course Outcomes:

- Upon completion of this course, students will be able to
- Demonstrate knowledge of the history, benefits, types, equipment, and safety of sports
- Demonstrate proficiency in the basic skills of indoor and outdoor games
- Understand the rules and regulations of selected sports
- Participate in sports competitions

Contents:

Sr. No.	Content	Hours
1	Football: History of Football: The earliest forms of football can be traced back to ancient China, Greece, and Rome. In England, the game of football developed in the 19th century, with different rules being used by different schools and organizations. In 1863, the Football Association (FA) was founded, and it standardized the rules of the game. The first international match was played between England and Scotland in 1872. Football became an Olympic sport in 1900, and the first World Cup was	12

	held in 1930. Today, football is the most popular sport in the world, with billions of fans around the globe. **Fundamental Skills of Football** Dribbling: Dribbling is the ability to move the ball with your feet while keeping control of it. It is a essential skill for all footballers, as it allows you to move past defenders and create scoring opportunities. Passing: Passing is the ability to accurately and effectively throw the ball to your teammates. It is another essential skill, as it allows you to move the ball up the field and create scoring opportunities. Shooting: Shooting is the ability to kick the ball with power and accuracy. It is the most important skill for scoring goals, and it is essential for all footballers to develop a good shot. Heading: Heading is the ability to use your head to control and direct the ball. It is a valuable skill for both attacking and defending, and it is important for all footballers to learn how to head the ball effectively. Tackling: Tackling is the ability to take the ball away from an opponent. It is an important skill for defenders, but it is also valuable for midfielders and attackers.	
2	**Basket Ball** **History of Basketball** Basketball was invented by James Naismith in 1891 at the International YMCA Training School in Springfield, Massachusetts. Naismith was a physical education instructor who was looking for a game that would be less injury-prone than football. He nailed two peach baskets to the lower rail of a balcony and used a soccer ball to play the game. The first game of basketball was played on December 21, 1891, with nine players on each team. The rules of basketball have evolved over time, but the basic premise of the game has remained the same. Today, basketball is one of the most popular sports in the world, with millions of players and fans around the globe. **Fundamental Skills of Basketball** Dribble: Dribbling is the ability to move the ball with your hands while keeping control of it. It is an essential skill for all basketball players, as it allows you to move past defenders and create scoring opportunities. Passing: Passing is the ability to accurately and effectively throw the ball to your teammates. It is another essential skill, as it allows you to move the ball up the court and create scoring opportunities. Shooting: Shooting is the ability to throw the ball through the hoop with power and accuracy. It is the most important skill for scoring points, and it is essential for all basketball players to develop a good shot. Rebounding: Rebounding is the ability to catch the ball after it has been missed by a shooter. It is an important skill for both offense and defense, as it allows teams to get second chances at scoring. Defense: Defense is the ability to prevent the other team from scoring points. It is an essential skill for all basketball players, as it is impossible to win a game without playing good defense.	10

3	**Volley Ball** **History of Volleyball** Volleyball was invented in 1895 by William G. Morgan, a physical education instructor at the Young Men's Christian Association (YMCA) in Holyoke, Massachusetts. Morgan was looking for a game that would be less vigorous than basketball, and he created volleyball as a way to keep his students active during the winter months. The original name of the game was "mintonette," but it was renamed "volleyball" in 1896. Volleyball quickly spread throughout the United States and around the world, and it became an official Olympic sport in 1964. **Fundamental Skills of Volleyball** Passing: Passing is the ability to receive the ball from the opponent and direct it to a teammate. It is an essential skill for all volleyball players, as it allows the team to keep possession of the ball and start an attack. Setting: Setting is the ability to control the height and direction of the ball so that a teammate can spike it. It is a critical skill for setters, as they are responsible for setting up the team's offense. Spiking: Spiking is the ability to hit the ball over the net with power and accuracy. It is the most important skill for scoring points in volleyball, and it is essential for all attackers to develop a good spike. Blocking: Blocking is the ability to prevent the opponent from spiking the ball over the net. It is an important skill for blockers, as they can prevent the other team from scoring points. Digging: Digging is the ability to prevent the opponent from scoring a point by returning the ball over the net. It is an important skill for all defenders, as they are responsible for preventing the other team from scoring points	10
4	**Kabaddi** **History of Kabaddi** Kabaddi is a contact team sport that originated in India. It is believed to have originated in the Indian subcontinent over 4,000 years ago. The game is mentioned in the Sangam literature of Tamil Nadu, which dates back to the 3rd century BC. Kabaddi was first played as a competitive sport in the Indian Olympic Games in 1938. It was included as a demonstration sport at the 1982 Asian Games in Delhi, and it became a full medal sport in the 1990 Asian Games in Beijing. Kabaddi is now played in over 100 countries around the world. **Fundamental Skills of Kabaddi** Dabki: Dabki is the act of entering the opponent's half of the court while chanting "kabaddi, kabaddi." It is a fundamental skill for all raiders, as it allows them to enter the opponent's half of the court without being tackled. Touch: Touching an opponent is the most important skill in kabaddi. It is how raiders score points for their team. There are many different ways to	08

	touch an opponent, such as touching their arm, leg, or torso. Tackling: Tackling is the act of preventing a raider from touching an opponent. It is a fundamental skill for all defenders, as it allows them to prevent the other team from scoring points. There are many different ways to tackle a raider, such as grabbing them, pushing them, or tripping them. Stamina: Stamina is essential for all kabaddi players, as the game is very physically demanding. Players need to be able to run, jump, and tackle for long periods of time. Agility: Agility is also important for kabaddi players, as they need to be able to change direction quickly and avoid being tackled.	
5	**Badminton** **History of Badminton** The game of badminton originated in ancient Greece, China, and India. It was brought to England in the 1870s by British army officers stationed in India. The first badminton club was founded in 1873 at Badminton House, the country estate of the Duke of Beaufort. The first official all-England badminton championships for men were held in 1899, and the first badminton tournament for women was arranged the next year. Badminton became an Olympic sport in 1992. Today, badminton is a popular sport played by millions of people around the world. **Fundamental Skills of Badminton** Grip: The grip is the most important fundamental skill in badminton. It allows you to control the racket and hit the shuttlecock with power and accuracy. There are many different grips, but the most common are the forehand grip and the backhand grip. Footwork: Footwork is essential for moving around the court and positioning yourself to hit the shuttlecock. There are many different footwork drills that you can practice to improve your footwork. Racket control: Racket control is the ability to hit the shuttlecock with power and accuracy. It is important to practice hitting the shuttlecock in different directions and with different levels of power. Timing: Timing is the ability to hit the shuttlecock at the right time. It is important to practice hitting the shuttlecock at the peak of its flight. Stamina: Stamina is essential for badminton, as it is a physically demanding sport. You need to be able to run, jump, and hit the shuttlecock for long periods of time.	08
6	**Soft Tennis** **History of Soft Tennis** Soft tennis is a racquet sport that originated in Japan in the early 20th century. It was created as a less dangerous alternative to lawn tennis, as the ball used in soft tennis is made of foam rubber and does not travel as fast as a regular tennis ball. The first soft tennis tournament was held in Japan in 1921, and the sport quickly spread to other countries in Asia.	

	Soft tennis was first introduced to the United States in the 1950s, and it has since become a popular recreational sport in the country. **Fundamental Skills of Soft Tennis** Grip: The grip is the most important fundamental skill in soft tennis. It allows you to control the racquet and hit the ball with power and accuracy. There are many different grips, but the most common are the forehand grip and the backhand grip. Footwork: Footwork is essential for moving around the court and positioning yourself to hit the ball. There are many different footwork drills that you can practice to improve your footwork. Racket control: Racket control is the ability to hit the ball with power and accuracy. It is important to practice hitting the ball in different directions and with different levels of power. Timing: Timing is the ability to hit the ball at the right time. It is important to practice hitting the ball at the peak of its flight. Stamina: Stamina is essential for soft tennis, as it is a physically demanding sport. You need to be able to run, jump, and hit the ball for long periods of time. **Here are some additional fundamental skills of soft tennis:** Ball control: The ability to control the direction and speed of the ball. Serve: The ability to serve the ball accurately and with power. Volley: The ability to hit the ball before it bounces. Overhead smash: The ability to hit the ball forcefully and accurately overhand. Drop shot: The ability to hit the ball softly and precisely so that it bounces low and close to the net.	
7	**Tennis** **History of Tennis** The origins of tennis can be traced back to a 12th–13th-century French handball game called jeu de paume (“game of the palm”), from which was derived a complex indoor racket-and-ball game: real tennis. The modern game of lawn tennis was invented in England in the 1870s by Major Walter Wingfield. Wing field created a set of rules and equipment for the game, and he called it "Sphairistike". The game quickly became popular, and it was renamed "lawn tennis" in 1874. The first lawn tennis tournament was held in 1877 at the All England Club in Wimbledon, England. Tennis became an Olympic sport in 1896. Today, tennis is a popular sport played by millions of people around the world. **Fundamental Skills of Tennis** Grip: The grip is the most important fundamental skill in tennis. It allows you to control the racket and hit the ball with power and accuracy. There are many different grips, but the most common are the forehand grip and the backhand grip. Footwork: Footwork is essential for moving around the court and positioning yourself to hit the ball. There are many different footwork	

	drills that you can practice to improve your footwork. Racket control: Racket control is the ability to hit the ball with power and accuracy. It is important to practice hitting the ball in different directions and with different levels of power. Timing: Timing is the ability to hit the ball at the right time. It is important to practice hitting the ball at the peak of its flight. Stamina: Stamina is essential for tennis, as it is a physically demanding sport. You need to be able to run, jump, and hit the ball for long periods of time.	
8	**Fencing** **History of Fencing** The earliest evidence of fencing dates back to ancient Egypt, Greece, and Rome. Fencing was used as a form of training for warfare and as a way to settle disputes. The modern sport of fencing developed in Italy in the 15th century. The first fencing competition was held in 1550 in Paris. Fencing became an Olympic sport in 1896, and it has been a part of every Games since then. Fencing is now a popular sport all over the world, and there are three main disciplines: foil, épée, and sabre. **Fundamental Skills of Fencing** Footwork: Footwork is essential in fencing, as it allows you to move quickly and efficiently around the piste. There are many different footwork drills that you can practice to improve your footwork. Bladework: Bladework is the ability to use the sword effectively. There are many different bladework techniques, and you need to practice them in order to become proficient. Parrying: Parrying is the ability to deflect an opponent's attack. There are many different parrying techniques, and you need to practice them in order to become proficient. Riposte: The riposte is the counterattack that follows a parry. It is an important skill in fencing, as it allows you to score points. Mental Focus: Mental focus is also an important skill in fencing, as it allows you to stay focused on the opponent and to avoid making mistakes.	
9	**Athletics** **History of Athletics** The history of athletics can be traced back to the ancient Olympic Games, which were held in Greece from 776 BC to 393 AD. The original events included running, jumping, throwing, and wrestling. The modern Olympic Games were revived in 1896, and athletics has been a part of every Games since then. Athletics is now a global sport, with competitions held at all levels, from local to international. **Fundamental Skills of Athletics** Running: Running is the most basic skill in athletics. It is the ability to move forward quickly and efficiently. There are many different types of running, including sprinting, distance running, and middle-distance	

	running. Jumping: Jumping is the ability to move upwards from the ground. There are many different types of jumping, including high jump, long jump, and triple jump. Throwing: Throwing is the ability to propel an object through the air. There are many different types of throwing, including shot put, discus throw, javelin throw, and hammer throw. Sprinting: Sprinting is a type of running that involves short bursts of speed. Sprinters need to be able to accelerate quickly and maintain their speed for a short period of time. Distance Running: Distance running is a type of running that involves running for long distances. Distance runners need to be able to pace themselves and maintain their energy levels for long periods of time. Middle-Distance Running: Middle-distance running is a type of running that involves running for distances between 800 meters and 1500 meters. Middle-distance runners need to be able to combine speed and endurance. High Jump: High jumping is a type of jumping that involves clearing a bar that is raised progressively higher. High jumpers need to be able to generate a lot of power in their legs and have good timing. Long Jump: Long jumping is a type of jumping that involves jumping as far as possible. Long jumpers need to have good speed and coordination. Triple Jump: Triple jumping is a type of jumping that involves jumping three times in a row. Triple jumpers need to have good speed, coordination, and power. Shot Put: Shot put is a type of throwing that involves throwing a heavy ball as far as possible. Shot putters need to have good upper body strength and technique. Discus Throw: Discus throw is a type of throwing that involves throwing a disc as far as possible. Discus throwers need to have good upper body strength and technique. Javelin Throw: Javelin throw is a type of throwing that involves throwing a spear as far as possible. Javelin throwers need to have good upper body strength and technique. Hammer Throw: Hammer throw is a type of throwing that involves throwing a heavy ball on a chain as far as possible. Hammer throwers need to have good upper body strength and technique.	
10	**Kho-Kho** **History of Kho-Kho** Kho-Kho is a tag game that originated in India. It is believed to have originated in the Indian subcontinent over 4,000 years ago. The game is mentioned in the Sangam literature of Tamil Nadu, which dates back to the 3rd century BC. Kho-Kho was first played as a competitive sport in the Indian Olympic Games in 1938. It was included as a demonstration sport at the 1982 Asian Games in Delhi, and it became a full medal sport in the 1990 Asian Games in Beijing. Kho-Kho is now played in over 100 countries around the world.	

	Fundamental Skills of Kho-Kho Touch: Touching an opponent is the most important skill in Kho-Kho. It is how raiders score points for their team. There are many different ways to touch an opponent, such as touching their arm, leg, or torso. Dive: Diving is a fundamental skill for all Kho-Kho players. It allows players to avoid being touched by the opponents. There are many different types of dives, such as front dive, side dive, and back dive. Stamina: Stamina is essential for all Kho-Kho players, as the game is very physically demanding. Players need to be able to run, jump, and dive for long periods of time. Agility: Agility is also important for Kho-Kho players, as they need to be able to change direction quickly and avoid being touched by the opponents. Teamwork: Teamwork is essential for Kho-Kho, as it is a team sport. Players need to be able to work together to score points and defend their territory.	
11	**Cricket** **History of Cricket** The history of cricket can be traced back to the 16th century in England. The game is believed to have originated from a game called "stoolball", which was played by children in the 15th century. The first recorded cricket match was played in 1611 between two teams of Kentish cricketers. Cricket became a popular sport in England during the 18th century, and it was first played in Australia in 1826. Cricket became an international sport in the 19th century, and the first Test match was played between England and Australia in 1877. Cricket is now played in over 100 countries around the world. **Fundamental Skills of Cricket** Batting: Batting is the act of hitting the ball with a bat. It is the most important skill in cricket, as it is how runs are scored. There are many different batting techniques, such as the defensive technique and the attacking technique. Bowling: Bowling is the act of delivering the ball to the batsman. There are many different bowling techniques, such as the fast bowling technique and the spin bowling technique. Fielding: Fielding is the act of catching the ball and preventing the batsman from scoring runs. It is an important skill for all cricketers, as it helps to prevent the other team from scoring runs. Running: Running is essential for scoring runs in cricket. Players need to be able to run quickly between the wickets to score runs. Stamina: Stamina is essential for all cricketers, as the game is very physically demanding. Players need to be able to run, jump, and field for long periods of time. Agility: Agility is also important for cricketers, as they need to be able to change direction quickly and avoid being run out. Teamwork: Teamwork is essential for cricket, as it is a team sport. Players need to be able to work together to score runs and prevent the other team from scoring runs.	

12	**Rifle Shooting** **History of Rifle Shooting** The history of rifle shooting can be traced back to the 16th century in Europe. The first recorded rifle shooting competition was held in 1533 in Zurich, Switzerland. Rifle shooting became a popular sport in Europe during the 18th century, and it was first introduced to the United States in the 1770s. Rifle shooting became an Olympic sport in 1896, and it has been a part of every Games since then. Rifle shooting is now a popular sport all over the world. **Fundamental Skills of Rifle Shooting** Accuracy: Accuracy is the most important skill in rifle shooting. It is the ability to hit the target with the bullet. There are many different factors that affect accuracy, such as the stance, the grip, the breathing, and the trigger control. Consistency: Consistency is also important in rifle shooting. It is the ability to hit the target with the same accuracy shot after shot. There are many different factors that affect consistency, such as the mental focus and the physical preparation. Stance: The stance is the position of the body when shooting. It is important to have a stable stance in order to be accurate. There are many different stances that can be used, such as the standing stance, the kneeling stance, and the prone stance. Grip: The grip is the way that the rifle is held. It is important to have a firm grip in order to be accurate. There are many different grips that can be used, such as the weaver grip and the isosceles grip. Breathing: Breathing is important in rifle shooting because it can affect the accuracy of the shot. It is important to breathe slowly and evenly before and after the shot. Trigger Control: Trigger control is the ability to pull the trigger smoothly and evenly. It is important to avoid jerking the trigger, as this can cause the shot to go off target.	
13	**Yoga** **History of Yoga** Yoga is a mind and body practice with a 5,000-year history in ancient Indian philosophy. The word "yoga" comes from the Sanskrit word "yuj," which means "to yoke" or "to unite." Yoga is a system of physical postures, breathing exercises, and meditation designed to help practitioners achieve physical, mental, and spiritual well-being. The earliest written records of yoga date back to the 2nd century BCE, and the practice has been evolving ever since. Yoga has spread to all corners of the world, and there are now many different styles of yoga practiced today. **Fundamental Skills of Yoga** Postures: The postures, or asanas, are the physical component of yoga. There are many different postures, and they can be practiced in a variety	

	of ways. Breathing: Breathing, or pranayama, is an important part of yoga. There are many different breathing techniques, and they can be used to help relax the body and mind. Meditation: Meditation is the mental component of yoga. There are many different meditation techniques, and they can be used to help focus the mind and achieve a state of peace.	
14	**Swimming** **History of Swimming** Swimming is one of the oldest sports in the world, with evidence of swimming dating back to 2500 BC. The earliest recorded swimming competitions were held in ancient Greece and Rome. Swimming became an Olympic sport in 1896, and it has been a part of every Game since then. Swimming is now a popular sport all over the world, and there are many different types of swimming, including freestyle, backstroke, breaststroke, butterfly, and individual medley. **Fundamental Skills of Swimming** Breathing: Breathing is one of the most important skills in swimming. It is important to be able to breathe efficiently while swimming in order to avoid getting tired. Body Position: Body position is another important skill in swimming. It is important to maintain a good body position in order to be hydrodynamic and to swim efficiently. Stroke Technique: Stroke technique is the way that the arms and legs are used to propel the body through the water. There are many different stroke techniques, and it is important to develop a good stroke technique in order to swim efficiently. Drills: Drills are exercises that can be used to improve swimming skills. There are many different drills, and they can be used to improve different aspects of swimming, such as breathing, body position, and stroke technique. Mental Toughness: Mental toughness is also an important skill in swimming. It is important to be able to stay focused and motivated during long swims, especially in competitions.	

Semester –II

Course Code: APS21BSL103 **Course Name:** Linear Algebra and Differential Equations **Course Category:**

Credits: 4 **Teaching scheme:** L-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university mathematics.

Course Objectives:

1. This course aims to make the students become familiar with the basic concepts of linear algebra with a thorough understanding of vector spaces, linear transformations and matrix operations enhancing the students' ability to reason mathematically and able to apply this knowledge to many fields in engineering, statistics and computer science.

2. Create and analyze mathematical models using differential equations.

Course Outcomes:

On completion of the course, the student should be able to:

1. Understand basic concepts such as vector spaces, linear dependence / independence of vectors, basis and linear maps.

2. Analyze and calculate eigen values, eigen vectors, rank nullity of a matrix / linear map.

3. Prove theorems, apply Gram-Schmidt process on inner product spaces, diagonalize special matrices.

4. Solve ordinary differential equations of first order and apply knowledge of differential of equations to solve engineering problems.

5. Find the solution of linear differential equations having their applications in mechanical and electrical systems.

Contents:

Unit	Content	Teaching Hours
1	**Matrices and Vector Spaces:** Basic properties of matrices, row operations and Gauss elimination, Inverse of a matrix. Basic concepts in linear algebra: vector spaces, subspaces, linear independence and dependence of vectors, bases, dimensions. Row and Column spaces, rank. Applications to systems of linear equations, Inverse transformation.	10
2	**Linear mappings and Diagonalization:** Linear mappings, representation by matrices, rank-nullity theorem, Diagonaliazation, Eigen values, Eigen vectors and their basic properties, Cayley Hamilton Theorem.	10
3	**Inner Product Spaces and Quadratic Forms:** Inner Product Spaces , Orthogonality, Gram-Schmidt process, Geometric Applications of Linear Transformation, Quadratic Forms: Positive Definiteness and applications	10
4	**First order ordinary differential equations and Applications:** Exact , Linear ,Bernoulli and separable differential equations, Applications to Population growth/decay, Mixing problems, Draining tank/Torricelli's Law problems, Newton's Law of Cooling, Electric circuits, Falling bodies.	10
5	**Ordinary differential equations of higher orders and Applications:** Linear differential equations with constant and variable coefficients, method of variation of parameters, Applications to mass spring systems and electrical circuits and Bending of beam and columns.	10

Text Books:

1. Introduction to Linear Algebra (2nd edition) by Serge Lang, Springer

2. Elementary Linear Algebra (10th edition) by Howard Anton and Chris Rorres, John Wiley and sons.

3. D.Poole, LinearAlgebra:AModern Introduction, 2nd Edition,Brooks/Cole,2005.

4. S. L. Ross, Differential Equations, 3rd Edition, Wiley India, 1984.

Reference Books:

1. Shanti Narayan, Differential Calculus, S. Chand & Co.

2. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, NewYork.

3. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, NewDelhi.

4. P. N. Wartikar, J. N. Wartikar, Applied Mathematics (Vol I & II) , Pune Vidyarthi Griha Prakashan,Pune.

5. Differential Equations with Applications and Historical notes by George Simmons, Tata McGraw Hill publishing company Ltd, New Delhi

6. K.D Joshi, Calculus for Scientists and Engineers, CRC Press.

7. Prasad and Reena Garg, Advanced Engineering Mathematics, Khanna Publishing Company Private Limited, New Delhi.

8. Schaum's outlines of Linear Algebra (5th edition) by Seymour Lipschutz, Marc Lipson, McGraw-Hill Education (India) Private Limited, New Delhi

MGMUNIVERSITY

Course Code:APS21BSL104**Course Name:**Engineering Chemistry **Course Category:**

Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, MSE-20, ESE–40

Pre-requisites: Fundamentals of basic chemistry.

Course Objectives:

1. The primary objective of an engineering chemistry course is to familiarize the students with new developments in engineering chemistry.
2. To impart a sound knowledge on the principles of chemistry involving the different application oriented topics required for all engineering branches.
3. The objectives of the laboratory sessions are to enable the learners to get hands-on experience on the principles discussed in theory sessions and to recognize the applications of these concepts in engineering.

Course Outcomes:

At the end of the course, students will be able to,

CO.1 : Illustrate the water quality parameters, water softening processes and causes of hard water in industries.

CO.2 : Demonstrate a comprehensive understanding of advanced concepts in polymer chemistry.

CO.3 : Apply fundamental concepts of corrosion science to solve problems arising in engineering applications.

CO.4 : Interpret physical, chemical properties and applications of fuels and lubricants.

CO.5 Describe the functions of batteries and applications of modern instrumental techniques like conductometry, pH metry, TLC, gas chromatography, UV-Visible and IR spectroscopy.

Contents:

Unit	Content	Teaching Hours
1	**Water Treatment:** Natural sources of water, Impurities in water, Water quality parameters and its BIS standards, Hardness- Definition, Types, Estimation of hardness by EDTA method, Numerical based on hardness calculation, Disadvantages of hard water, Boiler troubles (causes, effect on boiler operation and methods of prevention), Internal and external treatments, Alkalinity and its determination, Water softening: Ion exchange process, Sewage water analysis- Dissolved oxygen (DO) and its determination, Biological oxygen demand (BOD), Chemical oxygen demand (COD) and their significance, Sewage water treatment.	09
2	**Polymer Chemistry:** Introduction, Classification of polymers, Use and disposal of polymers, Polymerization and its types, Plastics and its types- Thermoplastic and thermosetting plastics, Preparation, properties and engineering applications of: PVC, PMMA, Bakelite and Epoxy resin, Moulding constituents of plastics, Methods for moulding of plastics into articles,	08

	Conducting polymers and Biopolymers (Introduction, types, examples and its applications.	
3	**Corrosion and its Control:** Introduction, Types of corrosion, Mechanism of dry & wet corrosion, Factors influencing on corrosion – Nature of metal &Nature of environment. Methods of corrosion control, Cathodic and anodic protection, Use of Inhibitors, Protective Coatings: a) Metallic coatings: Types of coatings methods of applications, (hot dipping, cladding and electroplating), b) Nonmetallic coatings: Chemical conversion coatings, Powder coatings.	08
4	**Fuels and Lubricants:** **Fuels:** Introduction, Classification of fuel, Calorific value of a fuel, Characteristics of a good fuel, Solid fuel- Coal, Various types of Coal, Analysis of coal- Proximate and Ultimate analysis, Numerical based on analysis of coals, Liquid fuel- Refining of Petroleum, Gaseous fuels- LPG and CNG. **Lubricants**: Introduction, Mechanism of lubrication, Classification of lubricants, Solid, Semi-solid and Liquid Lubricants, Properties of lubricants, Physical properties – Viscosity & Viscosity index, Surface tension, Flash and Fire point, Cloud and pour point. Chemical properties – Acid value, Saponification value, Aniline point.	10
5	**Electrochemistry and Instrumental Methods of Chemical Analysis**: **Electrochemistry**: Introduction - Basic concepts: Conductance, Specific Conductance, Equivalent conductance, Molecular conductance, Effect of dilution on conductance, Cell constant. Battery: Primary & secondary batteries, Ni-Cd cell, Lithium-air battery, Fuel cell- H_2-O_2 Cell. **Instrumental Methods of Chemical Analysis**: Basic principle, instrumentation and applications of pH metry, Conductometry, Thin layer chromatography, Gas Chromatography, Ultraviolet-Visible spectroscopy and Infra-Red Spectroscopy	10

Text Books:

1. Jain P.C & Jain Monica, Engineering Chemistry, Dhanpat Rai & Sons, Delhi.
2. O. G. Palanna , Engineering Chemistry, Tata McGraw-Hill Publication, New Delhi.
3. S. S. Dara, A textbook of Engineering Chemistry, McGraw-Hill Publication, New Delhi.
4. Shashi Chawla, Engineering Chemistry, Dhanpat Rai and Co Education and Technical Publishers.
5. Shikha Agrawal, Engineering Chemistry- Fundamentals and Applications, Cambridge Publishers

Reference Books:

1. Barrow G.M., Physical Chemistry, McGraw-Hill Publication, New Delhi.
2. Atkins, Physical Chemistry, OUP Oxford.
3. Bhal & Tuli, Text book of Physical Chemistry, S. Chand & Company, New Delhi.
4. V. R Gowarikar, Polymer Science, New Age International Ltd.
5. B. K. Sharma, Instrumental Methods of Chemical Analysis, Krishna Prakashan Medi

E-Resources:

Name of the website/ E-Journals/ Online Videos

1. NPTEL Basic Courses Engineering Chemistry (https://nptel.ac.in/courses/122/101/122101001/)
2. https://www.ncertbooks.guru/engineering-chemistry/
3. Coursera Chemistry Courses (https://www.coursera.org/browse/physical-science-and-engineering/chemistry?languages=en&page=2)
4. "Introduction to Polymer Physics" NPTEL Course (https://www.youtube.com/playlist?list=PLwdnzlV3ogoXe67WsgE8f1fOIWcc5GKKS)
5. "Introduction to Corrosion" NPTEL Course (https://www.youtube.com/playlist?list=PL8lylDWRkaW8BXestE4baRnN0699S11Lq)
6. "Tribology & Lubrication" NPTEL Course (https://www.youtube.com/playlist?list=PLLwnvFq-JAltJvWAfEVU4gcUK27hEA7FD)
7. "Electrochemistry" NPTEL Course (https://www.youtube.com/playlist?list=PLVFqK_9GOGXnnriQpsn0z1Rss96Rh0vsm)
8. "Modern Instrumental Methods of Analysis" NPTEL Course (https://www.youtube.com/playlist?list=PL400CAFBA72E94CF8)

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Course Code:APS21ESL103**Course Name:**EngineeringMechanics **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites:

1. CoordinateGeometry,Trigonometry,Sine&CosineRule,UnitConversions
2. FundamentalsofPhysics

Course Objectives:

Todevelopcapacitytopredictttheeffectofforceandmotioninthecourseofcarryingout thedesignfunctions of engineering.

Course Outcomes:

Studentsareableto,
CO1.Identifytheforcesystemforgivenconditionsbyapplyingthe basicsofmechanics.
CO2.Determinetheunknownforcesofdifferentengineeringsystemsbyapplyingequilibriu mconditions.
CO3.ApplytheprinciplesoffrictionandtolocateCenterofGravityandfindMomentof Inertiaofplanelamina.
CO4.Establishtherelationsbetweenkinematicparametersfordifferenttypesofmotion.
CO5.Formulatetherelevantequationsfortypesofmotioninkinetics.

Contents:

Unit	Content	Teaching Hours
1	**ForceSystem**: IntroductiontoMechanics,Lawsofmechanics,Newton'sLaws,LawofParallelogra m,Lawoftransmissibility,Characteristics of force, Systemof Forces, Method of resolution and compositionmomentof aforce, Law of Moments, Varignon's Theorem, Problemsonmoment,Resultant,Equivalentforce&couple,propertiesof couple	08
2	**Equilibrium:** IntroductiontoEquilibriumanditstypes,Equilibrant,ConceptofFBD,Analyticalc onditionsof equilibrium, Equilibriumofdifferentforcesystem,Lami'sTheorem,Typesofloads,beamsandsu pports.	07
3	**Friction,CentreofGravityandMomentofInertia:** Friction: Introductiontofriction,typesandapplication,Lawsoffriction,Angleoffriction,An gleofrepose, Cone of friction, Problems on horizontal &inclinedplane,block,andladder. CG&MI: Centroid of regular and composite plane	08

	lamina,MIanditsapplication,PerpendicularaxisTheorem,ParallelAxisTheorem,RadiusofGyration,Problemsonplaneandcompositelamina.	
4	**Kinematics:** Introduction and classification of dynamics, motionanditsclassification,RectilinearMotion,EquationofMotion,Motioncurves,CurvilinearMotion,rectangularandtangentialcomponentsofacceleration,ProjectileMotion:GeneralEquation ofProjectileMotion	04
5	**Kinetics**: Basicconceptsandlawsofmotion,D'AlembertsPrinciple, Problems on rectilinear motion,Curvilinearmotion,WorkEnergyPrinciple,WorkDonebyforce,WorkDoneby weight force,WorkDonebyfrictionalforce,WorkDonebyspringforce,KineticandPotentialenergyoftheparticle,ProblemsonallcasesofWorkDone,PrincipleofImpulse and Momentum, Principle ofConservationMomentum,Impact and its types,Coefficientof restitution, Problems on impact, Problemsonimpulseandmomentum,Kineticsof rigidbodyproblems.	03

TextBooks:

1. Beer,F.PandJohnstonJr.E.R.,"VectorMechanicsforEngineers(InSIUnits):StaticsandDynamics",8thEdition,Tata McGraw-HillPublishingcompany,NewDelhi(2004).
2. VelaMurali,"EngineeringMechanics",OxfordUniversityPress(2010).
3. S.Timoshenko,D.H.Young,"EngineeringMechanics",McGrawHill,1995.
4. TayalA.K.,"EngineeringMechanics",UmeshPublications,2010.
5. SingerF.L.,"EngineeringMechanics-Statics&Dynamics",HarperandRowPub.York.
6. KhurmiR.S.,"EngineeringMechanics",S.ChandPublications,N.Delhi.

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ReferenceBooks:

1. McLean,Nelson,"EngineeringMechanics",Schaum'soutlineSeries,McGrawHillBookCompany,N.Delhi,Publication.
2. Hibbeller,R.CandAshokGupta,"EngineeringMechanics:StaticsandDynamics",11thEdition,PearsonEducation(2010).
3. Bhavikatti,S.SandRajashekarappa,K.G.,"EngineeringMechanics",NewAgeInternational(P)LimitedPublishers,(1998).
4. IrvingH.ShamesandKrishnaMohanaRao.G.,"EngineeringMechanics– StaticsandDynamics",4thEdition,PearsonEducation(2006).
5. RajasekaranSandSankarasubramanianG.,"EngineeringMechanicsStaticsandDynamics",3rdEdition,VikasPublishing HousePvt.Ltd., (2005).
6. MeriamJ.L.andKraigeL.G.,"EngineeringMechanics-Statics-Volume1,Dynamics-Volume2",ThirdEdition,JohnWiley& Sons,(1993).
7. Kumar,K.L.,"EngineeringMechanics",3rd RevisedEdition,TataMcGraw-HillPublishingcompany,NewDelhi(2008).

GeneralInstructions:

E-Resources:

1. www.nptel.ac.in(LearningplatformfromIITprofessors)
2. http://www.asnu.com.au(ForEngineeringapplications)
3. www.discoveryforengineers.com(InvestigatingDiscoveries)

Course Code: APS21ESL104 **Course Name:** Building Programming logic in C **Course Category:**

Credits: 1 **Teaching scheme:** L-1 **Evaluation scheme:** CA–60, ESE–20

Pre-requisites: Pre-university mathematics.

Course Objectives:

1. The primary objective of an engineering chemistry course is to familiarize the students with new developments in engineering chemistry.
2. To impart a sound knowledge on the principles of chemistry involving the different application oriented topics required for all engineering branches.
3. The objectives of the laboratory sessions are to enable the learners to get hands-on experience on the principles discussed in theory sessions and to recognize the applications of these concepts in engineering.

Course Outcomes:

After the completion of this course, students will be able to:
CO1: Illustrate and explain the basic computer concepts and programming principles of C language.
CO2: Develop C programs to solve simple mathematical and decision making problems.
CO3: Develop C programs to solve simple engineering problems using looping constructs.
CO4: Develop C programs to demonstrate the applications of derived data types such as arrays, pointers, strings and functions.

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Contents:

Unit	Content	Teaching Hours
1	**Introduction To C Programming:** **Fundamentals of C Programming: Overview of C:** History of'C, Algorithm and Structure of C program. Keywords, Tokens, Data types, Constants, Literals and Variables. **Operators and Expressions:** Arithmetic operators, Relational operator, Logical operators, Expressions, Operator: operator precedence and associatively, Type casting, Console FO formatting, Unformatted I/O functions: getch(), getchar, getche(), getc(), putc(), putchar().	03
2	**Array and String:** **Control statements:** If-else, conditional operators, switch and break, nested conditional branching statements, loops: do while, while, for, Nested loops, break and continue, goto and label, exit function. **Array:** Array declaration, One and Two dimensional numeric and character arrays, Multidimensional arrays, operations on array. **String:** String declaration, initialization, string manipulation with/without using library function.	03
3	**Control Statements and Functions:** **Functions:** Definition, function components: Function arguments, return value, function call statement, function prototype, Types of function, Scope and lifetime of variable, Call by value,and call by reference. Function using arrays, function with command line argument. User defined function: maths and	03

	character functions, Recursive function.	
4	**Structure and Union:** **Structure:** Basics, declaring structure and structure variable, typedef statement, array of structure, array within structure, Nested structure; passing structure to function, function returning structure. **Union:** basics, declaring union and union variable, Difference between Structure and Union **Enum:** declaring enum and enum variable.	03
5	**Pointers**: **Pointer:** Definition of pointer, advantage and disadvantage using pointer, Pointer declaration, Using & and * operators. Void pointer, Pointer to pointer, Pointer in math expression, Pointer arithmetic, Pointer comparison, **Dynamic memory allocation functions**: malloc, calloc, realloc and free, Pointer vs. Array, Array of pointer, Pointer to array, Pointers to function, Function returning pointer, Passing function as Argument to function, Pointer to structure, Dynamic array of structure through pointer to structure.	03

Text Books:

1. Programming in ANSI C, E Balagurusamy, Tata McGraw-Hill, Third Edition.

2. Let Us C, YashwantKanetkar, Infinity Science Press, Eighth Edition.

3. Mastering C, K R Venugopa1,Tata McGraw-Hill.

Reference Books:

1. The C Programming Language, Brian W. Kernighan, Dennis M. Ritchie, Prentice Hall, 2nd Edition.
5.Applications Programming in ANSI C, R. Johnsonbaugh , Martin Kalin, Macmillan, 2nd Edition.

2. The Spirit of C, Mullish Cooper, Jaico publishing House.

3. How to solve it by Computer, R.G.Dromey, Pearson Education.

MGMUNIVERSITY

Course Code:APS21PCL101**Course Name:**BasicsofElectricaland ElectronicsEngineering
Course Category:

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university mathematics.

Course Objectives:

1. UnderstandElectricalcircuitsandclassifycircuitsasperlaws.
2. UnderstandMagneticcircuitsandapplythemintransformerdevices.
3. Understandtheworkingprincipleofsemiconductordevices.
4. Understand the basics and applications of digitalelectronics.

Course Outcomes:
Aftercompletionofthiscourse,studentswill beableto:

1. SolvesimpleDCandsingle-phaseACcircuitsusingKCL,KVLand networktheorems.
2. Explain fundamentals of magnetic circuit andtransformers.
3. Explaintheworkingofdiodesandtransistors.
4. Explain the importance of number systems and logicgates.

Contents:

Unit	Content	Teaching Hours
1	**Electrical Circuits:** Voltage and current sources: independent,dependent,idealandpractical;V-Irelationships of resistor, inductor, mutualinductor and capacitor; Kirchhoff's laws,meshandnodalanalysis,superposition,Thevenin's,maximumpowertransfertheorems.Alternatingvoltagesandcurrents, RMS, average, maximum values,Single Phase RL, RC, RLC series circuits,PowerinACcircuits,PowerFactor,Threephasebalanced systems.	08
2	**MagneticCircuits&Transformer:** Conceptsofm.m.f,flux,fluxdensity, reluctance, permeability and field strength,theirunitsandrelationship,right handthumbruleandcorkscrewrule, Faraday's law of electromagnetic induction, Fleming's right-hand rule, staticallyanddynamicallyinducedEMF's selfandmutualinductancecoefficientof coupling, energy stored in magnetic circuit, Single phase transformers: Construction,principleofworking,e.m.f. equations.	08
3	**SemiconductorDevices:** IntroductiontoSemiconductors,P-type andN-typeSemiconductors,P-NJunction	07

	Diode:Constructionandworking,V-I characteristicsofDiode, Reverse breakdownmechanism. Specialpurpose diodes:ZenerDiode, LightEmitting Diode(LED) and PhotoDiode - Construction,workingandapplications. BipolarJunctionTransistor(BJT):types, constructionandregions ofoperation.	
4	**DigitalElectronics:** Binary, Decimal, Octal, Hexadecimal number systems and their inter-conversion, Binary Addition, Binary Subtraction,One'scomplement,Two's complement, Logic Gates: Basic, Universal and Special. Boolean Laws (AND, OR, NOT) and Demorgan's theorems,Realizationofbasiclogicgates usinguniversalgates.	07

TEXTBOOKS:

1. V.N.MittalandArvindMittal,"BasicElectricalEngineering"McGrawHill
2. EdwardHughes,"ElectricalTechnology,",PearsonEducation
3. ElectricalTechnologyVol.1& Vol.4byB.L. Theraja&A.K.Theraja,S.ChandPublications
4. "PrinciplesofElectronics",V.K.Mehta,S.ChandPublications.
5. "ElectronicsDevices and Circuits",SSalivahanan,McGrawHillPublications.
6. "ModernDigitalElectronics",R.P.Jain,McGrawHillPublications.

REFERENCEBOOKS:

1. VincentDelToro,"ElectricalengineeringFundamentals",PHIsecondedition2011
2. RobertBoylestad,"ElectronicsDevicesandCircuitsTheory",PearsonEducation India
3. "ElectronicsDevicesand CircuitsTheory",RobertBoylestad,PearsonEducationIndia.

E-Resources:

1. OnlinecourseonNPTEL"BasicElectricalEngineering"
2. NPTELCourseon"IntroductiontoBasicElectronics",Prof. T.S.Natarajan,IITMadras.
3. NPTELCourseon"DigitalElectronicCircuits",Prof.GoutamSaha, IITKharagpur.

MGM UNIVERSITY

Course Code:APS21IKL1XX**Course Name:**Indian Knowledge Systems **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Nil

Course Objectives:

The objective of this course is
1. To make students understand foundational concepts in IKS for science, engineering and technology.
2. To explore ancient Indian pursuits and accomplishments in the various domains of engineering

Course Outcomes:

On completion of the course, the student should be able to:
CO 1: familiarise with key components of the IKS & develop appreciation for Indian philosophical systems.
CO 2: understand key features of Indian Numeral System, units of measurement and the framework for establishing the right knowledge.
CO 3:appreciate the unique & specific contributions of ancient Indian mathematicians in Arithmetic, Geometry & Trigonometry.
CO 4:develop awareness about engineering & technology heritage of India and understand ancient Indian contributions in various engineering domains.

Contents:

Unit	Content	Teaching Hours
1	**Introduction to IKS:** Importance of ancient knowledge, defining IKS, Classification framework for IKS, Historicity of IKS, Indian philosophical systems, Vedic schools of philosophy(Sankhy and yoga, Nyaya and Vaisesika, Purva- Mimamsa and Vedanta), Non- Vedic philosophical systems (Jain, Bauddha and Carvaka) , Wisdom through the ages: Issues of interest in the Puranas, Itihasa as a source of wisdom (Uniqueness of the two epics), Nitisastras.	08
2	**Foundational Concepts for Science and Technology:** Number system in India, salient features of the Indian numeral system, Measurements for Time, Distance and weight, The knowledge triangle, Prameya- a Vaisesikan approach to Physical Reality, Pramana – The means of valid knowledge, Framework for establishing valid knowledge.	07
3	**Science in IKS:** Mathematics: Great mathematicians and their contributions, Arithmetic (square of a number, square root, series and progressions), Geometry (Property of right angled triangle in Sulba- sutras, value of π), Trigonometry, Algebra, Binary Mathematics and combinatorial problems in Chandah- Sastra of Pingala	07
4	**Engineering and Technology in IKS:** The Indian Science and Technology Heritage, Mining and ore extraction, Metals and metal working technology, Iron and Steel in India, Lost wax casting of idols and artefacts, apparatuses used for extraction of metallic components. Literary sources for Science and Technology, Physical Structures in India, Irrigation & Water Management, Dyes and Painting Technology, Shipbuilding.	08

Textbooks:

- Mahadevan B.,Bhat Vinayak Rajat & Nagendra Pavana R.N. "Introduction to Indian Knowledge System Concepts and Applications"PHI,2023.
- Jha Amit "Traditional Knowledge System in India" Atlantic Publishers and Distributors (P) Ltd,2023

- Chauhan Bhag Chand "IKS: The Knowledge system of Bharata" Garuda Prakashan,2023
- Bag A.K., "Mathematics in Ancient and Medieval India"Chaukhambha Orientalia, 1979
- Sengupta Nirmal,"Traditional Knowledge in Modern India"Springer,2019

Reference Books:

- Bag A.K., "Historyof Technology in India,Vol.I" Indian National Science Academy,1997
- Kumar Alok, "Ancient Hindu Science" Jaico Publishing House,2019
- Datta B and Singh A.N."History of Hindu Mathematics:Parts I and II" Asia Publishing House,1962
- Kapoor Kapil, Singh Awdhesh Kumar "Indian Knowlege Systems Vol.-I & II" D.K. Print World Ltd,2005

E-Resources:

https://www.youtube.com/watch?v=-cBd6JYPWtY&list=PLRfu94TCePTtWtu0x145H_63WgoeYickE

https://www.youtube.com/watch?v=yvj5ROYbP2E&list=PLRfu94TCePTtLuEYSzmJXNYK_EnDSvY3N

https://iksindia.org/book-list.php

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Course Code:APS21VSP102**Course Name:**WorkshopPractices **Course Category:**

Credits: 2 **Teaching scheme:** L-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university English.

Course Objectives:

1.To know about the different carpentry tools and perform various carpentry operations to complete the job.

2. To understand different welding tools, joints, defects and perform welding operation to complete useful article/job.

3. To learn various types of pipes, plumbing tools, operations and perform thread cutting on GI pipes.

4. To know different sheet metal tools, operations, applications and perform various operations to complete job.

Course Outcomes:

On completion of the course, the student should be able to:

1. Perform basic carpentry operation on wood andprepare carpentry article.

2. Perform welding processes and prepare welding article by performing various welding operations.

3. Use plumbingtools, processesandperform threading onGI pipe.

4. Use sheet metal tools, processesand prepare sheet metal article.

Contents:

Unit	Content	Teaching Hours
1	**Carpentry Shop:** Carpentry shop: Types of woods, tools, joints, operations, applications, safety measures etc. Job: Exercises on wood involving operations marking, sawing, chiselling, planning, grooving etc to make useful wooden article/ job e.g. Wooden Trophies, Showpiece articles, Stools etc.	08
2	**Welding Shop:** Welding Shop: Types of welding, welding joints, tools, welding defects, applications, safety measures etc. Job: ExerciseinArcweldingtomake useful articles like Grills, Stools, Tree Guards, Flower pot stand, Shoe rack, Bag Stand, Showpiece Articles from Scrap etc.	07
3	**Plumbing Shop:** Plumbing shop: Study of types of pipe, pipe joints, operations, applications, safety measures etc. Job: Prepare threading on GI pipe to make useful items like Nipple, Pipe joints etc.	07
4	**Sheet Metal Shop:** Sheet Metal Shop: Sheet metal tools, operations, applications, safety measures etc. Job: Making an utility item using G I sheet involving development, marking, cutting , bending, spot welding/riveting Parts like i) Tray, ii) Funneletc.	08

- **Termwork:** Students shall maintain workshop diary which contains Job Drawing, information of tools, operations required to complete the jobs, records of job completions etc.

- **End Semester Exam:** ESE will be viva-voce based on jobs prepared by the students during the term.

Text Books:

2. Hazra and Chaudhary, Workshop Technology-I, Media promoters & Publisher private limited.

3. Hazra and Chaudhary, Workshop Technology-II, Media promoters & Publisher private limited.

Reference Books:

1. K. C. John, Mechanical Workshop Practice, Prentice Hall Publication, New Delhi, 2010.

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Course Code:APS21BSP102**Course Name:Engineering Chemistry Lab Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

Course Outcomes:

Upon completion of the Lab Experiments, the students should able to,

LO.1 **:** Perform the experiments as well as accurately record the readings and analyze the results of such experiments.

LO.2 : Estimate the impurities present in water using titrimetric and instrumental methods.

LO.3 : Employ the basic techniques used in chemistry laboratory for analysis such as volumetric titrations, complexometric titrations, Conductometry, pH metry, viscometer, Stalagmometer and TLC

Contents:

Sr. No.	List of Practical	Lab Hours
1	Determination of hardness of water sample by EDTA method.	02
2	Determination of chloride content in water sample by precipitation titration method.	02
3	Determination of dissolved oxygen in water by Iodometric method.	02
4	Determination of percentage purity of bleaching powder.	02
5	Determination of strength of acid / base using pH metric titration.	02
6	Determination of strength of acid / base using conductometric titration.	02
7	To determine the cell constant of the given conductivity cell.	02
8	To determine relative surface tension of unknown liquids by using stalagmometer.	02
9	To determine the viscosity of unknown liquids by using Ostwald / Redwood viscometer.	02
10	To determine acidity of given water sample.	02
11	Determination of acid value of an oil sample.	02
12	Determination of saponification number of an oil sample.	02
13	To determine alkalinity of given water sample.	02
14	Preparation of phenol- formaldehyde / urea-formaldehyde resin.	02
15	To find out the Rf value of given sample by thin layer chromatography.	02
16	To separate the given mixture using thin layer chromatography.	02
17	Proximate analysis of coal (moisture content, volatile matter, ash content).	02
18	To study factors influencing on rate of electrochemical corrosion.	02

19	To determine flash and fire point of given oil by Pensky-Marten flash point apparatus.	02
20	To determine cloud and pour point of lubricating oil.	02

Reference Books:

1. A Text book on Experiments and Calculations in Engineering Chemistry by Dr. S. S. Dara, S Chand Publication.
2. Laboratory manual on Engineering Chemistry by S. K. Bhasin & Sudha Rani, Dhanpat Rai Publishing Company.
3. Engineering Chemistry with Laboratory Experiments by M. S. Kaurav, Asia-Pacific the holdings Private Ltd.

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Course Code:APS21ESP104**Course Name:**Engineering Mechanics lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

Course Outcomes:
Upon completion of the Lab Experiments, the students should able to,
1.Describe the working principle of mechanics and correlate them with day to day engineering applications.
2.Formulate and solve mechanics problems based on law of moments, conditions of equilibrium.
3.Verify theoretical concepts through analytical, experimental and graphical methods.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Verification of law of Machine using Screw jack	02
2	Polygon law of coplanar forces.	02
3	Bell crank lever	02
4	Lami's theorem	02
5	Support reactions for beam	02
6	Problems on beam reaction by graphics statics method	02
7	Inclined plane (to determine coefficient of friction).	02
8	Centroid of irregular shaped bodies	02
9	Determine center of gravity for composite sections	02
10	Determine moment of inertia for composite sections	02
11	Moment of Inertia of fly wheel	02
12	Simple / compound pendulum	02

References:

1. "College of Engineering, Pune"
2. "University of Mumbai"

Course Code:APS21ESP105**Course Name:**Programming in C - LAB **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives: Nil

Course Outcomes: Nil

Contents:

Sr. No.	List of Practical	Lab Hours
1	a) Write a C program to find sum and average of three numbers. b) Write a C program to find the sum of individual digits of a given positive integer.	02
2	a) Write a C program to generate the first n terms of the Fibonacci sequence b) Write a C program to generate prime numbers between 1 to n. c) Write a C program to check if the given number is Armstrong or not	02
3	a) Write a C program to check whether the given number is perfect or not b) Write a C program to check whether the given number is strong or not	02
4	a) Write a C program to find the roots of a quadratic equation. b) Write a C program perform arithmetic operations using switch statement.	02
5	a) Write a C program to find factorial of a given integer using non-recursive function b) Write a C program to find factorial of a given integer using recursive function	02
6	a) Write C program to find GCD of two integers by using recursive function. b) Write C program to find GCD of two integers by using non-recursive function.	02
7	a) Write a C program to find the largest and smallest number in a list of integers. b) Write a C program to Sort the Array in an Ascending Order. c) Write a C program to find whether the given matrix is symmetric or not.	02
8	a) Write a C program to perform addition of two matrices. b) Write a C program using function to perform multiplication of two matrices.	02
9	a) Write a c program to use function to insert a sub-string in to given main string from a given position. b) Write a c program to swap the values of two variables using (i) call by value (ii) call by reference	02
10	a) Write a C program using user-defined functions to determine whether the given string is palindrome or not. b) Write a C program that displays the position or index in the main string S where the sub string T begins, or - 1 if S doesn't contain T.	02
11	a) Write C program to count the number of lines, words and characters in a given text. b) Write a C program to find the sum of integer array elements using	02

	pointers.	
12	a) Write a C Program to Calculate Total and Percentage marks of a student using structure.	02

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Course Code:APS21PCP101**Course Name:**Basics of Electrical andElectronicsEngineeringLab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

1. UnderstandtheDC circuitasperKCL&KVLandACcircuitsaspertheorems.
2. Understandtheworking of asingle-phasetransformer.
3. Understandthe working ofelectronicinstruments,components andlogicgates.
4. Understandthe working of aPNjunction diode.

Course Outcomes:

Aftercompletion ofthislab, students willbeable to:

1. Useof KCL&KVL tosolve DCcircuitsaswellasuseof ACtheoremstosolveACcircuits.
2. Demonstrateasingle-phasetransformer &itsworking.
3. Useelectronicinstruments,workingof electroniccomponents andlogics gates.
4. UseofPNJunction Diodeandits applications.

Contents:

Sr. No.	List of Practical	Lab Hours
1	VerificationofLoopAnalysisandNodalAnalysisforDCCircuits.	02
2	VerificationofThevenin'sTheoremforDCCircuits.	02
3	VerificationofMaximumPowerTransferTheoremforDCCircuits.	02
4	DeterminationofVoltage,CurrentandPowerFlowinSinglePhaseACCircuitincludingR, L and C with Combination.	02
5	DeterminationofMagneticMaterialTermsandEMFInduction.	02
6	Demonstration and Determination of Single-Phase Transformer Terms likeVoltageRatio and TurnsRatio.	02
7	StudyofElectronic Instruments.	02
8	Studyof ElectronicComponents.	02
9	Studyof LogicGates.	02
10	Studyof V-Icharacteristicsof aPNJunction Diodeusing V-Lab.	02
11	StudyofRectifierCircuitsusingEveryCircuitsimulationapplication.	02
12	Study ofworkingand troubleshootingofSmartphone, Computer&TV(CaseStudyApproach).	02

Course Code: MGM82CCP104 **Course Name:** National Service Scheme (NSS) **Course Category:**

Credits: 2 **Teaching scheme:** P-2 **Evaluation scheme:**

Pre-requisites: Nil

Course Objectives: Nil

Course Outcomes: Nil

Contents:

Sr. No.	List of Practical	Lab Hours
1	**IntroductiontoNationalServiceScheme(NSS):** **Emergence of NSS in India (Historical Background) and its development.**OrganizationalStructureofNationalServiceSchemefromNationalLeveltoCollegeLevel.Objectivesof National ServiceScheme(NSS) NationalServiceScheme(NSS)–Symbolanditsmeaning **SymbolofNSSanditsmeaning** MottoofNationalServiceScheme(NSS) Variousprayers,inspirationalsongstobeusedinNSSProgramme.	08
2	**NationalServiceScheme(NSS)RegularActivities:** **GuidelinesofDistributionofworkinghoursoracademicyear.** **ClassificationofRegularActivities intheSociety** • Rural • Urban • Campus • Needbasewithassociation **AssociationsinNSSActivities** • Govt.Organization • NGO ScopeforInnovation(Self-Generated)	08
3	**SocialIssuesin India:** **ConceptofSociety,Community(Stepsinvolvedinevaluationofsociety)** • FeaturesofIndianSociety • CommunitiesinIndia **BasicSocialIssuesin India** FamilySystem,Divisionoflabour,CastSysteminIndia,GenderIssues,RegionalImbalance	07
4	**IndianConstitutionandSocialJustice:** **IndianConstitution** • Preamble	07

	• Structure • Features • FundamentalRights&Duties **SocialJustice** • SocialJustice–theConceptanditsfeatures Contribution for Social Justice – Mahatma Jyotiba Phule, Dr. Babasaheb Ambedkar,ShahuMaharaj.	

References:

1. National Service Scheme Manual (Revised) 2006, Government of India, Ministry of Youth Affairs and Sports, New Delhi.
2. University of Mumbai National Service Scheme Manual 2009.
3. Avhan Chancellor's Brigade-NSS Wing, Training camp on Disaster Preparedness Guidelines, March 2012.
4. Rashtriya Seva YojanaSankalpana- Prof. Dr. SankayChakane, Dr. Pramod Pabrekar, Diamond Publication, Pune.
5. National Service Scheme Manual for NSS District Coordinators, National Service Scheme Cell, Dept. of Higher and Technical Education, Mantralaya,
6. Annual report of National Service Scheme (NSS) published by Dept. of Higher and Technical Education, Mantralaya,
7. NSS Cell, Dept. of Higher and Technical Education, Mantralaya, UTKARSHA- Socio and cultural guidelines.

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Course Code:MGM73CCP203 **Course Name:**Fine Art**Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The objective of this course is

- Students will be able to draw clean lines.
- Students will be able to explore shading from light to dark.
- Students will be able to draw perspective drawing.
- Students will be able to sketch landscape and portrait drawing.

Course Outcomes:
On completion of the course, the student should be able to:

- Students will be able to pay attention to core details in visualization.
- Students will be able to represent on paper what they have observed in terms of 3 and 2 dimensional objects and light and dark play of perspective.
- Students will be able to draw clean lines and neat figures which will gradually help them in fashion illustrations.

Contents:

Sr. No.	List of Practical	Lab Hours
1	**Introduction to Drawing:** • Explore pencil as medium for drawing and exploration of different types of grade of pencils. • Lines – Freeing hand movement by practicing various ways to draw lines (wavy/zigzag/diagonal/vertical/horizontal)	08
2	**3D objects with Shading:** • Exploring and sketching 3D objects with Shading/Hatching – Light to Dark (HB/2B/3B/4B/5B/6B) • Space Division – Studying of proportion, Perspectives- Coin / Box / Landscape.	07
3	**Sill Drawing:** • Introduction to landscape and portrait drawing. • Still drawing (perspective drawing of man- made object)	07
4	**Exploration of color mediums:** • Exploration of color mediums (chalk/ink/poster/pastels/water) • Landscape perspective of natural and man- made, charcoal rendering. • Still drawing of man- made and natural object by rendering and defining perspective.Ambedkar,ShahuMaharaj.	08

Reference Books:

- Mellem , Jeff; Sketching people & Live drawing basics.
- Belleville- Van Stone, France; Sketch: the Non-Artist’s guide to inspiration technique & drawing life

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Course Code:MGM73CCP204 **Course Name:**Visual Art **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The objective of this course is
- To understand the actual work process in advertising market.
- To understand the growth and necessity of advertising in market.

Course Outcomes:
On completion of the course, the student should be able to:
- Students will be able to choose a topic for campaign design.
- Students will understand that how campaign design is necessary
- Student will understand the actual work process in advertising market.
- Student will understand the process of designing.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Introduction to Campaign Design**:** • Topic for campaign design • It should be either product, service or social topic. • Mind mapping	08
2	**:** • Different types of media • New digital medias • Use of elective subject in campaign • Software use in designing	11
3	**Sill Drawing:** • Final layout • Printing in actual size media • Presentation on ppt of the topic includes artwork with rough work.	11

Reference Books:
- **Advertising Campaign Design Just the Essentials - By Robyn Blakeman**
- **Campaign Strategies and Message Design: A Practitioner's Guide from Start to Finish - by Mary Moffitt (Author)**
- **Student should refer the previous knowledge about medias and refer the internet for help.**

Suggested Tutorial / Practical:

Assignment I: Students will be required to submit mind map.

Assignment II: Students will be required to submit list of media selection.

Assignment III: Students will be required to submit different medias in digital form.

Assignment IV: Students will be required to submit different medias in print form.

Teaching Methods and learning activities-

- **Lecture with power point projections**
- **Demonstration with practical**
- **Documentaries and Presentation**
- **Use of different medium and art form to make a new concept of work**

Assessment Strategies-

Classroom Lecture

Power Point Presentations

Group Discussion

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Syllabus
Semester –I

Course Code:APS21BSL101 **Course Name:** Single and Multivariable Calculus **Course Category:**

Credits: 4 **Teaching scheme:** L-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university mathematics.

Course Objectives:
1. To provide the basic tools of calculus for the purpose of modelling the engineering problems mathematically and obtaining solutions.
2. To convey a sense of the utility of calculus and develop technical competence of the student.

Course Outcomes:
On completion of the course, the student should be able to:
CO1. Compute derivatives in engineering problems.
CO2. Compute definite integrals arise in the problems such as arc length, surface of revolution, Work and Fluid Forces.
CO3. Find the partial derivatives and apply the knowledge of partial differentiation to find maxima and minima of functions, Jacobians, estimating error and approximation.
CO4. Calculate Area, Volume, Center of mass and Gravity using Double and Triple integral.
CO5. Solve the problems on fundamental theorems of vector calculus such as Green's, Stokes and Divergence theorems.

Contents:

Unit	Content	Teaching Hours
1	**Differentiation and its Applications:** Limit ,Continuity and Differentiation, Rate of Change in sciences and Engineering, Chain Rule and implicit differentiation, Related rates, Extreme value theorem, Rolle's Theorem, Lagrange's Mean value theorems, Nth derivatives, Taylor and Maclaurin series Expansions, Linear approximations and Differentials.	10
2	**Integration and its Applications:** Integration, Reduction formulae, Beta and gamma function, Properties, Evaluation of integrals using Beta and gamma functions, Application of Definite integrals to volume, arc length, surface of revolution, Work and Fluid Forces.	10
3	**Partial Differentiation and its Applications:** Limit and Continuity, Partial derivatives of first and higher orders, Total differentials, Errors and Approximations, Total derivative, Extreme values and saddle points, Method of Lagrange multipliers, Jacobians, Vector differentiation, Gradient, Curl and Divergence, directional derivatives.	10
4	**Multiple Integrals and its Applications:** Double integrals ,Evaluation of Double integrals**,** Change of order of	10

	integration in double integrals,Change of variables (Cartesian to polar), Applications: Area and Volume, Root mean square value, Center of mass and Gravity (constant and variable densities); Triple integrals (Cartesian),Volume by triple integration.	
5	**Vector Integration and its Applications:** Line integrals, Surface integrals, Green's Theorem, Stokes Theorems, Divergence theorems.	10

Text Books:

1. James Stewart, Calculus Early Transcendental, 7th edition, Cengage.

2. George B. Thomas, Ross L. Finney, Calculus and Analytical Geometry, 9th edition, Pearson.

3. Howard Anton, Irl Bivens, Stephens Davis, Calculus, 10th Edition, Wiley.

Reference Books:

1. Shanti Narayan, Differential Calculus, S. Chand & Co.

2. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, New York.

3. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi.

4. P. N. Wartikar, J. N. Wartikar, Applied Mathematics (Vol I & II) , Pune Vidyarthi Griha Prakashan, Pune.

5. H. K. Das and Rajnish Verma, Higher Engineering Mathematics , S. Chand & CO. Pvt. Ltd., New Delhi.

6. K. D Joshi, Calculus for Scientists and Engineers, CRC Press.

7. Prasad and Reena Garg, Advanced Engineering Mathematics, Khanna Publishing Company Private Limited, New Delhi.

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Course Code:APS21BSL104**Course Name:**Engineering Chemistry **Course Category:**

Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, MSE-20, ESE–40

Pre-requisites: Fundamentals of basic chemistry.

Course Objectives:

4. The primary objective of an engineering chemistry course is to familiarize the students with new developments in engineering chemistry.
5. To impart a sound knowledge on the principles of chemistry involving the different application oriented topics required for all engineering branches.
6. The objectives of the laboratory sessions are to enable the learners to get hands-on experience on the principles discussed in theory sessions and to recognize the applications of these concepts in engineering.

Course Outcomes:

At the end of the course, students will be able to,

CO.6 : Illustrate the water quality parameters, water softening processes and causes of hard water in industries.

CO.7 : Demonstrate a comprehensive understanding of advanced concepts in polymer chemistry.

CO.8 : Apply fundamental concepts of corrosion science to solve problems arising in engineering applications.

CO.9 : Interpret physical, chemical properties and applications of fuels and lubricants.

CO.10 Describe the functions of batteries and applications of modern instrumental techniques like conductometry, pH metry, TLC, gas chromatography, UV-Visible and IR spectroscopy.

Contents:

Unit	Content	Teaching Hours
1	**Water Treatment:** Natural sources of water, Impurities in water, Water quality parameters and its BIS standards, Hardness- Definition, Types, Estimation of hardness by EDTA method, Numerical based on hardness calculation, Disadvantages of hard water, Boiler troubles (causes, effect on boiler operation and methods of prevention), Internal and external treatments, Alkalinity and its determination, Water softening: Ion exchange process, Sewage water analysis- Dissolved oxygen (DO) and its determination, Biological oxygen demand (BOD), Chemical oxygen demand (COD) and their significance, Sewage water treatment.	09
2	**Polymer Chemistry:** Introduction, Classification of polymers, Use and disposal of polymers, Polymerization and its types, Plastics and its types- Thermoplastic and thermosetting plastics, Preparation, properties and engineering applications of: PVC, PMMA, Bakelite and Epoxy resin, Moulding constituents of plastics, Methods for moulding of plastics into articles,	08

	Conducting polymers and Biopolymers (Introduction, types, examples and its applications.	
3	**Corrosion and its Control:** Introduction, Types of corrosion, Mechanism of dry & wet corrosion, Factors influencing on corrosion – Nature of metal &Nature of environment. Methods of corrosion control, Cathodic and anodic protection, Use of Inhibitors, Protective Coatings: a) Metallic coatings: Types of coatings methods of applications, (hot dipping, cladding and electroplating), b) Nonmetallic coatings: Chemical conversion coatings, Powder coatings.	08
4	**Fuels and Lubricants:** **Fuels:** Introduction, Classification of fuel, Calorific value of a fuel, Characteristics of a good fuel, Solid fuel- Coal, Various types of Coal, Analysis of coal- Proximate and Ultimate analysis, Numerical based on analysis of coals, Liquid fuel- Refining of Petroleum, Gaseous fuels- LPG and CNG. **Lubricants**: Introduction, Mechanism of lubrication, Classification of lubricants, Solid, Semi-solid and Liquid Lubricants, Properties of lubricants, Physical properties – Viscosity & Viscosity index, Surface tension, Flash and Fire point, Cloud and pour point. Chemical properties – Acid value, Saponification value, Aniline point.	10
5	**Electrochemistry and Instrumental Methods of Chemical Analysis**: **Electrochemistry**: Introduction - Basic concepts: Conductance, Specific Conductance, Equivalent conductance, Molecular conductance, Effect of dilution on conductance, Cell constant. Battery: Primary & secondary batteries, Ni-Cd cell, Lithium-air battery, Fuel cell- H_2-O_2 Cell. **Instrumental Methods of Chemical Analysis**: Basic principle, instrumentation and applications of pH metry, Conductometry, Thin layer chromatography, Gas Chromatography, Ultraviolet-Visible spectroscopy and Infra-Red Spectroscopy	10

Text Books:

6. Jain P.C & Jain Monica, Engineering Chemistry, Dhanpat Rai & Sons, Delhi.
7. O. G. Palanna , Engineering Chemistry, Tata McGraw-Hill Publication, New Delhi.
8. S. S. Dara, A textbook of Engineering Chemistry, McGraw-Hill Publication, New Delhi.
9. Shashi Chawla, Engineering Chemistry, Dhanpat Rai and Co Education and Technical Publishers.
10. Shikha Agrawal, Engineering Chemistry- Fundamentals and Applications, Cambridge Publishers

Reference Books:

6. Barrow G.M., Physical Chemistry, McGraw-Hill Publication, New Delhi.
7. Atkins, Physical Chemistry, OUP Oxford.
8. Bhal & Tuli, Text book of Physical Chemistry, S. Chand & Company, New Delhi.
9. V. R Gowarikar, Polymer Science, New Age International Ltd.
10. B. K. Sharma, Instrumental Methods of Chemical Analysis, Krishna Prakashan Medi

E-Resources:

Name of the website/ E-Journals/ Online Videos

9. NPTEL Basic Courses Engineering Chemistry (https://nptel.ac.in/courses/122/101/122101001/)
10. https://www.ncertbooks.guru/engineering-chemistry/
11. Coursera Chemistry Courses (https://www.coursera.org/browse/physical-science-and-engineering/chemistry?languages=en&page=2)
12. "Introduction to Polymer Physics" NPTEL Course (https://www.youtube.com/playlist?list=PLwdnzlV3ogoXe67WsgE8f1fOIWcc5GKKS)
13. "Introduction to Corrosion" NPTEL Course (https://www.youtube.com/playlist?list=PL8lylDWRkaW8BXestE4baRnN0699S11Lq)
14. "Tribology & Lubrication" NPTEL Course (https://www.youtube.com/playlist?list=PLLwnvFq-JAltJvWAfEVU4gcUK27hEA7FD)
15. "Electrochemistry" NPTEL Course (https://www.youtube.com/playlist?list=PLVFqK_9GOGXnnriQpsn0z1Rss96Rh0vsm)
16. "Modern Instrumental Methods of Analysis" NPTEL Course (https://www.youtube.com/playlist?list=PL400CAFBA72E94CF8)

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Course Code:APS21ESL101 **Course Name:**Python Programming**Course Category:**

Credits: 3 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

2. **Pre-requisites:** Basic Computer Knowledge &Knowledge of any programminglanguage(optional)

Course Objectives:

4. To understand fundamental concepts in Python Programming
5. To learn the different Conditional Loops and Iteration
6. To understand various data structures and packages

Course Outcomes:

CO1: Describe programming fundamentals of python.
CO2: Interpret the python syntax and semantics of control flow statements.
CO3: Identify the methods to create and manipulate programs with python data structures.
CO4: Use modular approach for problem solving.
CO5: Apply advanced features and packages of python programming required for data science.

Contents:

Unit	Content	Teaching Hours
1	**Python for everybody:** Why Program, Hardware Overview, Python as a Language, Why Python, Installation Python Jupyter notebook, Using the Python Playground ,how to write program and compile in Jupyter notebook. Writing input and output statements in Python, output formatting, Writing comments, keywords in Python .variables and variable assignments, Operators ,writing expressions	6
2	**Conditional Loops and Iteration:** Conditional Statements, in Python, Loops and Iteration, Definite Loops, Finding the Largest Value, Loop Idioms	6
3	**Data Structures in Python:** Strings, Manipulating Strings, Files, Processing Files, Dictionaries, sets, Tuples, Lists, Manipulating Lists, Lists and Strings, Strings, Manipulating Strings	6
4	**Functions, Modules and Packages:** Functions, Lambda functions, Recursive function, Types of functions, modules and packages.	6
5	**Packages in Python for Data Science:** Numpy introduction, Numerical operations on Numpy, Introduction of Matplotlib, getting started with Pandas, Data frames basics in Pandas, key operations on data frames.Introduction to Data Science, Binary search, finding elements in common in lists using Hash, Finding largest elements, Introduction of SQL	6

Text Books:

1. Kent D. Lee, “Python Programming Fundamentals”, SecondEdition, Springer Publication.

2. Wes McKinney, “Python for Data Analysis” O’Reilly Publication.

Reference Books:

1. The Python Language Reference: http://docs.python.org/2/reference/index.html

2. The Python Standard Library: http://docs.python.org/2/library/

3. https://docs.scipy.org/doc/scipy/reference/tutorial/stats.html

4. http://matplotlib.org/api/mlab_api.html#module-matplotlib.mlab

5. http://conference.scipy.org/proceedings/scipy2010/pdfs/seabold.pdf

6. http://seaborn.pydata.org

7. https://www.datacamp.com/community/data-science-cheatsheets

8. PEP 20 -- The Zen of Python: https://www.python.org/dev/peps/pep-0020/

9. https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html

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General Instructions:

The theory classes are to be conducted batch wise in Lab. Each class should be divided into four batches.

Course Code:APS21ESL103**Course Name:**EngineeringMechanics **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites:

3. CoordinateGeometry,Trigonometry,Sine&CosineRule,UnitConversions
4. FundamentalsofPhysics

Course Objectives:
Todevelopcapacitytopredictttheeffectofforceandmotioninthecourseofcarryingout thedesignfunctions of engineering.

Course Outcomes:
Studentsareableto,
CO1.Identifytheforcesystemforgivenconditionsbyapplyingthe basicsofmechanics.
CO2.Determinetheunknownforcesofdifferentengineeringsystemsbyapplyingequilibriu mconditions.
CO3.ApplytheprinciplesoffrictionandtolocateCenterofGravityandfindMomentof Inertiaofplanelamina.
CO4.Establishtherelationsbetweenkinematicparametersfordifferenttypesofmotion.
CO5.Formulatetherelevantequationsfortypesofmotioninkinetics.

Contents:

Unit	Content	Teaching Hours
1	**ForceSystem**: IntroductiontoMechanics,Lawsofmechanics,Newton'sLaws,LawofParallelogra m,Lawoftransmissibility,Characteristics of force, Systemof Forces, Method of resolution and compositionmomentof aforce, Law of Moments, Varignon's Theorem, Problemsonmoment,Resultant,Equivalentforce&couple,propertiesof couple	08
2	**Equilibrium:** IntroductiontoEquilibriumanditstypes,Equilibrant,ConceptofFBD,Analyticalc onditionsof equilibrium, Equilibriumofdifferentforcesystem,Lami'sTheorem,Typesofloads,beamsandsu pports.	07
3	**Friction,CentreofGravityandMomentofInertia:** Friction: Introductiontofriction,typesandapplication,Lawsoffriction,Angleoffriction,An gleofrepose, Cone of friction, Problems on horizontal &inclinedplane,block,andladder. CG&MI: Centroid of regular and composite plane	08

	lamina,MIanditsapplication,PerpendicularaxisTheorem,ParallelAxisTheorem,RadiusofGyration,Problemsonplaneandcompositelamina.	
4	**Kinematics:** Introduction and classification of dynamics, motionanditsclassification,RectilinearMotion,EquationofMotion,Motioncurves,CurvilinearMotion,rectangularandtangentialcomponentsofacceleration,ProjectileMotion:GeneralEquation ofProjectileMotion	04
5	**Kinetics**: Basicconceptsandlawsofmotion,D'AlembertsPrinciple, Problems on rectilinear motion,Curvilinearmotion,WorkEnergyPrinciple,WorkDonebyforce,WorkDoneby weight force,WorkDonebyfrictionalforce,WorkDonebyspringforce,KineticandPotentialenergyoftheparticle,ProblemsonallcasesofWorkDone,PrincipleofImpulse and Momentum, Principle ofConservationMomentum,Impact and its types,Coefficientof restitution, Problems on impact, Problemsonimpulseandmomentum,Kineticsof rigidbodyproblems.	03

TextBooks:

7. Beer,F.PandJohnstonJr.E.R.,"VectorMechanicsforEngineers(InSIUnits):StaticsandDynamics",8thEdition,Tata McGraw-HillPublishingcompany,NewDelhi(2004).
8. VelaMurali,"EngineeringMechanics",OxfordUniversityPress(2010).
9. S.Timoshenko,D.H.Young,"EngineeringMechanics",McGrawHill,1995.
10. TayalA.K.,"EngineeringMechanics",UmeshPublications,2010.
11. SingerF.L.,"EngineeringMechanics-Statics&Dynamics",HarperandRowPub.York.
12. KhurmiR.S.,"EngineeringMechanics",S.ChandPublications,N.Delhi.

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ReferenceBooks:

8. McLean,Nelson,"EngineeringMechanics",Schaum'soutlineSeries,McGrawHillBookCompany,N.Delhi,Publication.
9. Hibbeller,R.CandAshokGupta,"EngineeringMechanics:StaticsandDynamics",11thEdition,PearsonEducation(2010).
10. Bhavikatti,S.SandRajashekarappa,K.G.,"EngineeringMechanics",NewAgeInternational(P)LimitedPublishers,(1998).
11. IrvingH.ShamesandKrishnaMohanaRao.G.,"EngineeringMechanics– StaticsandDynamics",4thEdition,PearsonEducation(2006).
12. RajasekaranSandSankarasubramanianG.,"EngineeringMechanicsStaticsandDynamics",3rdEdition,VikasPublishing HousePvt.Ltd., (2005).
13. MeriamJ.L.andKraigeL.G.,"EngineeringMechanics-Statics-Volume1,Dynamics-Volume2",ThirdEdition,JohnWiley& Sons,(1993).
14. Kumar,K.L.,"EngineeringMechanics",3rd RevisedEdition,TataMcGraw-HillPublishingcompany,NewDelhi(2008).

GeneralInstructions:

E-Resources:

1. www.nptel.ac.in(LearningplatformfromIITprofessors)
2. http://www.asnu.com.au(ForEngineeringapplications)
3. www.discoveryforengineers.com(InvestigatingDiscoveries)

Course Code:MGM54AEL101**Course Name:**Communicative English **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: BasicknowledgeofEnglish

Course Objectives:
Thecourseaims at groomingthe professionalethicsofthe studentsthroughvariouspersonalitytraits andbehavioral patternsfocusingon communicationskills.

Course Outcomes:

CO.1 communicate formally with enhancedcommunicationCompetency

CO.2 to adapt professional nonverbal communication

CO.3construct English formal syntax and apply corporate vocabulary in written and verbal communication

CO.4 acquire listening and drafting skills with professional competency

Contents:

Unit	Content	Teaching Hours
1	**Communication andCommunicationProcess:** Introduction to Communication,Forms and functions ofCommunication, Barriers toCommunication and overcomingthem,Ways of EffectiveCommunication.	5
2	**Non-verbalCommunication And its types:** Kinesiccs, Occulesics, Appearance, Proxemics, Chronemics, Paralanguage, Qualities of effective speech	3
3	**English Grammar:** Overview of basic Mid-levelgrammar, Tenses&conceptoftime,Sentence construction, Corporate vocabulary, Difference between formal and informal sentences, phrases and words	3
4	**ListeningSkills and WritingSkills:** Listening : Active andPassive Listeningwritingstyles layouts BusinessLetters- job application, resignation, resume	4

Text Books/ Reference Books:

1 AshrafRizvi,CommunicationSkillsforEngineers,TataMcGrawHill

2 SanjayKumar, PushpLata,Communication Skills, Oxford UniversityPress, 2016.

3 MeenakshiRaman,SangeetaSharma,CommunicationSkills,Oxford UniversityPress,2017.

4 TeriKwalGamble,MichaelGamble,Communication Works,TataMcGraw HillEducation,2010.

5 Anderson, Kenneth. Joan Maclean and Tossny Lynch. Study Speaking: A Course in Spoken EnglishforAcademicPurposes.Cambridge: CUP, 2004.

6 Bellare,Nirmala.ReadingStrategies.Vols.1and2.NewDelhi.OxfordUniversityPress,1998

7 Bhasker,W.W. S&Prabhu, N.S.:English throughReading, Vols.1and2.Macmillan,1975

8 BoveeCourtland,Land Thrill,John V.BusinessCommunication, TodayMcGraw Hill, NewYork, TaxmanPublication(1989).

9 Murphy,Raymond.Essential EnglishGrammar, Cambridge:UniversityPress(2000)

10 HewingsMartinAdvancedEnglishGrammarCambridge:Universitypress(2003)

11 BansalHarrison.Spoken English

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Course Code:APS21PCL101**Course Name:**BasicsofElectricaland ElectronicsEngineering
Course Category:

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university mathematics.

Course Objectives:

5. UnderstandElectricalcircuitsandclassifycircuitsasperlaws.
6. UnderstandMagneticcircuitsandapplythemintransformerdevices.
7. Understandtheworkingprincipleofsemiconductordevices.
8. Understand the basics and applications of digitalelectronics.

Course Outcomes:
Aftercompletionofthiscourse,studentswill beableto:

5. SolvesimpleDCandsingle-phaseACcircuitsusingKCL,KVLand networktheorems.
6. Explain fundamentals of magnetic circuit andtransformers.
7. Explaintheworkingofdiodesandtransistors.
8. Explain the importance of number systems and logicgates.

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Contents:

Unit	Content	Teaching Hours
1	**Electrical Circuits:** Voltage and current sources: independent,dependent,idealandpractical;V-Irelationships of resistor, inductor, mutualinductor and capacitor; Kirchhoff's laws,meshandnodalanalysis,superposition,Thevenin's,maximumpowertransfertheorems.Alternatingvoltagesandcurrents, RMS, average, maximum values,Single Phase RL, RC, RLC series circuits,PowerinACcircuits,PowerFactor,Threephasebalanced systems.	08
2	**MagneticCircuits&Transformer:** Conceptsofm.m.f,flux,fluxdensity, reluctance, permeability and field strength,theirunitsandrelationship,right handthumbruleandcorkscrewrule, Faraday's law of electromagnetic induction, Fleming's right-hand rule, staticallyanddynamicallyinducedEMF's selfandmutualinductancecoefficientof coupling, energy stored in magnetic circuit, Single phase transformers: Construction,principleofworking,e.m.f. equations.	08
3	**SemiconductorDevices:** IntroductiontoSemiconductors,P-type andN-typeSemiconductors,P-NJunction	07

	Diode:Constructionandworking,V-I characteristicsofDiode, Reverse breakdownmechanism. Specialpurpose diodes:ZenerDiode, LightEmitting Diode(LED) and PhotoDiode - Construction,workingandapplications. BipolarJunctionTransistor(BJT):types, constructionandregions ofoperation.	
4	**DigitalElectronics:** Binary, Decimal, Octal, Hexadecimal number systems and their inter-conversion, Binary Addition, Binary Subtraction,One'scomplement,Two's complement, Logic Gates: Basic, Universal and Special. Boolean Laws (AND, OR, NOT) and Demorgan's theorems,Realizationofbasiclogicgates usinguniversalgates.	07

TEXTBOOKS:

7. V.N.MittalandArvindMittal,"BasicElectricalEngineering"McGrawHill
8. EdwardHughes,"ElectricalTechnology,",PearsonEducation
9. ElectricalTechnologyVol.1& Vol.4byB.L. Theraja&A.K.Theraja,S.ChandPublications
10. "PrinciplesofElectronics",V.K.Mehta,S.ChandPublications.
11. "ElectronicsDevices and Circuits",SSalivahanan,McGrawHillPublications.
12. "ModernDigitalElectronics",R.P.Jain,McGrawHillPublications.

REFERENCEBOOKS:

4. VincentDelToro,"ElectricalengineeringFundamentals",PHIsecondedition2011
5. RobertBoylestad,"ElectronicsDevicesandCircuitsTheory",PearsonEducation India
6. "ElectronicsDevicesand CircuitsTheory",RobertBoylestad,PearsonEducationIndia.

E-Resources:

4. OnlinecourseonNPTEL"BasicElectricalEngineering"
5. NPTELCourseon"IntroductiontoBasicElectronics",Prof. T.S.Natarajan,IITMadras.
6. NPTELCourseon"DigitalElectronicCircuits",Prof.GoutamSaha, IITKharagpur.

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Course Code:APS21VSP102**Course Name:**WorkshopPractices **Course Category:**

Credits: 2 **Teaching scheme:** L-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university English.

Course Objectives:

1.To know about the different carpentry tools and perform various carpentry operations to complete the job.

2. To understand different welding tools, joints, defects and perform welding operation to complete useful article/job.

3. To learn various types of pipes, plumbing tools, operations and perform thread cutting on GI pipes.

4. To know different sheet metal tools, operations, applications and perform various operations to complete job.

Course Outcomes:

On completion of the course, the student should be able to:

1. Perform basic carpentry operation on wood andprepare carpentry article.

2. Perform welding processes and prepare welding article by performing various welding operations.

3. Use plumbingtools, processesandperform threading onGI pipe.

4. Use sheet metal tools, processesand prepare sheet metal article.

Contents:

Unit	Content	Teaching Hours
1	**Carpentry Shop:** Carpentry shop: Types of woods, tools, joints, operations, applications, safety measures etc. Job: Exercises on wood involving operations marking, sawing, chiselling, planning, grooving etc to make useful wooden article/ job e.g. Wooden Trophies, Showpiece articles, Stools etc.	08
2	**Welding Shop:** Welding Shop: Types of welding, welding joints, tools, welding defects, applications, safety measures etc. Job: ExerciseinArcweldingtomake useful articles like Grills, Stools, Tree Guards, Flower pot stand, Shoe rack, Bag Stand, Showpiece Articles from Scrap etc.	07
3	**Plumbing Shop:** Plumbing shop: Study of types of pipe, pipe joints, operations, applications, safety measures etc. Job: Prepare threading on GI pipe to make useful items like Nipple, Pipe joints etc.	07
4	**Sheet Metal Shop:** Sheet Metal Shop: Sheet metal tools, operations, applications, safety measures etc. Job: Making an utility item using G I sheet involving development, marking, cutting , bending, spot welding/riveting Parts like i) Tray, ii) Funneletc.	08

- **Termwork:** Students shall maintain workshop diary which contains Job Drawing, information of tools, operations required to complete the jobs, records of job completions etc.

- **End Semester Exam:** ESE will be viva-voce based on jobs prepared by the students during the term.

Text Books:

2. Hazra and Chaudhary, Workshop Technology-I, Media promoters & Publisher private limited.

3. Hazra and Chaudhary, Workshop Technology-II, Media promoters & Publisher private limited.

Reference Books:

1. K. C. John, Mechanical Workshop Practice, Prentice Hall Publication, New Delhi, 2010.

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Course Code:APS21BSP102**Course Name:Engineering Chemistry Lab Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

Course Outcomes:
Upon completion of the Lab Experiments, the students should able to,

LO.4 **:** Perform the experiments as well as accurately record the readings and analyze the results of such experiments.

LO.5 : Estimate the impurities present in water using titrimetric and instrumental methods.

LO.6 : Employ the basic techniques used in chemistry laboratory for analysis such as volumetric titrations, complexometric titrations, Conductometry, pH metry, viscometer, Stalagmometer and TLC

Contents:

Sr. No.	List of Practical	Lab Hours
1	Determination of hardness of water sample by EDTA method.	02
2	Determination of chloride content in water sample by precipitation titration method.	02
3	Determination of dissolved oxygen in water by Iodometric method.	02
4	Determination of percentage purity of bleaching powder.	02
5	Determination of strength of acid / base using pH metric titration.	02
6	Determination of strength of acid / base using conductometric titration.	02
7	To determine the cell constant of the given conductivity cell.	02
8	To determine relative surface tension of unknown liquids by using stalagmometer.	02
9	To determine the viscosity of unknown liquids by using Ostwald / Redwood viscometer.	02
10	To determine acidity of given water sample.	02
11	Determination of acid value of an oil sample.	02
12	Determination of saponification number of an oil sample.	02
13	To determine alkalinity of given water sample.	02
14	Preparation of phenol- formaldehyde / urea-formaldehyde resin.	02
15	To find out the Rf value of given sample by thin layer chromatography.	02
16	To separate the given mixture using thin layer chromatography.	02
17	Proximate analysis of coal (moisture content, volatile matter, ash content).	02
18	To study factors influencing on rate of electrochemical corrosion.	02

19	To determine flash and fire point of given oil by Pensky-Marten flash point apparatus.	02
20	To determine cloud and pour point of lubricating oil.	02

Reference Books:

4. A Text book on Experiments and Calculations in Engineering Chemistry by Dr. S. S. Dara, S Chand Publication.
5. Laboratory manual on Engineering Chemistry by S. K. Bhasin & Sudha Rani, Dhanpat Rai Publishing Company.
6. Engineering Chemistry with Laboratory Experiments by M. S. Kaurav, Asia-Pacific the holdings Private Ltd.

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Course Code:APS21ESP101 **Course Name:**Python Programming Lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The Objective of this course is

Course Outcomes:
LO1 : Demonstrate python program using development environment.
LO2: Develop logical thinking to solve the problems using programming fundamental concepts.
LO3 : Construct python program using various data structures.
LO4 : Apply modularization approach for solving complex problem.
LO5: Make use of various packages in Python for data science.
LO6 : Implement different SQL commands in python.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Program to perform input/output operations Write a program to take input (integer, float, string) and print it.	02
2	Program based on operators 1.Write a program to simulate a simple calculator (+ - / * %) that takes two operands as input and displays the result 2.Write a program to find area and perimeter of geometric objects. 3.The distance between two cities (in km.) is input through the keyboard. Write a program to convert and print this distance in meters, feet, inches and centimeters. 4. Write a Program to interchange two numbers. 5.Write a program to compute Fahrenheit from centigrade	02
3	Programs based on Decision making. 1.Write a program to read marks from keyboard and your program should display equivalent grade according to following table(else-if) (ladder) Marks Grade 100 - 80 Distinction 79 - 60 First Class 59 - 40 Second Class < 40 Fail 2.Write a program to input basic salary of an employee and calculate gross salary according to given conditions. Basic Salary <= 10000 : HRA = 20%, DA = 80% Basic Salary is between 10001 to 20000 : HRA = 25%, DA = 90% Basic Salary >= 20001 : HRA = 30%, DA = 95% 3.If the ages of three brothers are input through the keyboard, write a C Program to determine the youngest and oldest of the three.	02

	4.Write a program to calculate overtime pay of employee. Overtime is paid at the rate of Rs. 12.00 per hour for every hour worked above 40 hours. Assume that employee do not work for fractional part of an hour. 5.Write a program for checking the speed of drivers. If speed is less than 70, it should print "Ok". Otherwise, for every 5km above the speed limit (70), it should give the driver one demerit point and print the total number of demerit points. For example, if the speed is 80, it should print: "Points: 2". If the driver gets more than 12 points, the function should print: "License suspended"	
4	Programs using while and for loops 1.WAP to find factorial of given number 2.WAP to check whether given number is Palindrome or not 3.WAP to check whether given number is Armstrong or not 4.WAP to print Fibonacci series 5.Write a Python program which iterates the integers from 1 to 50. For multiples of three print "Fizz" instead of the number and for the multiples of five print "Buzz". For numbers which are multiples of both three and five print "FizzBuzz". 6.WAP to check whether given number is Perfect number or not 7.WAP to check whether given number is Prime number or not 8.Write C program to print given star and number patterns and reverse it. • * 1 • ** 12 • *** 123 **** 1234	02
5	Programs on string 1.Write Python Program to find length of string without using len() function. 2.Count all letters, digits, and special symbols from a given string. 3.Python Program to Count the Number of Vowels in a String. 4.Python Program to Calculate the Number of Upper Case Letters and Lower Case Letters in a String. 5.Python Program to Check whether given string is palindrome or not	02
6	Programs on List and Tuple 1.Write a Python program to sum all the items in a list. 2.Write a Python program to multiply all the items in a list 3.Write a Python program to get the largest number from a list. 4.Write a Python program to get the smallest number from a list 5.Write a Python program to count all elements in list and count Occurrences Of A List Item In Python 6.Write a Python program to create a tuple with different data types 7. Write a Python program to check whether an element exists within a tuple 8.Write a Python program to reverse a tuple 9. Write a Python program calculate the product of all the numbers given in tuple. Original Tuple: (2, 4, 8, 8, 3, 2, 9) Product - multiplying all the numbers of the said tuple: 27648	02
7	Programs on set and dictionary	02

	1.Write a Python program to concatenate following dictionaries to create a new one. Sample Dictionary : dic1={1:10, 2:20} dic2={3:30, 4:40} dic3={5:50,6:60} Expected Result : {1: 10, 2: 20, 3: 30, 4: 40, 5: 50, 6: 60} 2.Write a Python program to check whether a given key already exists in a dictionary 3.Write a Python script to generate and print a dictionary that contains a number (between 1 and n) in the form (x, x*x) Sample Dictionary (n = 5) : Expected Output : {1: 1, 2: 4, 3: 9, 4: 16, 5: 25} 4.Write a Python program to merge two Python dictionaries 5.Write a Python program to get the maximum and minimum value in a dictionary 6.Write a Python program to create set difference, Union and intersection 7.Write a Python program to check if two given sets have no elements in common	
8	Programs using function 10. Write Functions to calculate your trip's costs: 11. Define a function called hotel_cost with one argument nights as input 12. Define a function called plane_ride_cost that takes a string, city, as input. 13. Define a function called rental_car_cost with an argument called days. 14. Define a function called trip_cost that takes two arguments, city and days. Like the example above, have your function return the sum of calling the rental_car_cost(days), hotel_cost(days), and plane_ride_cost(city) functions. 15. Write a program in to check a given number is even or odd using the function. 16. Write a function Exchange to interchange the values of two variables, say x and y. illustrate the use of this function in a calling function. 17. Write a program to find Sum of natural number using recursion. 18. Write a program to print Fibonacci series number using recursion	02
9	Program using NumPy, Matplotlib and Pandas library 1.Write a program to perform matrix addition, substraction, multiplication. 2.Plot all types of graph using Matplotlib. 3.Write a program which perform basic operation of Pandas.	02
10	Program on SQL Commands 4. Write a program of binary search 5. Write a program which perform basic SQL commands 6. Programs based on real life problems/GUI based programs	02

Course Code:APS21ESP104**Course Name:**Engineering Mechanics lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

Course Outcomes:
Upon completion of the Lab Experiments, the students should able to,
1.Describe the working principle of mechanics and correlate them with day to day engineering applications.
2.Formulate and solve mechanics problems based on law of moments, conditions of equilibrium.
3.Verify theoretical concepts through analytical, experimental and graphical methods.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Verification of law of Machine using Screw jack	02
2	Polygon law of coplanar forces.	02
3	Bell crank lever	02
4	Lami's theorem	02
5	Support reactions for beam	02
6	Problems on beam reaction by graphics statics method	02
7	Inclined plane (to determine coefficient of friction).	02
8	Centroid of irregular shaped bodies	02
9	Determine center of gravity for composite sections	02
10	Determine moment of inertia for composite sections	02
11	Moment of Inertia of fly wheel	02
12	Simple / compound pendulum	02

References:

3. "College of Engineering, Pune"
4. "University of Mumbai"

Course Code:MGM54AEP101**Course Name:**Communicative English Lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: BasicknowledgeofEnglish

Course Objectives:
Thecourseaims at groomingthe professionalethicsofthe studentsthroughvariouspersonalitytraits andbehavioural patternsfocusingon communicationskills.

Course Outcomes:
TheStudents willbeable to
LO.1 Introducethemselves formallyand informallythroughpractice.
LO.2 Pronounce English vowel and Consonant soundseffectively
LO.3 Participate effectively in G.Ds, Presentations, &Interviews
LO.4 Face Interviews competently
LO.5 Draft resume,businessletters,reports formally
LO.6 ComprehendthemeaningofEnglishtext bycomprehensiontechniques.

Contents:

Sr. No.	List of Practical	Lab Hours
1	SelfIntroduction	02
2	Pronunciation of Vowel sounds in English	02
3	Pronunciation of consonant sounds in English	02
4	Groupdiscussion	04
5	Presentation techniques	04
6	Interviewtechniques	04
7	Letter writing	04
8	Email writing, Agenda of the meeting and notices	02
9	Resume Writing	02
10	Report writing	02
11	Skimming & Scanning	02

Course Code:APS21PCP101**Course Name:**Basics of Electrical andElectronicsEngineeringLab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

5. UnderstandtheDC circuitasperKCL&KVLandACcircuitsaspertheorems.
6. Understandtheworking of asingle-phasetransformer.
7. Understandthe working ofelectronicinstruments,components andlogicgates.
8. Understandthe working of aPNjunction diode.

Course Outcomes:

Aftercompletion ofthislab, students willbeable to:

5. Useof KCL&KVL tosolve DCcircuitsaswellasuseof ACtheoremstosolveACcircuits.
6. Demonstrateasingle-phasetransformer &itsworking.
7. Useelectronicinstruments,workingof electroniccomponents andlogics gates.
8. UseofPNJunction Diodeandits applications.

Contents:

Sr. No.	List of Practical	Lab Hours
1	VerificationofLoopAnalysisandNodalAnalysisforDCCircuits.	02
2	VerificationofThevenin'sTheoremforDCCircuits.	02
3	VerificationofMaximumPowerTransferTheoremforDCCircuits.	02
4	DeterminationofVoltage,CurrentandPowerFlowinSinglePhaseACCircuitincludingR, L and C with Combination.	02
5	DeterminationofMagneticMaterialTermsandEMFInduction.	02
6	Demonstration and Determination of Single-Phase Transformer Terms likeVoltageRatio and TurnsRatio.	02
7	StudyofElectronic Instruments.	02
8	Studyof ElectronicComponents.	02
9	Studyof LogicGates.	02
10	Studyof V-Icharacteristicsof aPNJunction Diodeusing V-Lab.	02
11	StudyofRectifierCircuitsusingEveryCircuitsimulationapplication.	02
12	Study ofworkingand troubleshootingofSmartphone, Computer&TV(CaseStudyApproach).	02

Course Code:MGM82CCP101**Course Name:**National Cadet Corps **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
Nil

Course Outcomes:
Nil

Contents:

Unit No.	Content	Hours
1	**NCC General, National Integration and Awareness, Social Service and Community Development, Drill:** Aims, Objectives and Organization of NCC, Duties of NCC Cadet, National Integration Importance and Necessity, Factors Affecting National Integration, Foot Drill, Drill With Arms, Ceremonial Drill with Arms, Social Service and Community Development Activities- Pollution, Mission Indradhanush, Beti Bachao Beti Padhao, Tree Plantation, Digital Awareness.	12
2	**Personality Development, Disaster Management, Weapon Training, Map Reading :** Critical and Creative Thinking, Decision Making, Self Awareness, Public Speaking, Interview Skills, Types, Org, Capability and Role of NCC cadets, Initiative Trg, Organizing Skills, Do's & Don't , Introduction and Characteristics of .22 rifle, Handling of .22 rifle, Introduction to Map Reading, Conduct of MR- Google and Tourist Maps and Apps.	10
3	**Health and Hygiene, Environmental Awareness and Conservation, Adventure, Obstacle Training:** Hygiene & Sanitation (Personal & Camp Hygiene) Soch Vichar, First Aid in Common Medical Emergencies, Treatment & Care of Wounds, Introduction Yoga & Exercises, Water Conservation, Energy Conservation, Introduction Adventure Activities. Obstacle Course.	10
4	**Leadership, Introduction to Infantry Weapons and Equipments:** Traits, Indicators, Motivation, Ethics, Case Studies- Chhatrapati Shivaji Maharaj, Maharana Pratap, Jhasi ki Rani, Ratan Tata, Narayan Murty, Rabindra Nath Tagor, Organization of Infantry Battalion and its weapons.	08
5	**Armed Forces, Field Craft and Battle Craft, :** Armed Forces, Army, CAPF, Police, Modes of Entry to Army, CAPF, Police, Introduction to Field Craft, Indication of Landmark, Observation, Camouflage and Concealment, Fire and Move Capsule.	08

References:

- Cadet's Handbook- Common Subject, all wings by DG NCC, New Delhi.
- Cadet's Handbook- Common Subject by NCC Directorate- Bhubaneshwar.

- Cadet's Handbook- Specialised Subjects, Army, Navy, Air-Force by DG NCC, New Delhi.
- NCC OTA Precise by DG NCC, New Delhi.
- Chanakya's 7 Secrets of Leadership by Radhakrishanan Pillai and D. Shivnandhan.
- National Cadets Corps (India) by Lambert M. Suvarkar.

E-Resources:

1. National Cadet Corps, Youth in Action (Google eBook).

https://indiancc.nic.in/dg-ncc-lt-gen-gurbirpal-singh/

www.youtube.com

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Semester –II

Course Code: APS21BSL103 **Course Name:** Linear Algebra and Differential Equations **Course Category:**

Credits: 4 **Teaching scheme:** L-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university mathematics.

Course Objectives:

1. This course aims to make the students become familiar with the basic concepts of linear algebra with a thorough understanding of vector spaces, linear transformations and matrix operations enhancing the students' ability to reason mathematically and able to apply this knowledge to many fields in engineering, statistics and computer science.

2. Create and analyze mathematical models using differential equations.

Course Outcomes:

On completion of the course, the student should be able to:

1. Understand basic concepts such as vector spaces, linear dependence / independence of vectors, basis and linear maps.

2. Analyze and calculate eigen values, eigen vectors, rank nullity of a matrix / linear map.

3. Prove theorems, apply Gram-Schmidt process on inner product spaces, diagonalize special matrices.

4. Solve ordinary differential equations of first order and apply knowledge of differential of equations to solve engineering problems.

5. Find the solution of linear differential equations having their applications in mechanical and electrical systems.

Contents:

Unit	Content	Teaching Hours
1	**Matrices and Vector Spaces:** Basic properties of matrices, row operations and Gauss elimination, Inverse of a matrix. Basic concepts in linear algebra: vector spaces, subspaces, linear independence and dependence of vectors, bases, dimensions. Row and Column spaces, rank. Applications to systems of linear equations, Inverse transformation.	10
2	**Linear mappings and Diagonalization:** Linear mappings, representation by matrices, rank-nullity theorem, Diagonaliazation, Eigen values, Eigen vectors and their basic properties, Cayley Hamilton Theorem.	10
3	**Inner Product Spaces and Quadratic Forms:** Inner Product Spaces , Orthogonality, Gram-Schmidt process, Geometric Applications of Linear Transformation, Quadratic Forms: Positive Definiteness and applications	10
4	**First order ordinary differential equations and Applications:** Exact , Linear ,Bernoulli and separable differential equations, Applications to Population growth/decay, Mixing problems, Draining tank/Torricelli's Law problems, Newton's Law of Cooling, Electric circuits, Falling bodies.	10
5	**Ordinary differential equations of higher orders and Applications:** Linear differential equations with constant and variable coefficients, method of variation of parameters, Applications to mass spring systems and electrical circuits and Bending of beam and columns.	10

Text Books:

1. Introduction to Linear Algebra (2nd edition) by Serge Lang, Springer

2. Elementary Linear Algebra (10th edition) by Howard Anton and Chris Rorres, John Wiley and sons.

3. D.Poole, LinearAlgebra:AModern Introduction, 2nd Edition,Brooks/Cole,2005.

4. S. L. Ross, Differential Equations, 3rd Edition, Wiley India, 1984.

Reference Books:

1. Shanti Narayan, Differential Calculus, S. Chand & Co.

2. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, NewYork.

3. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, NewDelhi.

4. P. N. Wartikar, J. N. Wartikar, Applied Mathematics (Vol I & II) , Pune Vidyarthi Griha Prakashan,Pune.

5. Differential Equations with Applications and Historical notes by George Simmons, Tata McGraw Hill publishing company Ltd, New Delhi

6. K.D Joshi, Calculus for Scientists and Engineers, CRC Press.

7. Prasad and Reena Garg, Advanced Engineering Mathematics, Khanna Publishing Company Private Limited, New Delhi.

8. Schaum's outlines of Linear Algebra (5th edition) by Seymour Lipschutz, Marc Lipson, McGraw-Hill Education (India) Private Limited, New Delhi

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Course Code:APS21BSL102 **Course Name:**Engineering Physics **Course Category:**

Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Student should know Basic Physics and basic Mathematics

Course Objectives:

1. To impart knowledge in basic concepts of physics relevant to engineering applications
2. To introduce advances in technology for engineering applications.

Course Outcomes:

On completion of the course, the student should be able to:

CO.6 Summarize fundamentals of electron optics, modern physics and ultrasonic waves related to the engineering fields.

CO.7 Identify the importance of the optical phenomenon i.e. interference, diffraction and polarization in relevance with its engineering applications.

CO.8 Classify the material on the basis of electric conductivity as semiconductor and superconductors and dielectric materials this leads to their fascinating applications.

CO.9 Recognize the use of laser and optical fibers in various fields.

CO.10 Outline basics of crystallography and X- rays and demonstrate the applications of nano-materials relevant to engineering program.

Contents:

Unit	Content	Teaching Hours
1	**Modern Physics:** Electron Optics: e/m by Thomson's method, Positive ray, Bainbridge mass spectrograph. Quantum Mechanics: Role and concepts, De- Broglie's hypothesis, Uncertainty Principle, Fundamentals of quantum computing, Quantum features Ultrasonic Waves: Production of ultrasonic waves (Magnetostriction & Piezoelectric method), Applications. Numericals.	8
2	**Wave Optics:** Interference- Interference in thin films (reflected light), Newton's Rings, Engineering applications of Interference. Diffraction- Fresnel's and Fraunhofer Diffraction, Theory of plane transmission Grating. Polarization-Polarization by reflection and double refraction, Optical activity, Specific rotation, Construction and working of Laurent's half shade polarimeter, Engineering applications of Polarization. Numericals.	8
3	**Materials of Technological Importance:** Dielectric Materials: Introduction, Types of polarizations: Electronic and Ionic, Orientation Polarizations - Applications of Dielectrics Semiconducting Materials: Introduction, Fermi energy in Intrinsic	8

	semiconductors and extrinsic semiconductors, Hall effect, Applications of Semiconductors. Numericals. Superconducting Materials: Introduction, Type – I and Type – II superconductors, Meissner effect, BCS Theory, Application	
4	**Optoelectronic Materials and Devices:** LASER : Absorption, spontaneous and stimulated emission, population inversion n, pumping mechanism, Construction and working of Ruby laser, Construction and working of He-Ne laser. Lasers in various technological applications. Introduction to Optical Fibers-Introduction Acceptance Angle-Numerical Aperture, Applications of optical fibers. Numericals.	8
5	**Physics of Materials:** Crystal structure: Unit cell, Coordination number, atomic radius, packing density of cubic system. X-rays: Bragg's law, X-Ray Diffraction (XRD), Industrial Applications of X-Rays. Numericals. Particle detector: G.M. Counter Nano-Materials: Basic principles of nano science and technology, properties, applications of nanotechnology.	8

Text Books:

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5. Engineering Physics- H.K. Malik & A.K. Singh, McGraw Hill publication.
6. Engineering Physics - R.K. Gaur and S. L. Gupta. Dhanpat Rai Publications Pvt. Ltd.-New Delhi
7. M. N.Avadhanulu, P.G.Kshirsagar "A Text book of Engineering Physics"-S. Chand Publications.
8. B.K. Pandey and S. Chaturvedi, Engineering Physics, Cengage Learning, 2012.

Reference Books:

10. Fundamental of Physics - Halliday and Resnik. Willey Eastern Limited.
11. Introduction to Electrodynamics –David R. Griffiths.
12. Concept of Modern Physics – Arthur Beizer. Tata McGraw-Hill Publishing Company Limited.
13. Optics – AjoyGhatak.MacGraw Hill Education (India) Pvt. Ltd.
14. Science of Engineering Materials- C.M. Srivastava and C. Srinivasan. New Age International Pvt.Ltd.
15. Solid State Physics – A.J. Dekker. McMillan India –Limited.
16. The Feynman Lectures on Physics Vol I, II,III.
17. Introduction to solid state physics – Charles Kittel. John Willey and Sons
18. T Pradeep "A Text book of Nano Science and Nano Technology"-Tata Mc GrawHill 2019.

E-Resources:

1. https://nptel.ac.in/courses/122107035/ Engineering Physics
2. https://youtu.be/98xoZknQjl8 Polarization
3. https://youtu.be/yINtzw63KncMaxwell's Equations and EM theory
4. https://youtu.be/bwreHReBH2AMaxwell's Equations and EM theory

5. https://www.youtube.com/playlist?list=PLuv3GM6-gsE3-hVNaw-YEb7EeY5XVPZdz
6. Maxwell's Equations and EM theory(nptel)
7. https://nptel.ac.in/courses/115105120/ Experimental Physics
8. https://youtu.be/2CsMpEBl5QY Crystal Structure and X- rays
9. https://youtu.be/z_8aJPLr21E Crystal Structure and X- rays
10. https://youtu.be/_Ckh-60B6LY Condensed matter Physics
11. https://youtu.be/QQZ6EGf0Ju8 Magnetic Properties
12. https://youtu.be/DDLljK1ODeg Magnetic Materials
13. https://youtu.be/etjZmdmrjSU Dielectrics
14. https://youtu.be/k6ZxP9Yr02E Semiconductor
15. https://youtu.be/D-9M3GWoBrw Superconductivity
16. https://youtu.be/GglT1RoBPzg Superconductivity
17. https://youtu.be/VHp2Ff5N_bs Superconductivity
18. https://youtu.be/FNp81kkxj5c LASER
19. https://youtu.be/YvrwVK9ZqQY LASER
20. https://nptel.ac.in/courses/115107095/ Optic Fiber
21. https://youtu.be/cjBPnIXK60U Quantum Mechanics (Prof.H.C. Verma)
22. https://youtu.be/BDuqChhUhM0 Divergence and Curl(Prof.H.C. Verma)
23. https://youtu.be/sCviGSMaYfI Divergence and Curl (Prof.H.C. Verma)
24. https://youtu.be/SZCsFS9izfQ Divergence and Curl

And other related videos from following resources

25. www.nptel.ac.in ;www.swayam.gov.in;https://inflibnet.ac.in/
26. www.sciencedirect.com
27. http://vlabs.iitb.ac.in/vlab/
28. www.youtube.com

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Course Code:APS21SL102**Course Name:**Engineering Graphics **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Nil

Course Objectives:

8. Understand the basic principles of engineering graphics and improve the visualization skills
9. To gain knowledge on projection of points, straight lines, planes, solids.
10. To understand the real life objects through Drawings.
11. To know the principles of orthographic and isometric projections.

Course Outcomes:

CO 1. Identify basic concepts in drawing and its application.
CO 2. Plan and prepare neat orthographic drawings of points, straight lines, planes and solids.
CO 3 To visualize and draw orthographic and isometric projection of solids.
CO 4. Acquire skill to draw real life engineering objects by using the engineering drawing.

Contents:

Unit	Content	Teaching Hours
1	**Projections of Straight Lines:** Introduction to Engineering Graphics, Need of Engineering Drawing, Drawing Instruments, BIS code of practice for general engineering drawing, Projections of Points in Four Quadrants, Projections of Points in Reference Plane, Line Parallel to both the Plane, Line Parallel to One Plane and Perpendicular to the other, Line Inclined to One Plane and Parallel to The Other, Line Inclined to Both the Reference Planes, Traces of Line (Only first quadrant to be considered)	5
2	**Engineering Curves:** Curves used in Engineering Practice, Conic sections, Construction of conics by different methods, Rectangular-hyperbola, Cycloidal curves, Epi and hypo-cycloids, Involutes and Archimedean spiral.	5
3	**Projections of Planes:** Plane with Surface Parallel to One Plane and Perpendicular to other, Plane Inclined to One Plane and Perpendicular to other, Projections of Planes Inclined to both the Planes	5
4	**Projections of Solids:** Introduction to Solids: Prisms, Pyramid, Cylinder, Cone, Cube, Tetrahedron, Sphere, Projections of above Solids with Axis inclined to one plane, Projections of above solids with Axis inclined to both the Planes, Projection of composite solids (different arrangement of Spheres with above Solids).	5
5	**Orthographic Projections:** Orthographic projections of different Machine Parts (First Angle Projection method only)	5
6	**Isometric Projections ,Introduction to CAD:**	5

	Introduction to Pictorial views, Converting Orthographic Projections into Isometric Projections and Isometric views. Listing the computer technologies that impact on graphical communication, Demonstrating knowledge of the theory of CAD software [such as: The Menu System, Toolbars (Standard, Object Properties, Draw, Modify and Dimension), Drawing Area (Background, Crosshairs, Coordinate System), Dialog boxes and windows, Shortcut menus (Button Bars), The Command Line (where applicable), The Status Bar, Different methods of zoom as used in CAD, Select and erase objects	

Text Books:

1. N. D. Bhatt and Panchal V. M., "Engineering Drawing", Charotar Publishing House, Anand

2. P. J. Shah , " A Text Book of Engineering Drawing", S.Chand, New Delhi

3. Shah M.B. & Rana B.C, "Engineering Drawing & Computer Graphics", Pearson Publications,

4. Agrawal B.& Agrawal C.M, "Engineering Graphics", TMH Publications

12. Narayana K.L. & P. Kannaiah, "Engineering Drawing", Scitech Publications

13. P.I Vargese, "Engineering Graphics", Mcgraw Hill Publications

14. D.A.Hindoliya," Engineering Graphics", B. S. Publications

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Reference Books:

1.Dabhade M. L., "Engineering Graphics", Vol.-I and Vol.-II, Vision Publications, Pune

2.K.Venugopal, "Engineering Drawing and Graphics" ,New Age International Publishers

Course Code:APS21IKL1XX**Course Name:**Indian Knowledge Systems **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Nil

Course Objectives:

The objective of this course is
1. To make students understand foundational concepts in IKS for science, engineering and technology.
2. To explore ancient Indian pursuits and accomplishments in the various domains of engineering

Course Outcomes:

On completion of the course, the student should be able to:
CO 1: familiarise with key components of the IKS & develop appreciation for Indian philosophical systems.
CO 2: understand key features of Indian Numeral System, units of measurement and the framework for establishing the right knowledge.
CO 3:appreciate the unique & specific contributions of ancient Indian mathematicians in Arithmetic, Geometry & Trigonometry.
CO 4:develop awareness about engineering & technology heritage of India and understand ancient Indian contributions in various engineering domains.

Contents:

Unit	Content	Teaching Hours
1	**Introduction to IKS:** Importance of ancient knowledge, defining IKS, Classification framework for IKS, Historicity of IKS, Indian philosophical systems, Vedic schools of philosophy(Sankhy and yoga, Nyaya and Vaisesika, Purva- Mimamsa and Vedanta), Non- Vedic philosophical systems (Jain, Bauddha and Carvaka) , Wisdom through the ages: Issues of interest in the Puranas, Itihasa as a source of wisdom (Uniqueness of the two epics), Nitisastras.	08
2	**Foundational Concepts for Science and Technology:** Number system in India, salient features of the Indian numeral system, Measurements for Time, Distance and weight, The knowledge triangle, Prameya- a Vaisesikan approach to Physical Reality, Pramana – The means of valid knowledge, Framework for establishing valid knowledge.	07
3	**Science in IKS:** Mathematics: Great mathematicians and their contributions, Arithmetic (square of a number, square root, series and progressions), Geometry (Property of right angled triangle in Sulba- sutras, value of π), Trigonometry, Algebra, Binary Mathematics and combinatorial problems in Chandah- Sastra of Pingala	07
4	**Engineering and Technology in IKS:** The Indian Science and Technology Heritage, Mining and ore extraction, Metals and metal working technology, Iron and Steel in India, Lost wax casting of idols and artefacts, apparatuses used for extraction of metallic components. Literary sources for Science and Technology, Physical Structures in India, Irrigation & Water Management, Dyes and Painting Technology, Shipbuilding.	08

Textbooks:

- Mahadevan B.,Bhat Vinayak Rajat & Nagendra Pavana R.N. "Introduction to Indian Knowledge System Concepts and Applications"PHI,2023.
- Jha Amit "Traditional Knowledge System in India" Atlantic Publishers and Distributors (P) Ltd,2023

- Chauhan Bhag Chand "IKS: The Knowledge system of Bharata" Garuda Prakashan,2023
- Bag A.K., "Mathematics in Ancient and Medieval India"Chaukhambha Orientalia, 1979
- Sengupta Nirmal,"Traditional Knowledge in Modern India"Springer,2019

Reference Books:

- Bag A.K., "Historyof Technology in India,Vol.I" Indian National Science Academy,1997
- Kumar Alok, "Ancient Hindu Science" Jaico Publishing House,2019
- Datta B and Singh A.N."History of Hindu Mathematics:Parts I and II" Asia Publishing House,1962
- Kapoor Kapil, Singh Awdhesh Kumar "Indian Knowlege Systems Vol.-I & II" D.K. Print World Ltd,2005

E-Resources:

https://www.youtube.com/watch?v=-cBd6JYPWtY&list=PLRfu94TCePTtWtu0x145H_63WgoeYickE

https://www.youtube.com/watch?v=yvj5ROYbP2E&list=PLRfu94TCePTtLuEYSzmJXNYK_EnDSvY3N

https://iksindia.org/book-list.php

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Course Code: APS21ESL104 **Course Name:** Building Programming logic in C **Course Category:**

Credits: 1 **Teaching scheme:** L-1 **Evaluation scheme:** CA–60, ESE–20

Pre-requisites: Pre-university mathematics.

Course Objectives:

4. The primary objective of an engineering chemistry course is to familiarize the students with new developments in engineering chemistry.
5. To impart a sound knowledge on the principles of chemistry involving the different application oriented topics required for all engineering branches.
6. The objectives of the laboratory sessions are to enable the learners to get hands-on experience on the principles discussed in theory sessions and to recognize the applications of these concepts in engineering.

Course Outcomes:

After the completion of this course, students will be able to:
CO1: Illustrate and explain the basic computer concepts and programming principles of C language.
CO2: Develop C programs to solve simple mathematical and decision making problems.
CO3: Develop C programs to solve simple engineering problems using looping constructs.
CO4: Develop C programs to demonstrate the applications of derived data types such as arrays, pointers, strings and functions.

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Contents:

Unit	Content	Teaching Hours
1	**Introduction To C Programming:** **Fundamentals of C Programming: Overview of C:** History of'C, Algorithm and Structure of C program. Keywords, Tokens, Data types, Constants, Literals and Variables. **Operators and Expressions:** Arithmetic operators, Relational operator, Logical operators, Expressions, Operator: operator precedence and associatively, Type casting, Console FO formatting, Unformatted I/O functions: getch(), getchar, getche(), getc(), putc(), putchar().	03
2	**Array and String:** **Control statements:** If-else, conditional operators, switch and break, nested conditional branching statements, loops: do while, while, for, Nested loops, break and continue, goto and label, exit function. **Array:** Array declaration, One and Two dimensional numeric and character arrays, Multidimensional arrays, operations on array. **String:** String declaration, initialization, string manipulation with/without using library function.	03
3	**Control Statements and Functions:** **Functions:** Definition, function components: Function arguments, return value, function call statement, function prototype, Types of function, Scope and lifetime of variable, Call by value,and call by reference. Function using arrays, function with command line argument. User defined function: maths and	03

	character functions, Recursive function.	
4	**Structure and Union:** **Structure:** Basics, declaring structure and structure variable, typedef statement, array of structure, array within structure, Nested structure; passing structure to function, function returning structure. **Union:** basics, declaring union and union variable, Difference between Structure and Union **Enum:** declaring enum and enum variable.	03
5	**Pointers**: **Pointer:** Definition of pointer, advantage and disadvantage using pointer, Pointer declaration, Using & and * operators. Void pointer, Pointer to pointer, Pointer in math expression, Pointer arithmetic, Pointer comparison, **Dynamic memory allocation functions**: malloc, calloc, realloc and free, Pointer vs. Array, Array of pointer, Pointer to array, Pointers to function, Function returning pointer, Passing function as Argument to function, Pointer to structure, Dynamic array of structure through pointer to structure.	03

Text Books:

1. Programming in ANSI C, E Balagurusamy, Tata McGraw-Hill, Third Edition.

2. Let Us C, YashwantKanetkar, Infinity Science Press, Eighth Edition.

3. Mastering C, K R Venugopa1,Tata McGraw-Hill.

Reference Books:

1. The C Programming Language, Brian W. Kernighan, Dennis M. Ritchie, Prentice Hall, 2nd Edition.
5.Applications Programming in ANSI C, R. Johnsonbaugh , Martin Kalin, Macmillan, 2nd Edition.

2. The Spirit of C, Mullish Cooper, Jaico publishing House.

3. How to solve it by Computer, R.G.Dromey, Pearson Education.

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Course Code:APS21VSP101**Course Name:**Engineering Exploration**Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Nil

Course Objectives:

7. Understand the principles of Engineering Design and apply them to identify product requirements, objectives, and constraints through interactions with clients, users, and relevant stakeholders.
8. Develop critical thinking and problem-solving skills to analyze existing technologies, conduct surveys, study literature, and generate creative design concepts using various means and combinations.
9. Acquire proficiency in using appropriate tools and techniques to generate electronic and mechanical simulations, schematic diagrams, and product concept sketches or CAD models for effective communication of design ideas.
10. Demonstrate competency in categorizing inputs and outputs of systems in terms of materials, information, and energy, enabling a holistic understanding of engineering projects.
11. Cultivate effective teamwork and communication skills to collaborate with peers, mentors, and team members during the development and presentation of engineering prototypes.
12. Develop the ability to evaluate, defend, and communicate design decisions, progress, and project outcomes, showcasing awareness of course concepts and their application in real-world scenarios.

Course Outcomes:

On completion of the course, the student should be able to:

LO 7. Effectively interact with clients, users, and stakeholders to gather relevant information and derive product requirements, thereby demonstrating proficiency in Requirement Analysis and Client Interaction.

LO 8. Analyse and compare existing technologies, conduct surveys, and study literature to identify potential solutions, and develop objective trees and function trees, reflecting their competency in Conceptual Design and Analysis.

LO 9. Create product concept diagrams, combining different means, and prepare sketches or CAD models to present their design ideas aesthetically and coherently, indicating proficiency in Product Concept Development.

LO 10. Categorize system inputs and outputs in terms of materials, information, and energy, illustrating their understanding of System Categorization and Analysis.

LO 11. Exhibit hands-on skills in physical assembly, connection, and demonstration of engineering prototypes, showcasing Proficiency in Prototyping and Implementation.

LO 12. Explain and defend their design choices, project progress, and outcomes during presentations, demonstrating Communication Skills, Project Evaluation, and Awareness of Engineering Concepts.

Contents:

Sr. No.	List of Practical	Lab Hours
1	The course is conducted in the following modules: 8. Introduction to Engineering Exploration 9. Engineering Design 10. Platform Based Development 11. Mechanisms 12. Data acquisition and analysis 13. Engineering Ethics 14. Project Management	10
2	The following practical contents are delivered in an integrated mode along with theory: 14. Conceptualizing a product 15. Designing a product with constraints 16. Simulation of electronic circuits (at least 10 sets) 17. Implementation of electronic circuits (at least 5 sets) 18. Implementation of four bar chain mechanism 19. Conversion of problem statement to need statement 20. Identification of objectives, constraints and functions 21. Generation of black box, glass box and expanded class box 22. Generation of morphological chart 23. Generation of concepts 24. Comparison of concepts 25. Selection of concept 26. Implementation and testing of prototype	20

Reference Books:

- George E. Dieter and Linda C. Schmidt (2009), Engineering Design, 4ed, Mc Graw Hill Higher Education
- Clive L. Dym, Patrick Little, and Elizabeth J. Orwin (2014), Engineering Design: A project-based introduction, 4ed, John Wiley and Sons
- G. Pahl, W. Beitz, J. Feldhusen and K.-H. Grote (2007), Engineering Design: A systematic approach, 3ed, Springer, New York

Course Code:APS21BSP101 **Course Name:**Engineering Physics Lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Student should know the basic aspects of measurements like least count and range of instrument, scale identification, accuracy, error etc.

Course Objectives:
The Objective of this course is
1. To make the students gain practical knowledge to co-relate with the theoretical studies.
2. To achieve perfectness in experimental skills.
3. The study of practical applications will bring more confidence.

Course Outcomes:
At the end of the laboratory course, the students will be able to,
LO1: Plot the I-V characteristics of pn junction diode and determine the value of band gap energy which can be used for core engineering Courses.
LO2: Perform optical experiments; using light properties like interference, polarization, diffraction and their applications.
LO3: Interpret the results and analyze the data and use the experimental data to plot the graph for a best fit.
LO4: Discuss the characteristics of plateau region and determine operating voltage of G.M. counter
LO5: Determine the numerical aperture and bending loss of optical fibre cable

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Contents:

Sr. No.	List of Practical	Lab Hours
1	Thomson's Bar Magnetic method -Determination of e/m of electron	02
2	Newton's rings -Determination of radius of curvature of Plano convex lens.	02
3	Polarization - Half shade Polarimeter -Determination of specific rotation of optically active material.	02
4	Diffraction - Determination of wavelength of light by plane transmission grating.	02
5	Wedge Shaped film -Determination of thickness of thin wire	02
6	Semiconductors – To study of forward and reverse bias characteristic of Semiconductors diode	02
7	Fibre Optics – Determination of Bending losses in optical fibre	02
8	Fibre Optics – Determination of Numerical Aperture of given optical fibre	02
9	Determination of operating voltage of G.M. tube and characteristics of plateau region	02
10	Determination of wavelength of laser source	02

Course Code:APS21ESP102**Course Name:**Engineering Graphics Lab **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The Objective of this course is

Course Outcomes:
Students will be able to,
1. Develop competence in correct expression of the visualized objects
5. Dimension and annotate two-dimensional engineering drawings
6. Plan and prepare neat orthographic drawings of points, straight lines, planes and solids
4. Develop the ability to visualize and draw orthographic and isometric projection of solids

Contents:

Sr. No.	List of Practical	Lab Hours
1	Drawing three problems based on projections of lines on half imperial size drawing sheet	02
2	Drawing three problems based on engineering curves on half imperial size drawing sheet	02
3	Drawing three problems based on projections of planes on half imperial size drawing sheet	02
4	Drawing three problems based on projections of solids on half imperial size drawing sheet	02
5	Drawing three problems based on orthographic projections on half imperial size drawing sheet	02
6	Drawing three problems based on isometric projections on half imperial size drawing sheet	02
7	Demonstration of CAD software in CAD lab, drawing simple objects using various commands	02

Course Code:APS21ESP105**Course Name:**Programming in C - LAB **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives: Nil

Course Outcomes: Nil

Contents:

Sr. No.	List of Practical	Lab Hours
1	c) Write a C program to find sum and average of three numbers. d) Write a C program to find the sum of individual digits of a given positive integer.	02
2	d) Write a C program to generate the first n terms of the Fibonacci sequence e) Write a C program to generate prime numbers between 1 to n. f) Write a C program to check if the given number is Armstrong or not	02
3	c) Write a C program to check whether the given number is perfect or not d) Write a C program to check whether the given number is strong or not	02
4	c) Write a C program to find the roots of a quadratic equation. d) Write a C program perform arithmetic operations using switch statement.	02
5	c) Write a C program to find factorial of a given integer using non-recursive function d) Write a C program to find factorial of a given integer using recursive function	02
6	c) Write C program to find GCD of two integers by using recursive function. d) Write C program to find GCD of two integers by using non-recursive function.	02
7	d) Write a C program to find the largest and smallest number in a list of integers. e) Write a C program to Sort the Array in an Ascending Order. f) Write a C program to find whether the given matrix is symmetric or not.	02
8	c) Write a C program to perform addition of two matrices. d) Write a C program using function to perform multiplication of two matrices.	02
9	c) Write a c program to use function to insert a sub-string in to given main string from a given position. d) Write a c program to swap the values of two variables using (i) call by value (ii) call by reference	02
10	c) Write a C program using user-defined functions to determine whether the given string is palindrome or not. d) Write a C program that displays the position or index in the main string S where the sub string T begins, or - 1 if S doesn't contain T.	02
11	c) Write C program to count the number of lines, words and characters in a given text. d) Write a C program to find the sum of integer array elements using	02

	pointers.	
12	b) Write a C Program to Calculate Total and Percentage marks of a student using structure.	02

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Course Code:APS21ESP103**Course Name:**Recent Trends in Integrated Technologies Lab
Course Category:

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

1.To introduce students the basics of additive manufacturing/rapid prototyping and its applications in various fields, reverse engineering techniques.
2.To recognize industrial control problems suitable for Industrial Robotics.
3.To acquire basic skills in exploring the potential of the drone technology in professional activities
4.Ability to recognize industrial automation problems suitable for PLC control.

Course Outcomes:

LO1. Prepare 3D Model (slice & print) in either Stratasys or Zortrax & generate scan data through Hexagon portable scanning arm.
LO2.Describe basic industrial robotics & it's applications
LO3.Operate a small drone in a controlled environment
LO4.Explain principles of sensor, PLC & applications.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Scan to CAD, CAD to STL conversion& patching, machine setup & processes for printing	02
2	Programming the TATA Robot(pendant) for pick & place, programming the Yaskawa Robot(pendant) for motion planning, Demonstrate welding exercise by the instructor	02
3	Introduction to components of drone, Demonstration of assembly of drone, Demonstration of mission planning & flying the drone	02
4	Controller & sensor & their interfacing, basic ladder logic instructions, pneumatic& hydraulic actuator	02

Reference Books:

5. A Step-by-Step Guide For Beginners: Aircraft Design & Construction Design Guide by :Merlin Debrie
6. Industrial Automation & Robotics By A.K. Gupta & S.K.Arora
7. Additive Manufacturing Principles, Technologies & Applications By C.P.Paul (TMH)
8. Basics of unmanned aerial vehicle By Garvit Pandya (Motion press)

Course Code: MGM82CCP104 **Course Name:** National Service Scheme (NSS) **Course Category:**

Credits: 2 **Teaching scheme:** P-2 **Evaluation scheme:**

Pre-requisites: Nil

Course Objectives: Nil

Course Outcomes: Nil

Contents:

Sr. No.	List of Practical	Lab Hours
1	**IntroductiontoNationalServiceScheme(NSS):** **Emergence of NSS in India (Historical Background) and its development.**OrganizationalStructureofNationalServiceSchemefromNationalLeveltoCollegeLevel.Objectivesof National ServiceScheme(NSS) NationalServiceScheme(NSS)–Symbolanditsmeaning **SymbolofNSSanditsmeaning** MottoofNationalServiceScheme(NSS) Variousprayers,inspirationalsongstobeusedinNSSProgramme.	08
2	**NationalServiceScheme(NSS)RegularActivities:** **GuidelinesofDistributionofworkinghoursoracademicyear.** **ClassificationofRegularActivities intheSociety** • Rural • Urban • Campus • Needbasewithassociation **AssociationsinNSSActivities** • Govt.Organization • NGO ScopeforInnovation(Self-Generated)	08
3	**SocialIssuesin India:** **ConceptofSociety,Community(Stepsinvolvedinevaluationofsociety)** • FeaturesofIndianSociety • CommunitiesinIndia **BasicSocialIssuesin India** FamilySystem,Divisionoflabour,CastSysteminIndia,GenderIssues,RegionalImbalance	07
4	**IndianConstitutionandSocialJustice:** **IndianConstitution** • Preamble	07

	• Structure • Features • FundamentalRights&Duties **SocialJustice** • SocialJustice–theConceptanditsfeatures Contribution for Social Justice – Mahatma Jyotiba Phule, Dr. Babasaheb Ambedkar,ShahuMaharaj.	

References:

1. National Service Scheme Manual (Revised) 2006, Government of India, Ministry of Youth Affairs and Sports, New Delhi.
2. University of Mumbai National Service Scheme Manual 2009.
3. Avhan Chancellor's Brigade-NSS Wing, Training camp on Disaster Preparedness Guidelines, March 2012.
4. Rashtriya Seva YojanaSankalpana- Prof. Dr. SankayChakane, Dr. Pramod Pabrekar, Diamond Publication, Pune.
5. National Service Scheme Manual for NSS District Coordinators, National Service Scheme Cell, Dept. of Higher and Technical Education, Mantralaya,
6. Annual report of National Service Scheme (NSS) published by Dept. of Higher and Technical Education, Mantralaya,
7. NSS Cell, Dept. of Higher and Technical Education, Mantralaya, UTKARSHA- Socio and cultural guidelines.

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Course Code:MGM73CCP203 **Course Name:**Fine Art**Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The objective of this course is

- Students will be able to draw clean lines.
- Students will be able to explore shading from light to dark.
- Students will be able to draw perspective drawing.
- Students will be able to sketch landscape and portrait drawing.

Course Outcomes:
On completion of the course, the student should be able to:

- Students will be able to pay attention to core details in visualization.
- Students will be able to represent on paper what they have observed in terms of 3 and 2 dimensional objects and light and dark play of perspective.
- Students will be able to draw clean lines and neat figures which will gradually help them in fashion illustrations.

Contents:

Sr. No.	List of Practical	Lab Hours
1	**Introduction to Drawing:** • Explore pencil as medium for drawing and exploration of different types of grade of pencils. • Lines – Freeing hand movement by practicing various ways to draw lines (wavy/zigzag/diagonal/vertical/horizontal)	08
2	**3D objects with Shading:** • Exploring and sketching 3D objects with Shading/Hatching – Light to Dark (HB/2B/3B/4B/5B/6B) • Space Division – Studying of proportion, Perspectives- Coin / Box / Landscape.	07
3	**Sill Drawing:** • Introduction to landscape and portrait drawing. • Still drawing (perspective drawing of man- made object)	07
4	**Exploration of color mediums:** • Exploration of color mediums (chalk/ink/poster/pastels/water) • Landscape perspective of natural and man- made, charcoal rendering. • Still drawing of man- made and natural object by rendering and defining perspective.Ambedkar,ShahuMaharaj.	08

Reference Books:

- Mellem , Jeff; Sketching people & Live drawing basics.
- Belleville- Van Stone, France; Sketch: the Non-Artist’s guide to inspiration technique & drawing life

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Course Code:MGM73CCP204 **Course Name:**Visual Art **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The objective of this course is
- To understand the actual work process in advertising market.
- To understand the growth and necessity of advertising in market.

Course Outcomes:
On completion of the course, the student should be able to:
- Students will be able to choose a topic for campaign design.
- Students will understand that how campaign design is necessary
- Student will understand the actual work process in advertising market.
- Student will understand the process of designing.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Introduction to Campaign Design**:** • Topic for campaign design • It should be either product, service or social topic. • Mind mapping	08
2	**:** • Different types of media • New digital medias • Use of elective subject in campaign • Software use in designing	11
3	**Sill Drawing:** • Final layout • Printing in actual size media • Presentation on ppt of the topic includes artwork with rough work.	11

Reference Books:
- **Advertising Campaign Design Just the Essentials - By Robyn Blakeman**
- **Campaign Strategies and Message Design: A Practitioner's Guide from Start to Finish - by Mary Moffitt (Author)**
- **Student should refer the previous knowledge about medias and refer the internet for help.**

Suggested Tutorial / Practical:

Assignment I: Students will be required to submit mind map.

Assignment II: Students will be required to submit list of media selection.

Assignment III: Students will be required to submit different medias in digital form.

Assignment IV: Students will be required to submit different medias in print form.

Teaching Methods and learning activities-

- **Lecture with power point projections**
- **Demonstration with practical**
- **Documentaries and Presentation**
- **Use of different medium and art form to make a new concept of work**

Assessment Strategies-

Classroom lecture

Power point presentations

Group discussion

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Semester-III

Course Code: ELE24PCL201
Course Name: Electronic Device and Circuits
Course Category: PCC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme:CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Basic Components R, L & C, Elementary Mathematics, Basic Electrical Parameters, Semiconductor Basics.
Course Objectives: 1) To Understand the Working of Circuits having Diodes, Transistors and MOSFET as Circuit Elements. 2) To Understand the Working of Voltage Regulators, Multivibrators and Oscillators. 3) To Understand Power Amplifier. 4) To Understand Time Base Generator.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Analyze Simple BJT and MOSFET Circuits. **CO2:** Analyze Simple Rectifiers and Voltage Regulators Using IC. **CO3:** Apply Amplifiers as a Power Circuits. **CO4:**Understand Time Base Generators.

Contents:

Unit No.	Content	Teaching Hours
1.	**Electronic Devices** 1.1 Diode Parameters, Data Sheet, Bipolar Junction Transistor Construction, Symbols, Operation of a BJT, BJT Circuit Configurations, Current Gain of BJT in Common Base and Common Emitter Configuration, Input And Output Characteristics of BJT in CE Configuration 1.2 Frequency Response of CE Amplifier, BJT Datasheet, Heat Sink, LED, Photo Transistor. 1.3 MOSFET: Construction, Operation, Types, Circuit Symbol, Drain and Transfer Characteristics, 1.4 CMOS: Construction, CMOS as a Switch, Handling Precautions of CMOS Devices.	8
2.	**Electronic Circuits** 2.1 Transistor Switch, Rectifiers, Ripple Factor, Capacitor and LC Filters, Line and Load Regulation, Fixed and Variable IC Linear Voltage Regulator, Switching Voltage Regulator.	7

	2.2 Multivibrators: Astable & Monostable Multivibrators Using IC555 2.3 Oscillators: Principle of Positive Feedback, RC Phase Shift, Hartley and Crystal Oscillator.	
3.	**Power Amplifier** 3.1 Comparison Between Small Signal Amplifier and Power Amplifier 3.2 Classification: Class A, Class B, Class AB & Class C with respect to Operating Point on Load Line, Efficiency 3.3 Single Stage Class A, Power Amplifier Circuit: Operation, Input Output Waveforms. 3.4 Class B Push-Pull Amplifier, Operation ,Input Out Waveforms and its Advantages & Disadvantages, Applications of Power Amplifier.	8
4.	**Time Base Generators** 4.1 Unijunction Transistor (UJT): Construction, Working Principle & Characteristics. 4.2 Types of Time Base Generators: Free Running Time Base Generator, Working Principle Of UJT As Time Base Generators, (Relaxation Oscillator), Circuit Diagram and Working of Voltage Time Base Generator, Current Time Base Generator, Bootstrap Time Base Generator 4.3 Miller's Sweep Generator & Its Applications.	7

Text Books:

1. Applied Electronics by Dr. R.S. Sedha S. Chand Publication.
2. Network Analysis and Synthesis by Ravish Singh.
3. Electronic Devices and Circuit by S. Sallvahanan, 5th Edition.
4. Electronic Devices and Circuit by Dharmaraj Cheruku, 2nd Edition.

Reference Books:

1. Electronic Devices and Circuits: Jacob Millman & C Halkias, Tata McGraw Hill.
2. Network and Systems: D Roy Choudhary, New Age International Publishers.
3. Electronic Devices and Circuit Theory by Robert L. Boylestad and Louis Nashelsky, 11th Edition.
4. Electronic Devices and Circuits by Anil Maini and Varsha Agrawal , 2nd Edition.
5. Electronic Devices and Circuit by M.H. Rashid, 2nd Edition.

ICT / MOOKS: NPTEL / SWAYAM Lectures.

Course Code: ELE24PCP201
Course Name: Electronic Device and Circuits Lab
Course Category: PCC
Credits: 1
Teaching Scheme:P-2
Evaluation Scheme: TW:30, PR:20
Pre-requisites: Basic Components R, L & C, Elementary Mathematics, Basic Electrical Parameters, Semiconductor Basics.
Course Objectives:
1. To Understand the Working of Circuits having Diodes, Transistors and MOSFET as Circuit Elements.
2. To Understand the Working of Voltage Regulators, Multivibrators and Oscillators.
3. To Understand Power Amplifier.
4. To Understand Time Base Generator.
Course Outcomes: At the end of the course, the students will be able to –
CO1: Analyze Simple BJT and MOSFET Circuits.
CO2: Analyze Simple Rectifiers and Voltage Regulators Using IC.
CO3: Apply Amplifiers as a Power Circuits.
CO4: Understand Time Base Generators.

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Plotting V-I Characteristics of Voltage and Current Source and Determine its Source Resistance.	2
2.	Input Output Characteristics of Common Base and Common Emitter configuration.	2
3.	Plotting V-I Characteristics of a PN Junction Diode.	2
4.	Use BJT/ MOSFET as a Switch.	2
5.	Building a Monostable / Astable Multivibrator Using IC 555.	2
6.	Study of Hartley and RC Phase Shift Oscillator.	2
7.	Designing a Linear Variable Voltage Regulator.	2
8.	V-I Characteristics of UJT	2

9.	Draw Input and Output Wave Forms of Class B Push Pull Amplifier and Determine Efficiency.	2
10.	Determine T-on, T-off, Duty Cycle of Miller's Sweep Generator.	2

Text Books:

1. Applied Electronics by Dr. R.S. Sedha S. Chand Publication.
2. Network Analysis and Synthesis by Ravish Singh.
3. Electronic Devices and Circuit by S. Sallvahanan, 5th Edition.
4. Electronic Devices and Circuit by Dharmaraj Cheruku, 2nd Edition.

Reference Books:

1. Electronic Devices and Circuits: Jacob Millman & C Halkias, Tata McGraw Hill.
2. Network and Systems: D Roy Choudhary, New Age International Publishers.
3. Electronic Devices and Circuit Theory by Robert L. Boylestad and Louis Nashelsky, 11th Edition.
4. Electronic Devices and Circuits by Anil Maini and Varsha Agrawal , 2nd Edition.
   Electronic Devices and Circuit by M.H. Rashid, 2nd Edition.

ICT / MOOKS: NPTEL / SWAYAM Lectures.

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Course Code: ELE24PCL202
Course Name: Digital System Design
Course Category: PCC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Basic Knowledge of Electronic Circuits, Number System and Logic Gates.
Course Objectives: 1) Learn TTL and CMOS Logic Families. 2) Learn Various Reduction Techniques to Design Combinational Logic Circuits. 3) Learn 1-bit Memory Cell (FF) and its Conversion. 4) Learn Sequential Logic Circuits & PLDs.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Explain the Fundamental Concepts and Techniques used in Digital Electronics. **CO2:** Implement different functions and combinational circuits using Various Reduction Techniques and PLDs. **CO3:**Explain Operation of Flip Flops. **CO4:**Design Simple Sequential Circuits.

Contents:

Unit No.	Content	Teaching Hours
1.	**Digital Logic Families** 1.1 Classification of Logic Families, Characteristics of Digital ICs, TTL Logic, Operation of TTL NAND Gate 1.2 CMOS Logic – CMOS Inverter, NAND, NOR Gates.	7
2.	**Combinational Logic Design** 2.1 Standard Representations for Logic Functions, K-Map Representation of Logic Functions (SOP and POS Forms), Minimization of Logical Functions for min-terms and max-terms (up to 4 variables), don't care conditions. 2.2 Design Examples: Arithmetic Circuits, Code Converters (Binary to Gray, Gray to Binary, Binary to Excess3, Excess3 to Binary), Digital Comparator, Parity Generators / Checkers 2.3 Design of Multiplexers and Demultiplexers, Decoders.	8
3.	**Flip Flops** 3.1 1-Bit Memory Cell, Clocked SR, JK, MS J-K Flip Flop, D and T Flip-Flops. 3.2 Use of Preset and Clear Terminals, Excitation Table for Flip Flops, Conversion of Flip Flops.	7
4.	**Sequential Logic Design** 4.1 Application of Flip-Flops, Shift Registers, Counters (Ring Counters, Twisted	8

	Ring Counters), Ripple Counters, Up / Down Counters, Synchronous Counters, Sequence Generators. 4.2 Programmable Logic Devices (PLD), Study of PROM, PAL, PLA, Designing Combinational Circuits using PLDs, General Architecture of FPGA and CPLD.	

Text Books:

1. R.P. Jain - Modern Digital Electronics, 5th edition, 12th reprint Tata McGraw Hill Publication, 2007.
2. Anand Kumar, ―Fundamentals of Digital Circuits‖ 1st edition, Prentice Hall of India, 2001.
3. S.K. Kataria & Sons Digital System Design by Shipra Gupta.
4. Digital System Design by Dr. Raj Kishor Prasad.
5. Digital Logic & Design by Pratima Manhas and Shaveta Tharkal, Katson Books.

Reference Books:

1. Morris Mano ― Digital Logic and Computer Design‖ 4thedition, Prentice Hall of India, 2013.
2. Donald D. Givone - Digital Principals and Design. Tata McGrawHill Publication.
3. Digital System Design by S. Sallvahanan and S. Arivazghagan, 5th Edition.
4. Digital Design (Principles and Practices) by Johm F. Wakerly, Pearson Publication.
5. Digital Systems - Principles and Applications by Ronald J. Tocci, Pearson Publication.

ICT / MOOKS: NPTEL / SWAYAM Lectures.

MGMUNIVERSITY

Course Code: ELE24PCP202
Course Name: Digital System Design Lab
Course Category: PCC
Credits: 1
Teaching Scheme:P-2
Evaluation Scheme: TW:30, PR:20
Pre-requisites: Basic Knowledge of Electronic Circuits, Number System and Logic Gates.
Course Objectives:
1. Design Arithmetic Circuits using K-Maps.
2. Understand Code Conversion: Binary To Gray and Gray to Binary.
3. Understand Multiplexer and Demultiplexer Operations.
4. Understand Operation of Flip Flop using IC.
5. Understand Applications of Flip Flops.
Course Outcomes: At the end of the course, the students will be able to –
CO1: Construct Arithmetic Building Blocks.
CO2: Demonstrate Code Conversion Binary to Gray and Gray to Binary.
CO3: Verify Multiplexer and Demultiplexer.
CO4: Verify the Operation of Flip Flops.
CO5: Demonstrate the Ability to Work Independently and in a Group and to Prepare Lab Reports.

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	To Verify the Basic Logic Gates and Universal Gates.	2
2.	Design the Circuit for given Boolean Algebraic Equation (SOP and POS) Using K-Map.	2
3.	To Verify the Half Adder And Half Subtractor.	2
4.	To Verify the Binary to Gray Code Conversion.	2
5.	To Verify the Gray to Binary Code Conversion.	2
6.	To Verify the Multiplexer.	2
7.	To Verify the Demultiplexer.	2
8.	To Verify the Flip- Flops.	2

9.	To Verify the Counter.	2
10.	Study of PLD.	2

Text Books:

1. R.P. Jain - Modern Digital Electronics, 5th edition, 12th reprint Tata McGraw Hill Publication, 2007.
2. Anand Kumar, ―Fundamentals of Digital Circuits‖ 1st edition, Prentice Hall of India, 2001.
3. S.K. Kataria & Sons Digital System Design by Shipra Gupta.
4. Digital System Design by Dr. Raj Kishor Prasad.
5. Digital Logic & Design by Pratima Manhas and Shaveta Tharkal, Katson Books.

Reference Books:

1. Morris Mano ― Digital Logic and Computer Design‖ 4th edition, Prentice Hall of India, 2013.
2. Donald D. Givone - Digital Principals and Design. Tata McGrawHill Publication.
3. Digital System Design by S. Sallvahanan and S. Arivazghagan, 5th Edition.
4. Digital Design (Principles and Practices) by Johm F. Wakerly, Pearson Publication.
5. Digital Systems - Principles and Applications by Ronald J. Tocci, Pearson Publication.

ICT / MOOKS: NPTEL / SWAYAM Lectures.

MGMUNIVERSITY

Course Code:ELE24PCL203
Course Name: Data Structures
Course Category: PCC
Credits: 2
Teaching Scheme: L-2
Duration of ESE: 2 Hours
Evaluation Scheme: CA:60, ESE:40
Pre-requisites: Basics of C - Language.
Course Objectives: 1. The Course Aims to Teach Programming and Introduce Elementary Data Structures. 2. To Introduce the student to the Concept of Data Structures through Abstract Data Structures including Lists, Stacks, Queues and Graphs. 3. To Assess how the Choice of Data Structures and Algorithm Design Methods affects the Performance of Programs 4. To Choose the Appropriate Data Structure and Algorithm Design Method for a Specified Application.
Course Outcomes: At the end of the course, the students will be able to – **CO1:**Apply Knowledge of Basic Programming Constructs to Design Algorithms and Programs. **CO2:**To Use Appropriate Data Structures like Arrays, Linked Lists, Stacks and Queues to Solve Real-World Problems Efficiently. **CO3:**To Analyse Different Possible Solutions to a Program and Select the Most Efficient One. **CO4:**To Impart Familiarity with Various Sorting, Searching and Hashing Techniques and their Performance Comparison. **CO5:**Ability to Analyze Algorithms and Algorithm Correctness.

Contents:

Unit No.	Content	Teaching Hours
1.	**Introduction** 1.1 Introduction to Data & Data Structures. 1.2 Types of Data Structure - Primitive & Non- Primitive, Concept of Abstract Data Type, Arrays, Types of Arrays, Memory Representation of Array, Traversing Linear Array, Inserting and Deleting Elements in Array. Structures, Functions, Recursive Functions, Introduction to Time & Space Complexity. 1.3 Searching – Linear Search and Binary Search, Sorting – Bubble Sort , Selection Sort , Insertion Sort, Hashing, Hashing (Hash Function), Hash Collision, Collision Resolution, Chaining Open Addressing, Linear Probing, Quadratic Probing, Double Hashing .	8
2.	**Stack & Queues** 2.1 Stacks: Definition, Array Representation of Stacks, Linked representation of Stacks, Evaluation of a Prefix, Infix and Postfix Expression, Transforming Infix into Postfix Expression.	6

	2.2 Queues: Definition, Array Representation of Queues, Linked representation of Queues. 2.3 Types of Queues: Circular Queues, Double Ended Queue, Priority Queue, Stack using Two Queues, Queue using Two Stack.	
3.	**Linked List & Trees** 3.1 Linked List: Representation of Linked Lists in Memory, Insertion & Deletion from Linked List, Traversing & Searching a Linked List. 3.2 Types of Linked List: Circular Linked List and Doubly Linked Lists. 3.3 Trees– Terminology, Binary Trees, Properties of Binary Trees, Array and Liked Representation of Binary Trees, Traversals-In Order, Post Order, Preorder, Binary Search Trees- Definition, Insertion, Deletion, Traversal and Searching, Evaluation of Expression, AVL Trees.	8
4.	**Graph** 4.1 Definitions, Terminologies, Matrix and Adjacency List Representation of Graphs. 4.2 Traversal Methods: Breadth First Search and Depth First Search, Minimum Spanning Tree, Prim's & Krushal Algorithm.	8

Text Books:

1. Data Structures with C, S. Lipschutz, Mc Graw Hill.
2. Fundamentals of Data Structures in C++, E. Horowitz, Sartaj Sahni, D.Mehta, Universities Press.

Reference Books:

1. Data Structures using C, E. Balaguruswamy, McGraw Hill.
2. Introduction to Algorithms, T. H. Cormen, C. E. Leiserson, R. L. Rivest, C. Stein, MIT Press, 3rd Edition, 2009.

ICT/ MOOKS:

1. https://archive.nptel.ac.in/courses/106/106/106106127/
2. https://nptel.ac.in/courses/106102064

Course Code: ELE24PCP203
Course Name: Data Structures Lab
Course Category: PCC
Credits: 1
Teaching Scheme: P-2
Evaluation Scheme: TW:30, PR:20
Pre-requisites: Programming in C.
Course Objectives:
1. The Course Aims to Teach Programming and Introduce Elementary Data Structures.
2. To Introduce the student to the Concept of Data Structures through Abstract Data Structures including Lists, Stacks, Queues and Graphs.
3. To Use the Appropriate Data Structure for A Specific Application.
Course Outcomes: At the end of the course, the students will be able to –
CO1: Apply Basic Programming Skills such as Array, Functions.
CO2: Implement Different Programs for Linear Data Structures.
CO3: Implement Non Linear Data Structure.
CO4: Apply Hashing Techniques to Organize Data.

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Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Write a 'C' Program to Perform the following Operations on Array: Create, Insert, Delete and Display.	2
2.	Write a 'C' Program to Search a Particular Data from the Given Array of Strings using Linear Search Method.	2
3.	Write a 'C' Program to Search a Particular Data from the Given Array of Numbers using Binary Search Method.	2
4.	Write a 'C' Program to Sort an Array of Numbers using Bubble Sort Method.	2
5.	Write a 'C' Program to Sort an Array of Numbers using Selection Sort Method.	2
6.	Write a 'C' Program to Sort an Array of Numbers using Insertion Sort Method.	2

7.	Write a 'C' Program to Perform PUSH and POP Operations on Stack using an Array.	2
8.	Write a 'C' Program to Perform INSERT and DELETE Operations on Linear Queue using an Array.	2
9.	Write a 'C' Program to Perform INSERT and DELETE Operations on Circular Queue using an Array.	2
10.	Write a 'C' Program to Implement Singly Linked List with Operations: (i) Insert at Beginning (ii) Search (iii) Display.	2
11.	Write a 'C' Program to Implement BST (Binary Search Tree) and traverse in In-Order.	2
12.	Write a Program to Implement Graph Operations.	

Text Books:

1. Data Structures with C, S. Lipschutz, Mc Graw Hill.
2. Fundamentals of Data Structures in C++, E. Horowitz, Sartaj Sahni, D.Mehta, Universities Press.

Reference Books:

1. Data Structures using C, E. Balaguruswamy, McGraw Hill.
2. Introduction to Algorithms, T. H. Cormen, C. E. Leiserson, R. L. Rivest, C. Stein, MIT Press, 3rd Edition, 2009.

ICT/ MOOKS:

1. https://archive.nptel.ac.in/courses/106/106/106106127/
2. https://nptel.ac.in/courses/106102064

Course Code: ELE24HSL201
Course Name: Business Management & Financial Accounting
Course Category: EEMC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Basic Management.
Course Objectives: 1. To Help the Students to Analyze the Risk and Return of Alternative Sources of Financing. 2. To Enable Students to Describe the Estimated Cash Flows from a Project, including Operating, Net Working Capital and Capital Spending. 3. To Provide the Students to Understand the related Information of Business Finance. Students can prepare Basic Financial Statements. 4. To Enable Students to Prepare Final Financial Statements.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Analyze the Risk and Return of Alternative Sources of Financing. **CO2:** Estimate Cash Flows from a Project, including Operating, Net Working Capital and Capital Spending. **CO3:**Define Basic Terminology used in Finance and Accounts. **CO4:**Prepare Financial Statements.

Contents:

Unit No.	Content	Teaching Hours
1.	**Introduction to Business Management** 1.1Aims, Objective and Function of Business Management, Principles of Management. 1.2 Concept of Business Finance, Goals & Objectives of Financial Management, Sources of Financing-Long Term Financing and Short Term Financing.	7
2.	**Business Capital Management** 2.1Concept of Business Working Capital, Significance, Types. Adequacy of Working Capital, Factors Affecting Working Capital Needs. 2.2 Financing Approaches for Working Capital, Methods of Forecasting, Working Capital Requirements.	8
3.	**Basics of Financial Accounting** 3.1Meaning, Scope and Importance of Financial Accounting, Financial Accounting - Concepts and Conventions, Classification of Accounts.	7

	3.2 Rules and Principles Governing Double Entry Book-Keeping System, Meaning, Preparation of Journal, Ledger, Cash Book & Trial Balance.	
4.	**Financial Statement Preparation and analysis** 4.1Preparation of Financial Statements. Profit & Loss Account, Balance Sheet. 4.2 Ratio Analysis.	8

Text Books:

1. Financial Management by Khan & Jain, Text, Problem & Cases, Tata McGraw Hill Publication 5th Edition.
2. Tulsian Financial Management by Dr. P.C.Tulsian, S.Chand Publication 5th Edition.
3. Taxman's Financial Management by Ravi M. Kishore, Taxmann 2017 Edition.

Reference Books:

1. A Textbook of Financial, Cost & Management Accounting by Dr. P.Periasamy, Himalaya Publishing House.
2. Fundamentals of Financial Management by Bhabhtosh Banerjee, PHI publication, 2nd Edition.

ICT / MOOKS: NPTEL / SWAYAM Lectures.

MGMUNIVERSITY

Course Code: ELE24FPP201
Course Name: Community Engagement
Course Category: CEP/FP
Credits: 2
Teaching Scheme:P-4
Evaluation Scheme: TW:30, PR:20
Course Objectives:
1. To Develop an Appreciation of Rural Culture, Life-Style and Wisdom amongst Students
2. To Learn about the Status of Various Agricultural and Development Programmes.
3. To Understand Causes for Distress and Poverty faced by Vulnerable Households and Explore Solutions for the same
4. To Apply Classroom Knowledge of Courses to Field Realities and there by Improve Quality of Learning
Course Outcomes: At the end of the course, the students will be able to -
CO1: Gain an Understanding of Rural Life, Indian Culture & Ethos and Social Realities.
CO2: Develop a Sense of Empathy and Bonds of Mutuality with Local Community.
CO3: Appreciate Significant Contributions of Local Communities to Indian Society and Economy.
CO4: Learn to Value the Local Knowledge and Wisdom of the Community.
CO5: Identify Opportunities for Contributing to Community's Socio-Economic Improvements.

Contents –

Sr. No.	Content	Teaching Hours
1.	**Appreciation of Rural Society** 1.1 Rural Lifestyle, Rural Society, Caste and Gender Relations, Rural Values with Respect to Community, Nature and Resources, Elaboration of "Soul Of India Lies In Villages"(Gandhi), Rural Infrastructure. 1.2 Assignment - Prepare a Map (Physical, Visual or Digital) of the Village you Visited and Write an Essay about Inter-Family Relations in that Village.	12
2.	**Understanding Rural and Local Economy & Livelihood** 2.1 Agriculture, Farming, Land Ownership, Water Management, Animal Husbandry, Non-Farm Livelihoods and Artisans, Rural Entrepreneurs, Rural Markets, Migrant Labour. 2.2 Assignment - Describe Your Analysis of Rural Household Economy, Its Challenges and Possible Pathways to Address them.	16
3.	**Rural and Local Institutions** 3.1 Traditional Rural & Community Organizations, Self-Help Groups, Panchayat Raj Institutions (Gram Sabha, Gram Panchayat, Standing Committees), Nagarpalikas & Municipalities, Local Civil Society, Local Administration.	16

	3.2 Assignment - How Effectively are Panchayati Raj & Urban Local Bodies (ULBS) Institutions Functioning in the Village? What would you suggest to improve their effectiveness? Present a Case Study (Written or Audio-Visual).	
4.	**Rural & National Development Programmes** 4.1 History of Various / Development in India, Current National Programmes: Sarva Shiksha Abhiyan, Beti Bachao, Beti Padhao, Ayushman Bharat, Swatchh Bharat, PM Awas Yojana, Skill India, Gram Panchayat Decentralised Planning, Nrlm, Mnrega, Shram, Jal Jeevan Mission, Sfurti, Atma Nirbhar Bharat,Etc. 4.2 Assignment - Describe the Benefits Received and Challenges Faced in the Delivery of One of these Programmes in the Local Community, Give Suggestions about Improving Implementation of the Programme for the Poor.	16

Ref. Doc.:

1. 1906947_Revised-Draft-Framework-in-the-light-of-NEP-2020.pdf

2. https://www.ugc.gov.in/pdfnews/1906947_Revised-Draft-Framework-in-the-light-of-NEP-2020.pdf

MGMUNIVERSITY

Course Code: MGM56VEL102
Course Name: Constitution of India
Course Category: VEC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:30, ESE:20
Duration of ESE: 1 Hour
Pre-requisites: None.
Course Objectives: 1. Understand the Constitution and its importance. 2. Sensitize the students about Fundamental Rights and Duties Enshrined under Indian Constitution. 3. Familiarize students with the working of Indian Constitution.
Course Outcomes: At the end of the course, the students will be able to – **CO1:**Understand and Explain the Significance of Indian Constitution as the Fundamental Law of the Land. **CO2:**Exercise Fundamental Rights in Proper Sense at the Same Time Identify Responsibilities in National Building. **CO3:**Analyze the Indian Political System, Electoral Process, Emergency Provisions and Amendment Procedure. **CO4:**Understand the Powers and Functions of the Union, State and Local Governments in detail.

Contents:

Unit No.	Content	Teaching Hours
1.	**Historical Background and Philosophy of Indian Constitution** 1.1Societies before and after the Constitution Adoption, Introduction to the Indian constitution, Making of the Constitution. 1.2 Role of the Constituent Assembly, Preamble and Nature of Indian Constitution, Salient Features of Indian Constitution.	8
2.	**Fundamental Rights & Duties** 2.1Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion. 2.2 Cultural and Educational Rights, Right to Constitutional Remedies, Fundamental Duties.	8
3.	**Directive Principles of State Policy and Other Important Constitutional Provisions** 3.1Meaning and Significance of Directive Principles, Economic Principles, Political Principles. 3.2 Emergency Provisions under Indian Constitution, Election Commission.	8

4.	**Organs of Government** 4.1Centre and State Legislature, Centre and State Executive, Centre and State Judiciary, Local Self Government.	6

Text Books:

1. D. D. Basu, Introduction to the Constitution of India, Lexis Nexis, 2013.
2. Pravin Kumar, Basic Mechanical Engineering, Pearson Education; Second Edition, 2018.
3. M. P. Jain, Outline of Indian Legal and Constitutional History, Lexis Nexis, 2014.
4. Constitutional Law of India - J.N.Pandey.

Reference Books:

1. Austin, The Constitution of India, Cornerstone of a Nation. Oxford University press 1966.

MGMUNIVERSITY

Semester-IV

Course Code: ELE24PCL204
Course Name: Network Theory
Course Category: PCC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Basic Components R, L & C, Elementary Mathematics, Basic Electrical Parameters.
Course Objectives: 1.To Introduce the Fundamentals of Electrical Network Analysis using Graph Theory, Matrices, Differential Equations, Network Theorems, Symmetrical Component Analysis. 2.To Introduce Transient Network Analysis and their application. 3. To Introduce Multiport Network, Network Function and Network Synthesis Techniques.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Understand the Use of Network Topology and Network Solving Techniques for Solving Complex Electrical Networks. **CO2:** Understand the Importance of Transients in Electrical Circuits and can Apply to Real Life Problem. **CO3:** Students would be able to know various Two Port Parameters, Network Functions, Pole Zero Plot and the Time Domain Behavior of Electrical Networks. **CO4:** Students would be able to Design and Analyze Passive Electrical Circuits using Network Synthesis Techniques and the basic idea about Electrical Filters.

Contents:

Unit No.	Content	Teaching Hours
1.	**Network Theorems** 1.1 Series Parallel Combination of Resistors, Voltage Division Current Division, Voltage and Current Laws (KVL / KCL), Mesh Analysis, Node Analysis, Source Transformation. 1.2 Superposition, Reciprocity, Thevenin's, Norton's, Maximum Power Transfer Theorem for both AC and DC and its Problems.	8
2.	**Terminal Characteristics of Network and Resonance** 2.1 Z Parameters, Y Parameters, Series Resonance, Selectivity, Bandwidth, Q Factor 2.2 Circuit Magnification Factor, Parallel Resonance, Comparison of Series and Parallel Resonance.	7
3.	**Trigonometric and Exponential Fourier Series**	8

	3.1 Discrete Spectra and Symmetry of Waveform, Steady State Response of A Network to Non-sinusoidal Periodic Inputs, Power Factor, Effective Values. 3.2 Fourier Transform and Continuous Spectra, Three Phase Unbalanced Circuit and Power Calculation.	
4.	**Laplace Transforms and Properties** 4.1 Partial Fractions, Singularity Functions, Waveform Synthesis, Analysis of RC, RL and RLC Networks with and without Initial Conditions with Laplace Transforms Evaluation of Initial Conditions.	7

Text Books:

1. Applied Electronics by Dr. R.S. Sedha S. Chand Publication.
2. Network Analysis and Synthesis by Ravish Singh.
3. Network Theory by Dr. K.M. Soni, Katson Books.
4. Network Theory A Simplified Approach by Dr. Shylashree N., MedTech Publication, 3rd Edition.

Reference Books:

1. Electronic Devices and Circuits: Jacob Millman & C Halkias, Tata McGraw Hill
2. Network and Systems: D Roy Choudhary, New Age International Publishers.
3. Network Thoery by Shobha Agrwal, Khanna Publications.
4. Network Analysis and Synthesis by Smarajith Singh, PHI Publication.

ICT / MOOKS: NPTEL / SWAYAM Lectures.

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Course Code: ELE24PCP204
Course Name: Network Theory Lab
Course Category: PCC
Credits: 1
Teaching Scheme: P-4
Evaluation Scheme: TW:30, PR:20
Pre-requisites: Basic Components R, L & C, Elementary Mathematics, Basic Electrical Parameters.
Course Objectives:
1. To Introduce the Fundamentals Of Electrical Network Analysis using Graph Theory, Matrices, Differential Equations, Network Theorems, Symmetrical Component Analysis.
2. To Introduce Transient Network Analysis and their application.
3. To Introduce Multiport Network, Network Function and Network Synthesis Techniques.
Course Outcomes: At the end of the course, the students will be able to –
CO1: Apply Network Theorems to the Circuit to Find Currents and Voltages.
CO2: Evaluate Two Port Networks Parameters and Resonance Circuits.
CO3: Students would be able to know various Two Port Parameters, Network Functions, Pole Zero Plot and the Time Domain Behaviour of Electrical Networks.
CO4: Students would be able to Design and Analyze Passive Electrical Circuits using Network Synthesis Technique.

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Verification of Superposition Theorem.	2
2.	Verification of Reciprocity Theorem.	2
3.	Verification of Thevenin's Theorem.	2
4.	Verification of Norton's Theorem.	2
5.	Verification of Maximum Power Transfer Theorem.	2
6.	Plotting Frequency Response of a Series / Parallel Resonant Circuit.	2
7.	Determination of Two Port Network Parameters.	2
8.	Analysis of RC Networks.	2
9.	Analysis of RL Networks.	2

10.	Analysis of RLC Networks.	2

Text Books:

1. Applied Electronics by Dr. R.S. Sedha S. Chand Publication.
2. Network Analysis and Synthesis by Ravish Singh.
3. Network Theory by Dr. K.M. Soni, Katson Books.
4. Network Theory A Simplified Approach by Dr. Shylashree N., MedTech Publication, 3rd Edition.

Reference Books:

1. Electronic Devices and Circuits: Jacob Millman & C Halkias, Tata McGraw Hill
2. Network and Systems: D Roy Choudhary, New Age International Publishers.
3. Network Thoery by Shobha Agrwal, Khanna Publications.
4. Network Analysis and Synthesis by Smarajith Singh, PHI Publication.

ICT / MOOKS: NPTEL / SWAYAM Lectures.

MGMUNIVERSITY

Course Code: ELE24PCL205
Course Name: Analog Circuits
Course Category: PCC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Basic Components R, L & C, Elementary Mathematics, Basic Electronics Parameters.
Course Objectives: 1. To Introduce the Concept of Semiconductors and Devices along with Applications. 2. To Understand Power Supply Components, Amplifiers, Oscillators, Timers.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Understand the Use of Feedback Amplifiers. **CO2:** Understand the Importance of Waveform Generators. **CO3:** Students would be able to know various IC's and SMPS for different applications. **CO4:** Analyse Optoelectronic Devices.

Contents:

Unit No.	Content	Teaching Hours
1.	**Feedback Amplifiers** 1.1 Principle of Feedback Amplifier 1.2 Types of Feedback: Negative and Positive Feedback, Advantages and Disadvantages of Negative Feedback. 1.3 Types of Feedback Connections, Voltage Shunt, Voltage Series, Current Series and Current Shunt: Block Diagram, Circuit Diagram and Operation	7
2.	**Wave Form Generators** 2.1 Oscillators: Need, Oscillator and Amplifier 2.2 Condition for Oscillation (Barkhausen's Criteria), Classification of Oscillators. 2.3 Sine Wave Oscillator: RC Phase Shift Oscillator and Crystal Oscillator, Concept, Working and Applications. 2.4 Sweep Generator: Miller Sweep, Bootstrap Circuit, Current Time Base Generator.	8
3.	**IC Voltage Regulators and SMPS** 3.1 Types of IC Voltage Regulator: Fixed And Variable: 78XX, 79XX, Specification, Series and LM723. LM317, Line and Load Regulation. 3.2 SMPS Block Diagram, Working Principle, Specifications, Special	8

	Features, Advantages, Disadvantages and Applications. 3.3 Use of Heat Sink for Regulated Power Supply.	
4.	**Optoelectronic Devices** 4.1 Electron Emission: Types, Applications. 4.2 Symbols, Working and Applications Of Photo Diode, Optoisolator, Photo Voltaic Cell, LED, LDR, LCD, Opto Coupler.	7

Text Books:

1. Electronics Principles by Malvino, Mc. Graw Hill, Third edition.2000.
2. Electronics Analog and Digital by I. J. Nagrath, PHI Learning Pvt. Ltd., 2013 Edition.
3. Analog Circuits by S.Siddeswara Reddy.
4. Analog Electronics by J. B. Gupta, Katson Books.

Reference Books:

1. Linear Integrated Circuits by Ramakant A. Gayakwad, PHI Learning Pvt. Ltd., Fourth Edition.
2. Electronics Principles and Applications by Charles A Schuler and Roger L.Tokhiem, Sixth Edition, Mc. Graw Hill , 2008.
3. Fundamentals of Analog Circuits by Floyd and Buchia, Second Edition, Pearson Publication.
4. Analog Circuits by A. Rajkumar.

ICT / MOOKS: NPTEL / SWAYAM Lectures.

MGMUNIVERSITY

Course Code: ELE24PCP205
Course Name: Analog Circuit Lab
Course Category: PCC
Credits: 1
Teaching Scheme: P-2
Evaluation Scheme: TW:30, PR:20
Pre-requisites: Basic Components R, L & C, Elementary Mathematics, Basic Electronic Parameters.
Course Objectives:
1. To Introduce the Concept of Semiconductors and Devices along with Applications.
2. To Understand Power Supply Components, Amplifiers, Oscillators, Timers.
Course Outcomes: At the end of the course, the students will be able to –
CO1: Understand the Use of Feedback Amplifiers.
CO2: Understand the Importance of Waveform Generators.
CO3: Students would be able to know various IC's and SMPS for different applications.
CO4: Analyse Optoelectronic Devices.

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Build / Test the Performance of Audio Power Amplifier.	2
2.	Use Transistor to Build / Test Voltage Series Feedback Amplifier Parameters with and Without Feedback.	2
3.	Use Transistor to Built Test Voltage Shunt Feedback Amplifier Parameters With and Without Feedback.	2
4.	Test the Effect of Positive Feedback on the given Amplifier.	2
5.	Test the Effect of Negative Feedback on the given Amplifier.	2
6.	Build RC Phase Shift Oscillator and Measure the Generated Frequency using CRO.	2
7.	Build Crystal Oscillator and Measure the Generated Frequency using CRO.	2
8.	Simulate Hartley Oscillator using any Relevant Simulation software. (Like Spice, Multisim, Lab View, Ltspice, Octeva).	2

9.	Generate a Waveform using Miller's Sweep Generator and Measure Sweep Time and Retrace Time.	2
10.	Study of Optoelectronic Devices with its applications.	2

Text Books:

1. Electronics Principles by Malvino, Mc. Graw Hill, Third edition.2000.
2. Electronics Analog and Digital by I. J. Nagrath, PHI Learning Pvt. Ltd., 2013 Edition.
3. Analog Circuits by S.Siddeswara Reddy.
4. Analog Electronics by J. B. Gupta, Katson Books.

Reference Books:

1. Linear Integrated Circuits by Ramakant A. Gayakwad, PHI Learning Pvt. Ltd., Fourth Edition.
2. Electronics Principles and Applications by Charles A Schuler and Roger L.Tokhiem, Sixth Edition, Mc. Graw Hill , 2008.
3. Fundamentals of Analog Circuits by Floyd and Buchia, Second Edition, Pearson Publication.
4. Analog Circuits by A. Rajkumar.

ICT / MOOKS: NPTEL / SWAYAM Lectures.

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Course Code: ELE24PCL206
Course Name: Analog &Digital Communication
Course Category: PCC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Basics of Electronics Engineering.
Course Objectives: 1. To Understand the Basic Elements of Communication System and Amplitude Modulation. 2. To Develop Skill to Design FM Trans-Receiver. 3. To Interpret Importance of Sampling Theorem for Digital Communication System. 4. To be Aware of Noise in Communication Systems.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Demonstrate the Basic Concept of Communication System and Amplitude Modulation. **CO2:** Evaluate the Behavior of FM in Frequency Domain. **CO3:** Apply the Concept of Sampling Theorem to Design Pulse Analog and Digital Modulation. **CO4:** Compute Noise Parameters in the Communication Systems.

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Contents:

Unit No.	Content	Teaching Hours
1.	**Introduction to Communication System and Amplitude Modulation** 1.1 Elements of Communication System, Types of Communication System, Need of Modulation, Classification of Modulation. 1.2 Electromagnetic Spectrum & Its Applications, Multiplexing Techniques. 1.3 AM Definition, Mathematical Equation of AM, Modulation Index, Frequency Spectrum of AM, Bandwidth, Power Calculation. 1.4 AM Receiver and AM Detectors.	10
2.	**Frequency Modulation** 2.1 Definition, Mathematical Equation of FM, Frequency Spectrum. 2.2 FM Generation Methods by Reactance Modulators, Pre-Emphasis and De-Emphasis, FM Receiver. 2.3 FM Detectors, Slope Detector and Ratio Detector.	7
3.	**Pulse Modulation** 3.1 Classification of Pulse Analogue Modulation. 3.2 Sampling Theorem, Digital Modulation Techniques, Quantization and Encoding, Linear Prediction. 3.3 DPCM, Shift Keying Techniques.	7
4.	**Noise** 4.1 Noise in Analog and Digital Communication System, Sources of Noise. 4.2 SNR, Noise Figure, Noise Factor, Noise Resistance, Noise Temperature.	6

Text Books:

1. Electronic Communication, Dennis Roddy & Coolen, TMH.
2. Principles of Communication System, Taub Schilling, Mc Graw-HILL.
3. Digital Communication, Sanjay Sharma, Katariya & Sons.
4. Analog and Digital Communication by T.L. Singal.

Reference Books:

1. An Introduction to Analog and Digital Communication, Haykin Simon, John wiley.
2. Digital Communication, Prokis. J. G. Pearson Education.
3. Digital Communications Fundamentals and Applications, Sklar Bernard, Pearson Education.
4. Principles of Communication Engineering, Anokh Singh, S.Chand.
5. Introduction to Analog and Digital Communications by Simon Haykin, 2nd Edition, Wiley Publication.

ICT / MOOKS:

1. https://onlinecourses.nptel.ac.in/noc19_ee46/preview
2. http://nptel.ac.in/courses/117101051/

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Course Code: ELE24PCP206
Course Name: Analog & Digital Communication Lab
Course Category: PCC
Credits: 1
Teaching Scheme: P-2
Evaluation Scheme: TW:30, PR:20
Pre-requisites: Basics of Electronics Engineering.
Course Objectives:
1. To Understand the Usage of Various Equipments in Communication Engineering.
2. To Understand Various Modulation Techniques.
3. To Learn the Transmitters and Receivers in Electronic Communication.
4. To Study Multiplexing Techniques.
Course Outcomes: At the end of the course, the students will be able to –
CO1: Demonstrate the Use of Various Equipments in Communication Engineering.
CO2: Compare Various Modulation Techniques.
CO3: Interpret Pre-Emphasis and De-Emphasis.
CO4: Demonstrate the Multiplexing Techniques.

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Contents –

Sr. No.	List of Experiments	Lab Hours
1.	To Study RF Signal Generator.	2
2.	To Find MI of AM by Direct and Indirect Method.	2
3.	To Study the AM Transmitter and Receiver.	2
4.	To Study the FM Transmitter and Receiver.	2
5.	To Study Four Channel Analog TDM System.	2
6.	Sampling and Reconstruction of Lower Sampling Rate and Higher Sampling.	2
7.	To Perform PAM / Demodulator.	2
8.	To Perform PWM / Demodulator.	2
9.	To Perform PPM / Demodulator.	2
10.	To Plot Frequency Response of Pre-Emphasis and De-Emphasis.	2

Text Books:

1. Electronic Communication, Dennis Roddy & Coolen, TMH.
2. Principles of Communication System, Taub Schilling, Mc Graw-HILL.
3. Digital Communication, Sanjay Sharma, Katariya & Sons.
4. Analog and Digital Communication by T.L. Singal.

Reference Books:

1. An Introduction to Analog and Digital Communication, Haykin Simon, John wiley.
2. Digital Communication, Prokis. J. G. Pearson Education.
3. Digital Communications Fundamentals and Applications, Sklar Bernard, Pearson Education.
4. Principles of Communication Engineering, Anokh Singh, S.Chand.
5. Introduction to Analog and Digital Communications by Simon Haykin, 2nd Edition, Wiley Publication.

ICT / MOOKS:

1. https://onlinecourses.nptel.ac.in/noc19_ee46/preview
2. http://nptel.ac.in/courses/117101051/

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Course Code: ELE24VSP201
Course Name: Introduction of Microfabrication Lab
Course Category: VSEC
Credits: 2
Teaching Scheme:P-4
Evaluation Scheme: TW:60, PR:40
Pre-requisites: Basics of Physics and Chemistry.
Course Objectives:
1. Describe the Principles, Advantages and Disadvantages of Basic Lithography, Thin Film Deposition and Etching Techniques.
2. Determine an appropriate Sequence of Steps in Order to Design a Microsystems that can be used in Microelectronics, Micromechanics, Microelectromechanics, Photonics, Biosystems etc.
Course Outcomes: At the end of the course, the students will be able to –
CO1: Learning about Different Types of Micro-Fabrication Methods and Other Emerging Technologies.
CO2: Understand the Concept of Micromechanics.
CO3: Demonstrate the Micromechanics Transformations Techniques.
CO4: Select appropriately the correct fabrication process for a specific micro-device or Microsystems.

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Determination of Microfabrication Systems.	2
2.	To Perform Structural Behaviour Microfabrication.	2
3.	Test the Different Types of Microfabrication Technologies.	2
4.	Determination of Micromechanics Systems.	2
5.	Find Out Micromechanics Molecular Structure & Its Properties.	2
6.	Test the Micromechanics Transformations Techniques.	2
7.	Determination of Sacrificial Processes and Bonding Processes.	2
8.	Study of Special Machining.	2
9.	Analysis of Clean Room-Yield Model.	2

10.	Analysis of Layer Transfer Devices.	2

Text Books:

1. Fundamental of Micro Fabrication by Madore J. of CRC Press, 2002.
2. Microchip Fabrication by Peter Van Zant, McGraw Hill, 2004.
3. Microfluidics and Microfabrication by Suman Chakraborty, Springer Publication.

Reference Books:

1. Introduction to Micro Fabrication by Sami Franssila of John Wiley and Sons Ltd., UK
2. Micro Fabrication and Nano manufacturing by Mark J. Jackson of CRC Press, 2006.
3. Introduction to Microelectronic Fabrication by Richard C. Jaeger, 2nd Edition.
4. Introductory MEMS: Fabrication and Applications by Thomas M. Adams, Springer Publications.

ICT / MOOKS: NPTEL / SWAYAM Lectures.

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Course Code: ELE24HSL202
Course Name: Entrepreneurship Development
Course Category: EEMC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Pre-requisites: Knowledge of Planning.
Course Objectives: 1. Describe the Various Functional Areas of Management. 2.Essentials to Avoid Failure in Entrepreneurship.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** To Develop Skills related to Various Functional Areas of Management (Marketing Management, Financial Management, Operations Management, Personnel Management etc.). **CO2:** To Develop Skills related to Project Planning And Business Plan Development. **CO3:** To Determine and Compare Various Skills of Successful Entrepreneurs, Business Organizations and Business Laws. **CO4:** To Examine Essentials to Avoid Failure in Entrepreneurship.

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Contents:

Unit No.	Content	Teaching Hours
1.	**Introduction to Entrepreneurship Development** 1.1 Entrepreneurship Development, Evolution of Entrepreneurship, Meaning, Definition and Concept of Enterprise. 1.2 Concepts of Entrepreneurship, Role of Entrepreneurship in Economic Development, Factors Affecting Entrepreneurship. 1.3 Entrepreneurial Development models and Theories, Entrepreneurial Trait Tests, Business Opportunity Identification.	8
2.	**Entrepreneur** 2.1 Major Types of Entrepreneurship, Techno Entrepreneurship, Women Entrepreneurship, Social Entrepreneurship, Entrepreneurship (Corporate entrepreneurship), Rural Entrepreneurship, Family Business etc. 2.2 The Entrepreneurial Decision Process, Entrepreneurial Environment – Political, Legal, Technological, Natural, Economic, Socio – Cultural etc.	7
3.	**Project Planning, Skills for Successful Entrepreneurs & Business Organizations and Business Laws:** 3.1 Project Report, Project Appraisal, Setting Up an Industrial Unit – Procedure And Formalities in Setting Up an Industrial Unit; Business Plan Development. 3.2 Communication Skills, Creativity and Problem-Solving, Innovation,	8

	Negotiation Skills, Risk Management. 3.3 Types of Business Organizations – Sole, Proprietorship, Joint Hindu Family Business, Partnership, Limited Liability Partnership (LLP), Corporate Governance, Franchising, Business Laws in India to Start Business.	
4.	**Concepts of Entrepreneurship Failure** 4.1 Issues of Entrepreneurial failure, Reasons of Entrepreneurial Failure, Essentials to Avoid Failure in Entrepreneurship. 4.2 Case Study of Failure in Entrepreneurship.	7

Text Books:

1. Dynamics of Entrepreneurship Development by Vasant Desai.
2. Entrepreneurship Development and small business management by Poornima M. Charantimath.
3. Dynamics of Entrepreneurship Development by Vasant Desai.

Reference Books:

1. Innovation and Entrepreneurship by Peter F. Drucker
2. Kathleen R Allen, Launching New Ventures, An Entrepreneurial Approach, Cengage Learning, 2016.
3. Vijay Sathe, Corporate Entrepreneurship, Cambridge, 2009.

ICT / MOOKS: NPTEL / SWAYAM Lectures.

MGMUNIVERSITY

Course Code: MGM21VEL101
Course Name: Environment Studies
Course Category: VEC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:30, ESE:20
Duration of ESE: 1 Hour
Pre-requisites: Basic Science.
Course Objectives: 1. To Study the Environment and Ecosystems. 2. To Study Different Types of Natural Resources. 3. To Study Concept of Biodiversity and its Conservation. 4. To Study the Concept of Causes, Effects and Control of Different Types of Environmental Pollution. 5.To Study Population Growth and Its Impact on Environment.
Course Outcomes: At the end of the course, the students will be able to – **CO1:**Describe Environment and Ecosystem. **CO2:**Explain Environmental Impacts of Human Activities on Natural Resource. **CO3:**Identify the Importance of Conservation of Biodiversity. **CO4:**Describe the Problems of Environmental Pollution, Its Impact on Human and Ecosystem and Control Measures. **CO5:** Identify the Impact of Increased Population on Environment.

Contents:

Unit No.	Content	Teaching Hours
1.	**Ecosystem and the Environment** 1.1 Definition, Scope and Importance of Environmental Studies, Need for Public Awareness. 1.2 Concepts of an Ecosystem, Structure and Functions of an Ecosystem. 1.3 Producers, Consumers and Decomposers in an Ecosystem.	6
2.	**Natural Resources** 2.1 Renewable and Non-Renewable Resources, Role of Individuals in Conservation of Natural Resources for Sustainable Lifestyles. 2.2 Use and Over Exploitation of Forest Resources, Deforestation, Timber Extraction, Mining, Dams and their Effects on Forest and Tribal People. 2.3 Use and Over Exploitation of Surface and Ground Water Resources, Floods, Drought, Conflicts over water, Dams- Benefits and Problems.	6
3.	**Biodiversity and its Conservation** 3.1Definition, Genetic, Species and Ecosystem Diversity, Bio-Geographical Classification of India.	6

	3.2 Value of Biodiversity: Consumptive Use, Productive Use, Social, Ethical, Aesthetic and Option Values.	
4.	**Environmental Pollution** 4.1 Definition, Causes, Effects and Control Measures of Air Pollution. Water Pollution and Soil Pollution. 4.2 Causes, Effects and Control Measures of Marine Pollution, Noise Pollution, Thermal Pollution, Nuclear Hazards.	6
5.	**Population Issues and the Environment** 5.1 Population Growth, Variation Among Nations, Population Explosion-Family Welfare Programmes, Environment and Human health, Human Rights, Value Education, HIV/AIDS. 5.2 Woman and Child Welfare, Role of Information Technology in Environment and Human Health, Case Studies	6

Text Books:

1. Bharucha Erach (ed) Text Book of Environmental Studies., University Press (India) Pvt. Ltd.
2. Bharucha Erach, 2003. The Biodiversity of India, Mapin Publishing Pvt. Ltd, Ahmedabad – 380013.
3. Bharucha Erach, 2003. The Biodiversity of India, Mapin Publishing Pvt. Ltd, Ahmedabad – 380013.
4. Deswal, S. and Deswal, A. 2005. A Basic Course in Environmental Studies, Dhanpat Rai & Co ltd., Delhi.

Reference Books:

1. IndiaSantra S.C, (ed), Environmental Science, New Central Agency Pvt. Ltd. Kolkata, India.
2. Botbin, D., and Keller, E.1995. Environmental Science, John Wiley and Sons, USA.
3. Cunningham, W.P. and Saigo, B.W. 1997. Environment Science, McGraw-Hill, USA.
4. Lal, R. and Stewart, B.A. 1992. Advances in Science: Soil Restoration Springer Verlag, New York.

ICT / MOOKS: NPTEL / SWAYAM Lectures.